INCH-POUNDS
MIL-DTL-62421(AT)
!6 AUGUST 2002

DETAIL SPECIFICATION

CONTROL, UNIT, ELECTRONIC, DIGITAL

This specification is approved for use by the U.S. Army Tank-automotive and Armaments Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the system and acceptance requirements for a Digital Electronic Control Unit (DECU), herein referred to as the "assembly". The assembly provides fuel handling and vehicle interface capabilities for the AGT 1500 gas turbine engine. The AGT 1500 is used in the M1, M1A1, M1A2 and Heavy Assault Bridge (HAB) applications.

1.2 Classification. This specification applies to the following classification types of assemblies (see 6.3).

Type I - 12324815 includes space for a MIL-STD-1553 data bus, an integral backup battery and a diagnostic window (M62421-1).
Type Ia - 12324815-1 includes space for a MIL-STD-1553 data bus, a removeable backup battery, and a diagnostic window (M62421-2).
Type II - 12324820 includes a MIL-STD-1553 data bus, a removeable backup battery, and does not include a diagnostic window (M62421-3).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/AEIT, 6501 E. 11 Mile Road, Warren, MI 48397-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963

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2. Applicable documents.

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirement documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-52557	- Fuel Oil, Diesel, for Posts, Camps and Stations.
A-A-59133	- Cleaning Compound, High Pressure (Steam) Cleaner.

DEPARTMENT OF DEFENSE

MIL-DTL-5624	- Turbine Fuel, Aviation, Grades JP-4, JP-5, and JF-5/JP-8 ST.
MIL-PRF-6083	- Hydraulic Fluid, Petroleum Base, for Preservation and Operation.
MIL-DTL-15024	- Plates, Tags, and Bands for Identification of Equipment, General Specification for.
MIL-F-16884	- Fuel, Naval Distillate.
MIL-PRF-19500	- Semiconductor Devices, General Specification for.
MIL-S-22473	- Sealing, Locking and Retaining Compounds: (Single Component).
MIL-C-22520/36	- Compression Tool, Hand, Type 2 For Semirigid Coaxial Cable (Category F Connectors).
MIL-PRF-46170	- Hydraulic Fluid, Rust Inhibited, Fire-Resistant, Synthetic Hydrocarbon Base.
MIL-DTL-83133	- Turbine Fuels, Aviation, Kerosene Types, NATO F-34(JP-8), NATO F-35, and JP-8 +100.

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STANDARDS

FEDERAL

FED-STD-595 - Colors Used in Government Procurement.

DEPARTMENT OF DEFENSE

MIL-STD-130 - Identification Marking of U.S. Military Property.
MIL-STD-171 - Finishing of Metal and Wood Surfaces.
MIL-STD-202 - Test Methods for Electronic and Electrical Component Parts.
MIL-STD-252 - Wired Equipment Classification of Visual and Mechanical Defects.
MIL-STD-461 - Requirements for the Control of Electromagnetic Interference Characteristics of Subsystems and Equipment.
MIL-STD-464 - Electromagnetic Environmental Effects Requirements for Systems.
MIL-STD-810 - Environmental Engineering Considerations and Laboratory Tests.
MIL-STD-883 - Microcircuits.
MIL-STD-1553 - Digital Time Division Command/Response Multiplex Data Bus.
MIL-STD-38510 - Microcircuits, General Specification for.

HANDBOOKS

DEPARTMENT OF DEFENSE

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment.
MIL-HDBK-454 - General Guidelines for Electronic Equipment.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service, 700 Robins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.2.2 Other Government documents, drawings, and publications. The following other documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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DRAWINGS

ARMY

12324815	- Digital Electronic Control Unit (type I).
12324815-1	- Digital Electronic Control Unit (type Ia)
12324820	- Digital Electronic Control Unit (type II).
12344343	- Exterior CARC paint specification.

(Copies of drawings are available from the U.S. Army Tank-automotive and Armaments Command, ATTN: AMSTA-TR-E/AEIT, 6501 E. 11 Mile Road, Warren, MI 48397-5000.)

PUBLICATIONS

NATIONAL BUREAU OF STANDARDS (NBSS)

NBS Circular 561

DEPARTMENT OF DEFENSE

DI-IPSC-81433	- Software Requirements Specification (SRS).
DI-IPSC-81435	- Software Design Description (SDD).
DI-IPSC-81439	- Software Test Description (STD).
DI-IPSC-81441	- Software Product Specification (SPS).
DI-IPSC-81442	- Software Version Description (SVD).

(Copies of publications are available from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues for the documents cited in the solicitation (see 6.2).

ALLIEDSIGNAL ENGINES AND SYSTEMS

21-9187	- Data Packet Specification for DECU.

(Application for copies should be addressed to the AlliedSignal Engines and Systems, 111 S. 34th Street, Phoenix, AZ 85034.)

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/MIL-STD-1815 - ADA Programming Language.

(Application for copies should be addressed to the American National Standards Institute, 25 W. 43rd Street, 13th Floor, New York, NY 10036 or website: www.ansi.org.)

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME-Y14.100 - Engineering Drawings Practices.

(Application for copies should be addressed to the American Society of Mechanical Engineers, 3 Park Avenue, M/S 10D, New York, NY 10016 or website: www.asme.org)

AMERICAN SOCIETY OF TESTING AND MATERIALS (ASTM)

ASTM D975 - Oils, Diesel Fuel (DoD adopted).
ASTM D1655 - Aviation Turbine Fuels (DoD adopted).
ASTM D4814 - Fuel, Automotive, Spark-Ignition Engine (DoD adopted).

(Application for copies should be addressed to the American Society of Testing and Materials, 100 Barr Harbor Drive, West Conshohocken, PA 19428-2959 or website: www.astm.org)

ELECTRONIC INDUSTRIES ALLIANCE (EIA)

EIA-232 - Interface Between Data Terminal Equipment and Data Circuit-Termination Equipment Employing Serial Binary Data Interchange.

(Application for copies should be addressed to the Electronic Industries Alliance, 2500 Wilson Blvd., Arlington, VA 22201-3834 or website: www.eia.org.)

GENERAL DYNAMICS LAND SYSTEMS (GDLS)

LS 198800000160(S) - Nuclear Hardening Criteria.

(Application for copies should be addressed to the General Dynamics Land Systems, 38500 Mound Road, Sterling Heights, MI 48310-3200 or website: www.gdls.com)

INSTITUTE OF ELECTRICAL AND ELECTRONIC ENGINEERS (IEEE)

IEEE 12207 - Standard for Information Technology – Software Life Cycle Processes (DoD adopted).

(Application for copies should be addressed to the Institute of Electrical and Electronic Engineers, IEEE Service Center, 455 Hoes Lane, P.O. Box 1331, Piscataway, NJ 08855-1331 or website: www.ieee.org)

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SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

SAE-AS478 - Identification Marking Methods (DoD adopted).
SAE-AMS-DTL-23053/8 - Insulation Sleeving, Electrical, Heat Shrinkable, Polyvinylidene Fluoride, Semi-Rigid, Crosslinked.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096 or website: www.sae.org)

2.4 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.1.1.

3.2 Design, materials, and manufacturing processes. Unless otherwise specified, the design, materials and manufacturing process selection is the prerogative of the contractor as long as all articles submitted to the government fully meet the operating, interface, support and ownership, and operating environment requirements specified.

3.2.1 Recycled, recovered, or environmentally preferable materials. Recycled, recovered, or environmentally preferable materials should be used to the maximum extent possible provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.2 Dissimilar metals. Dissimilar metals shall not be used in direct contact with one another unless protected to prevent galvanic corrosion, with the exception where assemblies use stainless steel screws for mounting and grounding.

3.2.3 Configuration.

3.2.3.1 Type I. The assembly configuration shall conform to Drawing 12324815.

3.2.3.2 Type Ia. The assembly configuration shall conform to Drawing 12324815-1.

3.2.3.3 Type II. The assembly configuration shall conform to Drawing 12324820.

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3.2.4 Design and construction.

3.2.4.1 Surface treatment and painting. Unless otherwise specified (see 6.2), surfaces other than ground/bonding strap mounting shall be treated in accordance with MIL-STD-171 and Drawing 12344343. Paint color shall be FED-STD-595, type I, class 17875 (insignia white).

3.2.4.2 Soldering. Soldering shall be of good commercial practice and quality, and shall not adversely affect the operational, environmental and maintainability of the overall assembly.

3.2.4.3 Crimp contacts. Crimp type connector contacts shall be crimped with tools and procedures specified in accordance with MIL-C-22520/36, or equivalent.

3.2.4.4 Wire wrap connections. Wire wrap connections shall be of good commercial practice and quality, and shall not adversely affect the operational, environmental and maintainability of the overall assembly.

3.2.4.5 Heat shrink tubing. Heat shrinking of tubing shall be in accordance with SAE-AMS-DTL-23053/8. Tubing shall be uniformly heat shrunk until the tubing configuration reflects the contour of the surface to be insulated.

3.2.4.6 Wiring stripping. Removal of outer jacket insulation from all wires shall be of good commercial practice and quality, and shall not adversely affect the operational, environmental and maintainability of the overall assembly.

3.2.4.7 Thread locking compound. Thread locking compound shall meet the requirements of MIL-S-22473, or equivalent.

3.2.4.8 Insulating and impregnating compounds. Insulating and impregnating compounds shall not support rapid combustion. Impregnating compounds shall not lessen the dielectric strength of, or cause injury or deterioration to, adjacent metal or plastic parts, upon original application or through aging.

3.2.5 Software programming. The software development environment shall be defined in the System Level documents listed in 3.2.5.1. These requirements shall apply, as a minimum, to the assembly's software. The software shall be designated as a Computer Software Configuration Item (CSCI), and may be redefined by the supplier by partitioning into additional CSCIs. Software documentation for each CSCI shall be as listed in 3.2.5.1.

3.2.5.1 Software documentation. Each CSCI documentation shall include the following:

a. DI-IPSC-81433 - Software Requirements Specification (type II)
b. DI-IPSC-81435 - Software Design Description (type II)
c. DI-IPSC-81439 - Software Test Description (type II)
d. DI-IPSC-81441 - Software Product Specification (type II)
e. DI-IPSC-81442 - Software Version Description

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3.2.5.2 Software language. Software shall be programmed in the programming language ADA in accordance with ANSI/MIL-STD-1815 (see 6.5.1).

3.2.5.3 Software location. Operational software and software for initialization and Built-In-Test (BIT) shall be located in the Electrically Erasable Programmable Ram Only Memory (EEPROM).

3.2.6 Central Processing Unit (CPU). The CPU shall use a 16-bit Complementary Metal-Oxide Semiconductor (CMOS) microprocessor with instruction set architecture specified herein. There shall be 20 percent (%) excess throughput capacity of current cycle time at the time of production. Cycle time shall be defined as 20 milliseconds (ms).

3.2.7 Memory.

3.2.7.1 EEPROM. The non-volatile EEPROM shall be connected to the local CPU bus. The EEPROM shall be capable of 10 000 write cycles per byte with data retention.

a. For types I and Ia, the assembly shall be configured for a minimum of 16K words (16 bits per word) of excess program memory at initial production configuration.
b. For type II, the assembly shall be configured for a minimum of 14K words (16 bits per word) of excess program memory at initial production configuration.

3.2.7.1.1 Storage EEPROM.

3.2.7.1.1.1 Fault EEPROM (types I and Ia). Types I and Ia assemblies shall be configured with a minimum of 2 kilobyte (K) words for fault EEPROM to support tables I and II.

3.2.7.1.1.2 Fault EEPROM (type II). Type II assemblies shall be configured with a minimum of 4K words for fault EEPROM to support AlliedSignal Engines document 21-9187.

3.2.7.2 Random Access Memory (RAM). The on-board RAM shall be connected to the local CPU bus and shall be configured for available memory to support the throughput requirements.

3.2.7.3 Serial communication link. The internal clock for the baud rate generator shall be software programmable (see 3.3.1.6.1). A Universal Asynchronous Receiver/Transmitter (UART) shall be provided to allow the CPU to communicate with the EIA-232 interface. The assembly shall allow use of standard byte or bit-oriented protocols in accordance with EIA-232 in an asynchronous mode.

3.3 Operating requirements. Each assembly shall provide the following functional, operational, and performance capabilities.

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3.3.1 Performance.

3.3.1.1 Real-time clock. The assembly shall be provided with a real-time clock. The clock shall be set to National Institute of Standards and Technology Radio station WWV Time plus or minus (±) 1 minute. The clock shall have a dedicated battery with a minimum life of 5 years, and shall maintain an accuracy of ± 2 minutes per month at 20 to 120 degrees Fahrenheit (°F) (-7 to +49 degrees Celsius (°C)) (see 4.3.3.1.1).

3.3.1.2 Backup battery. The assembly shall be equipped with a self-contained, rechargeable backup battery that shall operate as follows:

a. Provide a 30 minute energy supply for all engine operation levels at or above 20°F (-7°C)
b. Function as a supplemental energy source to the assembly during a start, and,
c. Meet the requirements of 3.6 at the time of start (see 4.3.3.1.2).

The battery shall be tested for a minimum shelf life of 18 months at room temperature of 73 ± 18°F (23 ± 10°C) without maintenance. For a battery discharged to 18.1 volt (V) direct current (dc), the recharging time shall be a maximum of 8 hours with a recharging current of 0.5 amp (A). The assembly shall disable the backup battery if battery voltage drops below 16 V. Type I assemblies shall be equipped with an integral backup battery. Types Ia and II assemblies shall be equipped with a removable backup battery.

3.3.1.2.1 Backup battery during engine operation.

3.3.1.2.1.1 Backup battery during engine operation (types I and Ia). If the assembly switches to the backup battery after the power turbine (NPT) governor is satisfied (see 3.3.2.2), the fault lamp (see 3.3.1.8.3.4c) shall flash on and off at 0.5-second intervals. There shall be a 5-second delay from the time the assembly switches to backup power to the time the fault lamp starts to flash. The fault lamp shall stop flashing if the primary power to the assembly is restored. A fault shall be stored in EEPROM (see 4.3.3.1.2.1 and table III).

3.3.1.2.1.2 Backup battery during engine operation (type II). If the assembly switches to the backup battery after the NPT governor is satisfied (see 3.3.2.2), a signal shall be sent over the MIL-STD-1553 (hereafter known as “1553”) data bus (see 4.3.3.1.2.1).

3.3.1.2.2 Backup battery operation on start.

3.3.1.2.2.1 Backup battery operation on start (types I and Ia). During a start attempt, if the assembly switches to the backup battery, and the battery voltage falls below 18 Vdc the assembly shall signal the fault lamp (see 3.3.1.8.3.4c) for ten seconds and a fault (see protective mode (PM), PM IV and 3.3.5.5.1h) shall be stored in EEPROM (see 4.3.3.1.2.2 and table III).

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3.3.1.2.2.2 Backup battery operation on start (type II). During a start attempt, if the assembly switches to the backup battery, and the battery voltage falls below 18 Vdc, a signal shall be sent over the 1553 data bus (see AlliedSignal Engines document 21-9187 and 4.3.3.1.2.2).

3.3.1.3 Nuclear Event Detector (NED). A nuclear event detector shall be provided which shall remove power from the assembly for a minimum of 100 ms when a nuclear event is detected. Total interruption of power shall not exceed 300 ms. Upon return to power, the assembly shall resume prior operating conditions if the compressor speed (NH) is greater than 40% and the turbine inlet temperature (T_7) is greater than 400°F (204°C). The assembly shall shut down the engine if NH is less than 40% or T_7 is less than 400°F (see 4.3.3.1.3).

3.3.1.3.1 NED (types I and Ia). In types I and Ia configuration, a nuclear event message shall not be reported (see 4.3.3.1.3.1).

3.3.1.3.2 NED (type II). In the type II configuration a nuclear event shall be reported across the 1553 data bus. The data commands shall be in accordance with AlliedSignal Engines document 21-9187 (see 4.3.3.1.3.2).

3.3.1.4 Diagnostic window.

3.3.1.4.1 Diagnostic window (types I and Ia). The assembly shall have a diagnostic window to display fault messages. The window display chip shall have at least twelve characters of at least 0.15 inch (in.) (3.8 millimeters (mm)) in height. The window shall be readable with a minimum illuminous intensity of 0.1 Power Spectral Density/Segment at 5 Vdc and a viewing angle of ± 20 degrees from a line normal to the display (see 4.3.3.1.4).

3.3.1.4.2 Diagnostic window (type II). Type II assemblies shall not be equipped with a diagnostic window (see 4.3.3.1.4).

3.3.1.5 Power supply voltage. The assembly shall contain a hardware circuit which measures power supply voltage and protects the assembly. The input voltage of the assembly shall be 18 to 30 Vdc. Power interruption shall not affect the performance of the assembly. The backup battery shall provide power during interruption (see 3.3.1.2). If the power supply voltage fails, the assembly shall cause a CPU reset (see 4.3.3.1.5).

3.3.1.5.1 Power consumption. The peak steady state power consumption from 18 to 30 Vdc input shall not exceed 85 watts (W) (see 4.3.3.1.5.1).

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3.3.1.6 Data bus communications.

3.3.1.6.1 EIA-232. The EIA-232 throughput shall be at a rate of 19 200 baud (see 4.3.3.1.6.1). Data commands shall be in accordance with Appendix A.

3.3.1.6.2 1553 (type II only). The assembly shall meet the throughput requirements of MIL-STD-1553 for remote terminal (RT) mode. Data commands shall be in accordance with 3.3.1.3.2, 3.3.1.7.5 and 3.3.1.10 (see 4.3.3.1.6.2).

3.3.1.7 Signal inputs.

3.3.1.7.1 Engine inlet air temperature (T_1). The assembly shall be capable of processing resistance probe signals with the characteristics as specified in table IV. The assembly shall compensate for a nominal cable resistance of less than or equal to 0.5 ohms (Ω) (see 4.3.3.1.7.1).

3.3.1.7.2 Power turbine inlet temperature (T_7). The assembly shall be capable of processing thermocouple signals with the following (see 4.3.3.1.7.2):

a. Type K thermocouple electromotive force outputs in accordance with National Bureau of Standards (NBS) Circular 561.
b. Harness line resistance (including thermocouples) between 50 to 100 Ω.
c. Insulation resistance between any line and the engine frame of 1 MΩ minimum at 500 Vdc.
d. Time constant of the thermocouples of 0.2 seconds at idle and 0.05 seconds at maximum power.

3.3.1.7.3 Linear Variable Differential Transformer (LVDT) and Rotary Variable Differential Transformer (RVDT) secondary signals. The signal outputs from the LVDT and RVDT secondary shall be converted to direct current voltage by a standard demodulator circuit (see 4.3.3.1.7.3).

3.3.1.7.3.1 Power Turbine Stator (PTS) position feedback. The PTS position feedback shall be provided by a RVDT. The assembly shall measure the RVDT secondary signal of the PTS (see 4.3.3.1.7.3.1 and tables I and V).

3.3.1.7.3.2 Inlet Guide Vane (IGV) position feedback. The IGV position feedback shall be provided by a RVDT. The assembly shall measure the RVDT secondary signal of the IGV (see 4.3.3.1.7.3.2 and tables I and V).

3.3.1.7.3.3 Fuel flow (WF) metering valve position. The position of the fuel metering valve shall be sensed by a LVDT. The assembly shall measure the LVDT secondary signal of the fuel flow (see 4.3.3.1.7.3.3, table I and figure 1).

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3.3.1.7.3.4 Power level angle (PLA). The assembly shall measure the RVDT secondary signal of the PLA (see 4.3.3.1.7.3.4, tables I and VI and figure 2).

3.3.1.7.4 Frequency signal inputs. The assembly shall be equipped with input electronics to convert frequency sensor inputs to a 12-bit parallel digital format. The high pressure NH and the NPT are each sensed by two magnetic pickups with the characteristics as shown in tables I and VII (signals are assumed sinusoidal) (see 4.3.3.1.7.4).

3.3.1.7.5 1553 data bus input (type II only). The following input shall be received across the 1553 data bus in accordance with AlliedSignal Engines document 21-9187, data packet 24-01 (see 4.3.3.1.7.5):

a. Inlet air pressure (PI) pounds per square inch area (psia)
b. Tactical idle (flag)
c. Low oil pressure (LOP) flag)
d. Nuclear, Biological, Chemical (NBC) command (flag)
e. Second shot fire shutdown (flag)
f. Pivot neutral (flag)
g. NH trim, hertz (Hz)

3.3.1.7.6 Discrete signal inputs. The assembly shall have input electronics to accommodate the discrete inputs and shall register a “logic 1” when a voltage on a discrete input pin is greater than the threshold defined in table II, and a “logic 0” when the voltage is less than the threshold voltage (see 4.3.3.1.7.6).

3.3.1.7.6.1 Discrete input signal processing.

3.3.1.7.6.1.1 Purge. The purge command shall be acknowledged if NH is less than 5%. As long as the purge discrete input is present, the cutoff solenoid and starter pilot relay will be energized. The starter pilot relay will be de-energized when the purge discrete input is no longer present. When NH decreases below 5% after the purge discrete input is no longer present, a 45-second cutoff timer will be activated which will de-energize the cutoff solenoid after time out (see 4.3.3.1.7.6.1.1).

3.3.1.7.6.1.1.1 Purge (type II only). After a successful start has occurred, the purge discrete input shall be used to activate the auto cooldown logic (see 3.3.6.4.2) if the pivot neutral discrete input is present, signifying that the assembly is installed in a Heavy Assault Bridge (HAB) vehicle (see 4.3.3.1.7.6.1.1.1).

3.3.1.7.6.1.2 Start.

3.3.1.7.6.1.2.1 Start (types I and Ia). The start command shall be accepted if the discrete input is present for at least 0.3 seconds, NH is less than 5%, and the voltage that represents the

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current through the cutoff solenoid is greater than 0.9 V. Upon acceptance of the start command, the starter pilot relay and ignition exciter are energized, the 60-second timer is activated (see 3.3.2.1 for fuel scheduling). During starting, zero current shall be delivered to the IGV and PTS solenoids. When NH increases above 50%, a successful start is achieved, the starter pilot relay and ignition exciter shall be de-energized, and the run time meter shall be energized. In the event NH has not reached 50% when the start timer times out, then the start shall be aborted. The abort of the start can be overridden as long as the start discrete input is present after the start timer has timed out. In the event a PM I fault occurs or either stop discrete input is received during the start sequence, the start shall be aborted. For protective mode PM I fault conditions (see 3.3.5.2.1) and for 3.3.1.7.6.1.3 stop commands (see 4.3.3.1.7.6.1.2.1).

3.3.1.7.6.1.2.2 Start (type II). The start logic is the same as for types I and Ia (see 3.3.1.7.6.1.2.1) with the following exception. Upon acceptance of the start command, a start flag shall be transmitted on the 1553 data bus (see AlliedSignal Engines document 21-9187, data packet 25) and there shall be a 0.5-second delay before the starter pilot relay and ignition exciter are energized and the 60-second start timer is activated (see 4.3.3.1.7.6.1.2.2).

3.3.1.7.6.1.3 Stop. The stop command shall initiate the shutdown sequence during a start and auto cooldown after a successful start if the stop discrete input is present for a minimum of 40 ms. The shutdown sequence energizes the cutoff solenoid and de-energizes the starter pilot relay and ignition exciter if they are energized. When NH decreases below 5%, the run time meter will be de-energized and a 45-second cutoff timer will be activated which will remove power to the assembly by de-energizing the cutoff solenoid when it times out. For type II only, the stop signal shall be sent from the assembly on the 1553 data bus (see AlliedSignal Engines document 21-9187, data packet 25-01 and 4.3.3.1.7.6.1.3).

3.3.1.7.6.1.4 Fault reset.

3.3.1.7.6.1.4.1 Manual fault reset. When a fault reset is commanded, the assembly shall exit the assumed fault mode and return to normal operation assuming no further fault exists. This shall only be applicable to PM II and PM III conditions (see 3.3.5.3.1 and 3.3.5.4.1). The fault reset discrete input shall also be used as an actuation parameter for the test cell PTS routine (see 3.3.6.2) and the engine health check routine (see 3.3.6.3). For type II only, the discrete input shall also be used as a deactivation parameter for the auto cooldown routine (see 4.3.3.1.7.6.1.4.1).

3.3.1.7.6.1.4.2 Automatic fault reset. An automatic fault reset routine shall be incorporated that will automatically clear PM II and PM III conditions once per power up. The routine shall be active if the following conditions are satisfied:

a. 50% < NH < 70%
b. PLA < 10 degrees (°)
c. PM II and/or PM III is true

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d. Manual fault reset is false
e. Any two of the following three are true
   (1) IGV > 60°
   (2) PTS < 10°
   (3) WF error < 20 pounds per hour (lb/h) (9 kilograms per hour (kg/h) for PM II or WF < 60 lb/h (27 kg/h) for PM III

A 15-second delay shall be incorporated between the time the criteria are satisfied and the auto fault reset execution (see 4.3.3.1.7.6.1.4.2).

3.3.1.7.6.1.5 Tactical idle. When the tactical idle discrete input is in the normal idle position (see tables III and VIII), the minimum NPT shall be between 25 and 30% based on the pivot neutral discrete input (see 3.3.1.7.6.1.6), specifically 30% from drive and a programmable value, with a nominal value of 30%, shall be stored in storage EEPROM and may be changed through the EIA-232 interface (see 3.3.1.6.1 and Appendix A). The tactical idle discrete input in the tactical idle position will increase the minimum NPT to 43.2%. For type II only, tactical idle can be commanded via the 1553 data bus (see AlliedSignal Engines document 21-9187, data packets 24-01 and 25-01 and 4.3.3.1.7.6.1.5).

3.3.1.7.6.1.6 Pivot neutral. When the pivot neutral discrete input is in the pivot neutral position (logic 1), the maximum NPT shall be 80%, while in the drive position the maximum NPT shall be 103.3%. In the event that the T1 biased NH underspeed governor has been selected and the PLA is less than 10°, then the maximum NPT shall be 45.6%. For type II only, pivot neutral can be commanded via the 1553 data bus (see AlliedSignal Engines document 21-9187, data packets 24-01 and 25-01 and 4.3.3.1.7.6.1.6).

3.3.1.7.6.1.7 NBC.

3.3.1.7.6.1.7.1 NBC (types I and Ia). The NBC logic shall be commanded "on" when both the tactical idle (see 3.3.1.7.6.1.5) and pivot neutral (see 3.3.1.7.6.1.6) discrete inputs are true. Minimum NH speed shall be governed by the T1 bias NH idle (see 3.3.2.3.2 and 4.3.3.1.7.6.1.7.1).

3.3.1.7.6.1.7.2 NBC (type II). The NBC logic shall be commanded "on" when both the NBC command and the pivot neutral signals are received over the 1553 data bus, (see AlliedSignal Engines document 21-9187). When the NBC command is received, the assembly shall automatically turn tactical idle "on". Minimum NH speed shall be governed by the T1 bias NH idle (see 3.3.2.3.2). If tactical idle is subsequently turned "on" and then "off", the standard NH governor shall become active (see 3.3.2.3.1 and 4.3.3.1.7.6.1.7.2).

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3.3.1.8 Output signals.

3.3.1.8.1 Output drivers. The assembly shall have output driver circuitry for each of the PTS, IGV, and fuel solenoids. The circuitry shall convert the difference of request voltage and transducer feedback voltage to a current that shall position the solenoid. There shall be two connector pins for each solenoid. One connector pin shall be for current output (solenoid HI) and the other for current return (solenoid LO) (see 4.3.3.1.8.1 and table IX).

3.3.1.8.1.1 PTS driver (PTS solenoid HI and PTS solenoid LO). The PTS solenoid shall operate a valve in the hydromechanical unit (HMU) that directs fuel flow to the cylinder of a piston. This piston shall move both the PTS vanes and a RVDT. The solenoid coil resistance is 5 to 15 Ω. At 425 lb/h (193 kg/h) fuel flow, the nominal relationship between PTS solenoid current and fuel flow to the PTS piston shall be as follows (see 4.3.3.1.8.1.1):

a.	0 milliamperes (mA)	Maximum flow to open PTS vanes
b.	350 mA	Zero flow
c.	900 mA	Maximum flow to close PTS vanes

The gain and phase relationships of this servo are shown in figure 3.

3.3.1.8.1.2 IGV driver (IGV solenoid HI, IGV solenoid LO). The IGV solenoid shall operate a valve in the HMU that directs fuel flow to the cylinder of a piston. This piston shall move both the IGV vanes and a RVDT. The solenoid coil resistance is 5 to 15 Ω. At 425 lb/h (193 kg/h) fuel flow, the nominal relationship between IGV solenoid current and fuel flow to the IGV piston shall be as follows (see 4.3.3.1.8.1.2):

a.	0 mA	Maximum flow to close IGV vanes
b.	350 mA	Zero flow
c.	900 mA	Maximum flow to open IGV vanes

The gain and phase relationships of this servo shall be as shown in figure 4.

3.3.1.8.1.3 WF driver (fuel solenoid LO). The WF solenoid shall operate a metering valve in the HMU that directs fuel flow to the fuel nozzle between 40 and 800 lb/h (18 and 363 kg/h), proportional to the solenoid position. A LVDT attached to the base of the metering valve plunger shall provide feedback voltage. The solenoid coil resistance shall be 5 to 15 Ω. The nominal relationship between WF solenoid current and fuel flow to the fuel nozzle shall be as follows (see 4.3.3.1.8.1.3):

a.	0 mA	Maximum reduction in fuel flow
b.	350 mA	Zero change in fuel flow
c.	900 mA	Maximum increase in fuel flow

The gain and phase relationships of this servo shall be as shown in figure 5.

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3.3.1.8.2 NPT meter. The assembly shall drive a NPT meter. Output voltage shall be proportional to speed (5 V – 120%). The indicator minimum resistance shall be 1500 Ω nominal (see 4.3.3.1.8.2).

3.3.1.8.3 Discrete output. The assembly shall have driver circuitry to accommodate the discrete outputs and shall have current-limiting circuitry (see 4.3.3.1.8.3 and table X).

3.3.1.8.3.1 Engine starter relay. The engine starter relay shall require vehicle battery voltage, 1.4 A at 70 millihenrys (mH) and 20 millijoules (mJ) (see 4.3.3.1.8.3.1).

3.3.1.8.3.2 Main fuel cutoff solenoid. The main fuel cutoff solenoid shall be spring-loaded open and shall close upon application of voltage. Nominal current required to close the main cutoff solenoid shall be 950 milliamperes (mA). Coil resistance of the solenoid shall be 25 to 30 Ω (see 4.3.3.1.8.3.2).

3.3.1.8.3.3 Engine run-time meter. The assembly shall power the engine run-time meter when NH is greater than 50%. For type II only, the engine run time shall be reported across the 1553 data bus. Data commands shall be in accordance with AlliedSignal Engines document 21-9187 (see 4.3.3.1.8.3.3).

3.3.1.8.3.4 Vehicle indicator lamps. The assembly shall illuminate the 0.3-A indicator lamps (see 4.3.3.1.8.3.4):

a. NBC prioritization (formerly T_7 flag).
b. NPO (power turbine overspeed).
c. Fault alarm.

3.3.1.8.3.4.1 NBC prioritization. This signal shall be activated when T7 is less than 18°F (10°C) below the T7 limit shown in figure 6. This signal is used by the vehicle to reduce the bleed flow to the NBC system in order to reduce T7. The signal shall remain active until T7 falls to 38°F (21°C) below the limit shown in figure 6. This signal shall only be active after a successful start and below 78% NH (see 4.3.3.1.8.3.4.1 and 6.5.3).

3.3.1.8.3.5 Ignition exciter. The assembly shall supply 10 V minimum voltage and 2 A average current to the ignition exciter. The assembly shall limit current to 2.8 A maximum. The ignition exciter has an inductance of 6 mH and a power consumption of 12 W (see 4.3.3.1.8.3.5).

3.3.1.8.3.6 Backup fuel cutoff solenoid. The backup fuel cutoff solenoid is spring-loaded open and shall close upon the application of voltage. Coil resistance of the solenoid shall be 35 to 60 Ω (see 4.3.3.1.8.3.6).

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3.3.1.9 Differential transformer excitation. The assembly shall have an excitation circuit for the LVDT and RVDTs as follows:

a. WF LVDT primary HI and LO
b. PTS RVDT primary HI and LO
c. IGV RVDT primary HI and LO
d. PLA RVDT primary HI and LO

The excitation circuit shall consist of a 20 V, peak-to-peak (p-p), 3500 Hz triangle wave oscillator driving a power amplifier. There shall be two connector pins for each primary signal. One connector pin shall be for excitation (HI) and the other for return (LO) (see 4.3.3.1.9).

3.3.1.10 1553 data bus output (type II only). The following output data shall be made available to the 1553 data bus in accordance with AlliedSignal Engines document 21-9187 (see 4.3.3.1.10):

a. Auto cooldown (flag)
b. Ignition exciter (flag)
c. Tactical idle (flag)
d. T7 over temperature (flag)
e. Backup battery operation (flag)
f. Main fuel cutoff solenoid (flag)
g. Starter pilot relay (flag)
h. Health check active (flag)
i. NPO (flag)
j. NH (%)
k. Engine run-time (hours)
l. Engine output shaft speed (revolutions per minute (rpm))
m. T1 (°F)
n. NH trim value (Hz)

3.3.2 Fuel scheduling.

3.3.2.1 Starting. When NH increases above 5%, the cutoff solenoid shall be de-energized and fuel flow shall be set to 50 lb/h (22.7 kg/h). The fuel shall be ramped up at a rate of 5 lb/h per second (2.27 kg/h per second). Fuel shall be limited to 90 lb/h (40.8 kg/h) for NH less than 30% and 165 lb/h (74.8 kg/h) for NH greater than 30%. The start schedule shall be maintained until the requirements of 3.3.2.3.1 and 3.3.2.3.2 are met (see 4.3.3.2.1 and figure 7).

3.3.2.2 NPT governing. The NPT governor (see figure 8) shall consist of an underspeed governor and a fuel schedule directly proportional to the PLA position (see 3.3.1.7.3.4). The vehicle acceleration shall be controlled by the NPT governor. The fuel request based on PLA shall be in accordance with figure 3 (see 4.3.3.2.2).

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3.3.2.2.1 NPT idle (pivot/neutral). The minimum NPT, when the assembly is in tactical idle mode (see 3.3.1.7.6.1.5), shall be programmable via the EIA-232 port. The programmable range for this speed shall be from 25 to 30% and the assembly shall be programmed initially to 30% (see 4.3.3.2.2.1).

3.3.2.2.1.1 NPT idle (drive). A constant 30% NPT shall be maintained whenever the pivot neutral discrete is low (see 3.3.1.7.6.1.6 and 4.3.3.2.2.1.1).

3.3.2.2.1.2 NPT idle (tactical idle). When the assembly is in tactical idle mode (see 3.3.1.7.6.1.5), the minimum NPT shall be 45.6% (see 4.3.3.2.2.1.2).

3.3.2.2.1.3 Maximum NPT governing. The assembly shall schedule the maximum NPT by modulating fuel flow in accordance with figure 8. Maximum NPT shall be 62% when the assembly is in the T1 biased NH idle setting (see 3.3.2.3.2) and PLA is <10°. Maximum NPT shall be 80% when the assembly is in the pivot/neutral mode (see 3.3.1.7.6.1.6), and 103.3% when not in the pivot/neutral mode (see 4.3.3.2.2.1.3).

3.3.2.3 NH governing.

3.3.2.3.1 Standard NH idle. Minimum NH shall be 53.5% in both low and tactical idle modes (see 4.3.3.2.3.1 and figure 9).

3.3.2.3.2 T1 biased NH idle. When NBC is commanded “on” (see 3.3.1.7.6.1.7), minimum NH speed shall be a function of T1 (see 3.3.1.7.1) according to the following equation:

$$NH = (61.4452 + T1 * 0.1001569)$$

Where NH is in %, and T1 is in °F

T1 shall be filtered to stabilize the governor output and must change by ± 1°F (± 0.6°C) before a new governor value is calculated (see 4.3.3.2.3.2).

3.3.2.3.3 Maximum NH governing. Maximum nominal NH speed shall be set at 97.40% for T_1 of 97°F (36°C), and shall vary with T_1 when T_1 is less than 97°F. The maximum speed curve shall be adjustable between the limits shown in figure 10. The methods for adjusting the limit curve shall be in accordance with 3.3.6.6 (see 4.3.3.2.3.3 and figure 10).

3.3.2.4 Acceleration schedule. The acceleration schedule shall limit fuel as a function of rate of change of NH (NH dot) and maximum T_7 limit (see 4.3.3.2.4).

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3.3.2.4.1 Acceleration command. Acceleration shall be defined as a NH dot greater than 5.0% per second or a PLA rate of greater than 50° per second. Maximum allowable NH rate shall be 10.5% per second after a successful start (see 4.3.3.2.4.1).

3.3.2.5 Maximum T_7 control. A compensated value of T_7, (T_7I), shall be calculated to account for the delay inherent in the thermocouples. The algorithm shall be as shown in figure 11. During all operating conditions, the assembly shall modulate fuel flow to maintain T_7I below the maximum limits specified in figure 6. When the PTS vanes are open, fuel flow shall be modulated to maintain T_7I below 1430°F (777°C). If T_7I reaches 1620°F (882°C), the assembly shall schedule fuel flow in accordance with 3.3.2.6 until T_7I is 200°F (93.3°C) below the schedule in figure 6. At that time, normal fuel control logic shall resume (see 4.3.3.2.5).

3.3.2.6 Deceleration schedule. Deceleration shall be any negative NH rate. The assembly shall schedule fuel flow according to figure 10, as a function of NH with no T_1 bias (see 4.3.3.2.6 and 6.5.4).

3.3.2.7 Temperature gradient control (T_7 lag function). Fuel flow shall be limited by an estimation of recuperator temperature (T_7 lag). T_7 lag shall be the greater of:

a. 500°F (260°C) or
b. If NH < 5%, T_7 lag = T_7 or
c. If NH < 5%, T_7 lag (z) = T_7 lag (z-1) + 0.002 * [T_7I(z) - T_7lag (z-1)] (°F)
   z = Is defined as the current microprocessor time.

After a successful start, if T_7I - T_7lag > 250°F (121°C), fuel flow shall be reduced in accordance with 3.3.2.6 until the temperature difference is reduced below 250°F (see 4.3.3.2.7).

3.3.2.8 NPO. When NPT speed exceeds 108%, fuel shall be scheduled in accordance with 3.3.2.6, the NPO lamp (see 3.3.1.8.3.4b) shall be energized, and the PTS shall be held open. When NPT drops below 95%, the assembly shall resume control mode logic. The NPO lamp shall not be deactivated until a fault reset command (see 3.3.1.7.6.1.4) has been received (see 4.3.3.2.8).

3.3.2.9 Secondary fuel flow (WF secondary). The WF secondary shall be used when a PM II fault (see 3.3.5.3) is present and shall be a function of the PLA according to the first order lag equation (see 4.3.3.2.9):

$$WF\ secondary = PLA * (3.367/(0.85s+1)) + 76.533$$

3.3.3 PTS control. During a start, the PTS shall remain open until the NPT governor is satisfied (see 3.3.2.2). In steady state operation, the PTS angle shall be scheduled as the smaller of the minimum power turbine nozzle area angle and the angle that maintains T_7I below the maximum steady state T_7 schedule (see 6.5.5 and figure 12). Figure 13 shall be trimmed by the

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PTS adaptive routine logic (see 3.3.6.2) for NH between 52 and 75%. The equation of the line intersecting the forward and backward slopes shall be:

$$T_7 = 33.2 * NH - 1638$$

For NH less than 75%, the PTS shall be opened if T_7 is within 50°F (28°C) of the maximum power turbine inlet temperature schedule (see 3.3.2.5). The PTS shall remain open until the T_7 is 20°F (11°C) less than the forward slope of figure 12, at which time the above scheduling shall engage. During acceleration, the PTS shall be held open until the difference between T_7 and T_7 lag is less than 100°F (56°C). A 500°F (278°C) margin of hysterisis shall be incorporated before the reinitialization of the 100°F requirement. Each of the following shall be a reason for the PTS to go full open (see 4.3.3.3):

a. The selection of tactical idle (see 3.3.1.7.6.1.5).
b. The transition from neutral to drive (see 3.3.1.7.6.1.6).
c. The activation of the cooldown routine (see 3.3.6.4).
d. The presence of PM II fault (see 3.3.5.3).
e. The presence of PM III fault (see 3.3.5.4).

3.3.4 IGV control. During a start, the IGV shall remain closed until NH has reached 47%. In steady state operation, where steady state is specified as NH dot less than 3% per second for 60 seconds, the IGV angle shall be scheduled to the lower of 40° or the schedule specified in figure 14. The schedule specified in figure 14 shall be followed during any of the following conditions (see 4.3.3.4):

a. Acceleration
b. Deceleration
c. Selection of tactical idle (see 3.3.1.7.6.1.5)
d. Transition from neutral to drive (see 3.3.1.7.6.1.6)

3.3.4.1 IGV prioritization. For NH less than 70%, the IGV shall be commanded full closed when one or more of the following conditions occur (see 6.5.6):

a. NH dot is greater than 3% per second
b. Transition of tactical idle from one position to the other (see 3.3.1.7.6.1.5)
c. Transition of pivot neutral from one position to the other (see 3.3.1.7.6.1.6)
d. The activation of the cooldown routine (see 3.3.6.4)
e. The presence of a PM II fault (see 3.3.5.3)
f. The presence of a PM III fault (see 3.3.5.4)

The idle fuel flow request changes at least ± 1 lb/h for 5 consecutive time frames with a total delta (Δ) of greater than ± 5 lb/h total. When the IGV are commanded full closed, and NHC is

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less than 70%, fuel flow shall be held constant until the IGV feed back has reached 57°. This shall be referred to as IGV prioritization logic (see 4.3.3.4.1 and figure 15).

3.3.5 PMs. The assembly shall have a PM selected depending on the failure detected and on the operating conditions present at the time. Detection criteria and the appropriate fault message shall be stored in EEPROM (see 4.3.3.5).

a. For types I and Ia assemblies, the EEPROM data shall support tables III and XI.
b. For Type II assemblies, the EEPROM data shall support tables III and XI and AlliedSignal Engines document 21-9187.

3.3.5.1 PM 0 - no crank (type II only). A PM 0 fault shall be acknowledged only during the start sequence. When a PM 0 fault is present, the assembly shall abort the start (see 3.3.1.7.6.1.3) and the fault alarm lamp shall be energized (see 4.3.3.5.1).

3.3.5.1.1 PM 0 fault conditions. Any of the following conditions shall result in a PM 0 fault (see 4.3.3.5.1.1):

a. The voltage representing the main fuel cutoff solenoid current is < 0.9 Vdc for 1.6 seconds.
b. The voltage representing the backup fuel cutoff solenoid current is ≤ 0.8 Vdc for 360 ms.
c. Ground or ± 10 Vdc reference voltages exceeding nominal value by ± 0.25 Vdc for 100 ms (see 3.5.5.1.1.1).
d. Any of the 8 most significant bits of either Stuck Bit Test Word (± 15 Vdc) being the opposite polarity of what it should be (see 3.5.5.1.1.3).
e. Any of the three Digital-to-Analog (D/A) wraparound check voltages exceeding their nominal value by ± 1 Vdc for 160 ms (see 3.5.5.1.1).
f. Triangle wave voltage ≤ 10 Vp-p (see 3.3.5.8).

3.3.5.2 PM I - engine abort. A PM I fault shall be acknowledged only during the start sequence. When a PM I fault is present, the assembly shall abort the start and the fault alarm lamp shall be energized (see 3.3.1.7.1.6.1.3 and 4.3.3.5.2).

3.3.5.2.1 PM I fault conditions. Any of the following conditions, at 7.5 seconds after a start is initialized, shall result in a PM I fault (see 4.3.3.5.2.1):

a NH1 and NH2 both less than 1% for 40 ms
b. Difference between WF and WF request > 100 lb/h (45.4 kg/h) for 0.5 seconds
c. WF < 15 lb/h (6.8 kg/h) for 100 ms
d. Power supply voltage < 8 Vdc and backup battery voltage > 18 Vdc
e. Power supply voltage < 10.8 Vdc and backup battery voltage < 18 Vdc

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f. T7 < 250°F (121°C) and NH > 30% for 40 ms
g. T7 > 1800°F (982°C) for 40 ms
h. 1360°F (738°C) < T7 < 1800°F, NH > 5% for 40 ms (Fault alarm lamp shall not be energized)
i. 60 second start timer times out and NH < 50% (Fault alarm lamp shall not be energized)

3.3.5.3 PM II. A PM II fault shall be acknowledged only after a successful start. When a PM II fault is present, the secondary fuel flow schedule (see 3.3.2.9) shall be followed, and the assembly shall command the following (see 4.3.3.5.3):

a. IGV: fully closed
b. PTS: fully opened
c. Fault alarm lamp: energized

3.3.5.3.1 PM II fault conditions. Any of the following conditions shall result in a PM II fault (see 4.3.3.5.3.1):

a. T7 < 250°F (121°C) or T7 > 1800°F (982°C) for 40 ms, no shutdown in progress.
b. PTS > 68° for 40 ms.
c. Difference between IGV and IGV request > error level for 1 second, no shutdown in progress. See figure 16 for error limits.
d. NPT1 and NPT2 both less than 1% for 40 ms and one of the following conditions is true:
   (1) NH > 65%.
   (2) NPT has been > 27%.
   (3) NH1 and NH2 both less than 1% for 40 ms.
e. Ground or ± 10 Vdc reference voltages exceeding nominal value by ± 0.25 Vdc for 100 ms (see 3.5.5.1.1.1).
f. Any of the 4 least significant bits of either Stuck Bit Test Word (± 15 Vdc) being the opposite polarity of what it should be (see 3.5.5.1.1.3).
g. Triangle wave voltage ≤ 10 Vp-p (see 3.3.5.8).

3.3.5.4 PM III. A PM III fault shall be acknowledged only after a successful start. When a PM III fault is present, the assembly shall command the following (see 4.3.3.5.4):

a. Backup fuel cutoff solenoid: de-energized (see 3.3.1.8.1.3.6).
b. Backup battery: disabled (see 3.3.1.2).
c. Fuel flow: commanded to 65 lb/h (29.5 kg/h).
d. IGV: commanded fully closed.
e. PTS: commanded fully opened.
f. Fault alarm lamp: energized.

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3.3.5.4.1 PM III fault conditions. Any of the following conditions shall result in a PM III fault (see 4.3.3.5.4.1):

a. PM II is present and pivot neutral discrete is in the neutral position (see 3.3.1.7.6.1.6).
b. Difference between WF and WF request > 100 lb/h for 0.5 seconds.
c. WF < 15 lb/h for 100 ms.
d. Power supply voltage < 10.8 Vdc and backup battery voltage < 17 Vdc.
e. Any of the three D/A wraparound check voltages exceeding their nominal value by ± 1 Vdc for 160 ms (see 3.5.5.1.8.3).
f. Ground or ± 10 Vdc reference voltages exceeding nominal value by ± 1 Vdc for 100 ms (see 3.5.5.1.1.1).
g. Any of the 8 most significant bits of either Stuck Bit Test Word (± 15 Vdc) being the opposite polarity of what it should be (see 3.5.5.1.1.3).
h. The assembly enters the tight loop mode (see 3.3.5.4.2.1).

3.3.5.4.2 Tight loop mode. The assembly shall be provided with a tight loop mode. When in tight loop mode, the assembly shall disable all outputs and execute an infinite null calculation loop. A PM III fault shall be generated when the assembly enters the tight loop mode (see 4.3.3.5.4.2).

3.3.5.4.2.1 Tight loop mode conditions. Any of the following conditions, specific to an assembly failure, shall cause the assembly to enter the tight loop mode (see 3.3.5.4.2 and 4.3.3.5.4.2.1):

a. CPU test fault (see 3.5.5.1.2)
b. RAM function fault (see 3.5.5.1.3)
c. EEPROM check fault (see 3.5.5.1.8)
d. Background iteration check fault (see 3.5.5.1.5)

3.3.5.4.3 PM III fault removal. The PM III fault shall be removed if the transmission is shifted from neutral to drive provided the PM III was the result of shifting from drive to neutral (see 4.3.3.5.4.3).

3.3.5.5 PM IV. A PM IV fault shall be acknowledged during and after start, with the exception of the speed level faults which shall only be acknowledged after a successful start has occurred. When a PM IV fault is present, the fault lamp shall be energized (see 4.3.3.5.5).

3.3.5.5.1 PM IV fault conditions. Any of the following conditions shall result in a PM IV fault (see 4.3.3.5.5.1):

a. NH1 < 1% for 0.5 seconds with NH2 > 50%
b. NH2 < 1% for 0.5 seconds with NH1 > 50%

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c. NPT1 < 1% for 0.5 seconds with NH > 50%
d. NPT2 < 1% for 0.5 seconds with NH > 50%
e. *T1 < -100°F (-73°C) or T1 > 200°F (93°C) for 0.5 seconds with NH > 5% (assembly shall default to T1 = 125°F (52°C))
f. EEPROM fault assembly failure (see 3.5.5.1.8) (types I and Ia only)
g. PTS error (types I and Ia only)
h. Battery voltage (see 3.3.1.2.2.1) (types I and Ia only)
i. Fuel flow restriction (see 3.5.5.2.3.1) (types I and Ia only)

3.3.5.6 PM V (type II only). A PM V fault shall be acknowledged when no activity is observed on the 1553 data bus. This logic shall become active 15 seconds after a successful start has occurred (see 4.3.3.5.6).

3.3.5.7 CPU reset mode. The assembly shall be provided with a CPU reset mode. During a CPU reset, the assembly shall disable all outputs and resume program execution at its initial memory location (see 4.3.3.5.7).

3.3.5.7.1 CPU reset mode conditions. Any of the following conditions, specific to an assembly failure, shall cause a CPU reset (see 4.3.3.5.7.1):

a. Power supply unit (see 3.3.1.5)
b. CPU test (see 3.5.5.1.2)
c. Real time task time out (see 3.5.5.1.6)
d. Watchdog timer (see 3.5.5.1.7)

3.3.5.8 Transformer fault protection. The assembly shall be equipped with logic that checks the primary insulation resistance to protect the 20 Vp-p triangle-wave excitation signal. If the primary voltage falls below 10 Vp-p, the assembly shall enter PM 0 during a start (see 3.3.5.1) or PM II at any other time (see 3.3.5.3 and 4.3.3.5.8).

a. For types I and Ia, faults shall be stored in EEPROM to support tables III and XI.
b. For type II, faults shall be stored in EEPROM to support AlliedSignal Engines document 21-9187.

3.3.6 Algorithms.

3.3.6.1 PTS test cell logic. When the NH speed is set 85 to 90% and FAULT RESET and START are activated simultaneously, the PTS shall be set to 5°. While maintaining the START command, the requested PTS shall be changed by 1° for every actuation of the FAULT RESET button. The PTS shall move in 1° increments each time the FAULT RESET button is actuated. The PTS shall first move closed until 15° is reached and then shall move open until 5° is reached. When NH is lowered below 80%, normal PTS logic (see 3.3.3) shall resume (see 4.3.3.6.1 and figure 17).

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3.3.6.2 PTS adaptive routine logic. A software routine shall be provided, allowing the PTS voltage request setting to be externally programmable (see figure 18). Any of the following conditions shall override the PTS convergence routine:

a. PLADT ≥ 50° per second (driver override) PLA rate of change.
b. NHdot ≥ 5% (NH acceleration).
c. NPO
d. T7 – T_7PTSs < 100 (T_7 has exceeded T_7PTS schedule).
e. NH < 51.5% or NH > 75%.
f. A limit of 47° WFPSA (programmed PTS position).

The PTS request from the PTS trimming routine shall supersede the PTS request from the temperature controlled PTS schedule (see figure 12) while between 52 and 75% NH, and while overrides are not in effect. Upon convergence of the routine, a flag shall be set, bypassing the routine until the assembly has been depowered or a manual NH trim is initiated. The routine shall be run once to completion on a power-up (see 4.3.3.6.2).

3.3.6.3 Engine health check. The engine health check routine shall calculate the percentage of power available based on the prevailing ambient temperature; hereafter, referred to as "day power" (see figures 19 and 20). Five valid calculations shall be conducted in order to complete the routine. A calculation shall be considered valid if PM II (see 3.3.5.3) or PM III (see 3.3.5.4) is not present. The final calculation, the corresponding mission capability power (MCP) number and the current date shall be stored in EEPROM if no PM II or PM III is present. The MCP number shall represent the day power referenced to an 87°F (31°C) day rounded in 5% increments as follows (see 4.3.3.6.3):

MCP	87°F day power
1	95% < power ≤ 100%
2	90% < power ≤ 95%
3	85% < power ≤ 90%
4	80% < power ≤ 85%
5	75% < power ≤ 80%
6	70% < power ≤ 75%
7	65% < power ≤ 70%
8	60% < power ≤ 65%
9	55% < power ≤ 60%
10	50% < power ≤ 55%
11	45% < power ≤ 50%
12	40% < power ≤ 45%

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a. Once the routine is activated, the following conditions shall be satisfied for 5 seconds in order for the calculation to occur:
   (1) Tactical idle true (see 3.3.1.7.6.1.5).
   (2) Vehicle in drive (see 3.3.1.7.6.1.6).
   (3) -10°F (-23°C) < T7 – T7 lag < 25°F (-4°C) (see 3.3.2.5 and 3.3.2.7).
   (4) NPT < 41.88% (see 3.3.2.2.1).
   (5) Fault reset is true (see 3.3.1.7.6.1.4).
b. In addition to the above conditions, the fuel flow request shall have stabilized to within -0.13 to + 0.25 lb/h (-0.06 to + 0.11 kg/h). The health check routine shall stop if one or both of the following conditions is satisfied:
   (1) PLA > 20°.
   (2) Tactical idle is disabled.

3.3.6.3.1 Engine health check (types I and Ia). While the calculation is taking place, the fault lamp shall be illuminated. If the calculation is taking place while a PM IV is present (see 3.3.5.5) the fault lamp shall be de-energized until the calculation is complete. The results of the health check shall be displayed in the diagnostic window (see 3.3.1.4) during BIT (see 3.5.5.2) in the following order: percent day power, date, and MCP number (see 4.3.3.6.3.1).

3.3.6.3.2 Engine health check (type II). While the calculation is taking place, a message shall be transmitted from the assembly on the 1553 data bus (see AlliedSignal Engines document 21-9187). The results of the health check shall be transmitted over the 1553 data bus after completion of the routing and when BIT (see 3.5.5.2) has been activated. The health check shall not activate nor shall stop processing if a PM V is present (see 4.3.3.6.3.2).

3.3.6.4 Cooldown logic.

3.3.6.4.1 Cooldown logic types I and Ia (M1 and M1A1 vehicles). Upon pressing the purge button (see 3.3.1.7.6.1.1) after a successful start, a cooldown routine shall be initiated when tactical idle is false (see 3.3.1.7.6.1.5) and pivot neutral is true (see 3.3.1.7.6.1.6). At that time, the assembly shall command the PTS to the full open position, set the PLA to 4°, and cause the fault reset lamp to flash on and off at 0.5-second intervals (see figure 21). At the end of 2 minutes, the lamp shall stop flashing and the assembly shall return to normal operational mode. If, during this routine, any of the following conditions are met, the routine shall be disabled (see 4.3.3.6.4.1):

a. The fault reset button is pressed (see 3.3.1.7.6.1.4).
b. Tactical idle is switched to true (see 3.3.1.7.6.1.5).
c. Pivot neutral is switched to false (see 3.3.1.7.6.1.6).

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3.3.6.4.2 Cooldown logic (type II).

3.3.6.4.2.1 Cooldown logic type II (M1A2 Abrams vehicle). Upon receiving the stop command via the 1553 data bus (see AlliedSignal Engines document 21-9187) after a successful start, a cooldown routine shall be initiated when tactical idle is false (see 3.3.1.7.6.1.5) and pivot neutral is true (see 3.3.1.7.6.1.6). At that time, the assembly shall command the PTS to the full open position, set the PLA to 4°, and broadcast a message on the 1553 data bus (see figure 22). At the end of 2 minutes, the engine shall automatically shut down. If, during the routine, any of the following conditions are met, this routine shall be disabled (see 4.3.3.6.4.2.1):

a. The fault reset is initiated (see 3.3.1.7.6.1.4).
b. Tactical idle is commanded (see 3.3.1.7.6.1.5).
c. Pivot neutral is switched to false (see 3.3.1.7.6.1.6).
d. PM V is true (see 3.3.5.6).
e. NBC is commanded "on" (see 3.3.1.7.6.1.7).

3.3.6.4.2.2 Cooldown logic type II (HAB). The assembly shall function as specified in 3.3.6.4.1, except the purge discrete input shall be used to activate the routine (see 4.3.3.6.4.2.2).

3.3.6.5 External stop and LOP command. When the external stop and LOP command is received, the assembly shall respond according to the following (see 4.3.3.6.5 and figure 23):

a. If NH < 50%, the assembly conducts a shutdown (see 3.3.1.7.6.1.3).
b. If NH is between 50% and 60%, the assembly shall interpret this as the LOP command and shall increase the NH idle speed one percent per second until the command goes away or 60% NH is reached.
c. If NH ≥ 60%, the assembly conducts a shutdown (see 3.3.1.7.6.1.3).

3.3.6.5.1 External stop and LOP command (types I and Ia). The EEPROM message specified in 3.3.6.5 shall support tables III and XI (see 4.3.3.6.5.1).

3.3.6.5.2 External stop and LOP command (type II). The EEPROM message specified in 3.3.6.5 shall support AlliedSignal Engines document 21-9187 (see 4.3.3.6.5.2).

3.3.6.6 Manual NH trim.

3.3.6.6.1 Manual NH trim (types I and Ia). The manual NH trim algorithm shall be activated when the assembly receives a fault reset signal (see 3.3.1.7.6.1.4) followed by the tactical idle signal (see 3.3.1.7.6.1.5), while power is supplied to the assembly, and while NH is less than 10%. Pulsing the fault reset button shall cause the assembly to increment the NH maximum speed limit curve from the minimum curve to the maximum curve in increments of 0.2% (4.33 Hz). By removing the tactical idle signal, the number displayed (in Hz) shall be stored as

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the manually programmed value. At the time of manual trim activation, the control shall capture the PTS feedback value for PTS adaptive logic (see 3.3.6.2) and reset the fuel flow trimming valve (see 3.3.6.8.1) to zero (see 4.3.3.6.6.1).

3.3.6.6.2 Manual NH trim (type II). The activation procedure from 3.3.6.6.1 shall be in place to support 3.3.6.2 and 3.3.6.8.1. The maximum NH shall be programmable via the 1553 data bus (see AlliedSignal Engines document 21-9187 and 4.3.3.6.6.2).

3.3.6.7 Rollback protection. If NH drops below 45%, after a successful start, T7 is within 10°F (5.6°C) of the limit (see 3.3.2.5), and the PTS is less than 6°, a shutdown shall be commanded (see 3.3.1.7.6.1.3) assuming a stop has not already been commanded. The assembly shall store this fault in EEPROM (see 4.3.3.6.7).

a. Types I and Ia EEPROM data shall support tables III and XI.
b. Type II EEPROM data shall support AlliedSignal Engines document 21-9187.

3.3.6.8 Flameout detection and auto relight. When the assembly detects a NH of less than 50% after a successful start, unless a stop command has been issued, the ignition exciter shall be energized until NH is greater than 51%, or less than 45%. Upon activation of the exciter, PLA request (PLAR) shall be set to 4°, ignoring the PLA feedback. The PLA shall resume control when the PLA feedback is less than 10° and NH is greater than 50%. If the engine reaches 45% NH and fuel flow request is greater than 170 lb/h (77 kg/h), the assembly shall store this condition in EEPROM as a flameout. The ignition exciter shall also be energized based on the rate of change of high pressure compressor speed (NH dot), rate of change NH dot (NH double dot), and the rate of change of T7 (T7 dot). If the values of NH dot, NH double dot, or T7 dot exceed the levels shown in table XII, the exciter shall be energized for 10 seconds. If a PM II or a PM III condition exists, this logic shall be disabled (see 4.3.3.6.8).

a. Types I and Ia condition stored in EEPROM shall support tables III and XI.
b. Type II condition stored in EEPROM shall support AlliedSignal Engines document 21-9187.

3.3.6.8.1 WF trimming. There shall be WF trimming to adjust fuel request and eliminate system steady state error, and to compensate for a system null current shift (see figure 24). The value shall be stored in EEPROM during each power-down sequence. During the manual NH trim procedure (see 3.3.6.6), the value shall be reset to zero (see 4.3.3.6.8.1).

3.4 Interface requirements. Each DECU shall accommodate the following inputs and interfaces.

3.4.1 Mechanical. The assembly mount configuration shall be as depicted in Drawings 12324815, 12324815-1 and 12324820 (see 4.3.4.1).

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3.4.1.1 Weight. The weight of the assembly shall not exceed 33 pounds (lb) (15 kilograms (kg)) (see 4.3.4.1.1).

3.4.2 Electrical.

3.4.2.1 Connectors and receptacles. The connectors and receptacles configuration shall be as specified in table XIII (see 4.3.4.2.1).

3.4.2.2 Pins. The electrical pins interface shall be as specified in table VIII see 4.3.4.2.2).

3.4.2.3 Power. The primary power shall be provided through the J3 connector as specified in table VIII (see 4.3.4.2.3).

3.4.3 1553 data bus (type II only). Two external connectors shall provide the assembly dual 1553 data bus interface (see 4.3.4.3 and tables VIII and XIII).

3.4.4 1553 RT interface board (type II only). The 1553 data bus RT board shall have dual port RAM that is connected to the local bus. The dual port RAM shall be configured with a minimum of 2K words of memory. The interface response time for RT mode shall be 8 ± 4 microseconds, in accordance with MIL-STD-1553 (see 4.3.4.4).

3.4.5 EIA-232. An EIA-232 interface (see 3.3.1.6.1) shall be provided to download program software into EEPROM (see 3.2.7.1). The assembly shall provide access to this interface through the pin connections on the J1 connector, as specified in table VIII (see 4.3.4.5).

3.4.6 Bonds and grounds. Bonding of grounds for equipment, electromagnetic interference, (EMI), EMP, and shock hazards shall be installed so that expansion, contraction, or movement incident to normal service use and maintenance will not break or loosen connections. Electrical connector receptacles shall be bonded to the assembly enclosure in a manner that will prevent the dc bond resistance from connector to enclosure from exceeding 50 millivolts (mV). External ground straps shall be bonded to the enclosure in a manner that will prevent the dc bond resistance from strap to enclosure from exceeding 2.5 milliohms (mΩ). Bond and ground requirements shall be in accordance with MIL-STD-464, except for stainless steel screws for mounting and grounding (see 3.2.2 and 4.3.4.6).

3.4.6.1 Chassis isolation. The input power returns of the assembly and assembly chassis shall be isolated from each other. Direct current resistance measurement between the assembly (28 Vdc return) and the chassis (bond strap) for acceptance, qualification, first article and control test requirements shall be as specified in 3.4.6.1.1 and 3.4.6.1.2 (see 4.3.4.6.1).

3.4.6.1.1 Chassis isolation – acceptance tests. Resistance shall be a minimum of 10 MΩ with a voltage of less than 10 Vdc at 73 ± 18°F (23 ± 10°C) (see 4.3.4.6.1.1).

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3.4.6.1.2 Chassis isolation – qualification, first article and control tests. Resistance shall be a minimum of 4.5 MΩ with a voltage source of 40 ± 5 Vdc measured at 130°F (54°C) (see 4.3.4.6.1.2).

3.4.7 J1 connector cap.

3.4.7.1 J1 connector cap (types I and Ia). The assembly shall have a connector cap which mates with the J1 receptacle as shown in table XIII. The cap shall protect the J1 port when diagnostic equipment is not interfaced with the assembly. The assembly shall be disabled (see table VIII) if there is not continuity between the J1 receptacle and the mating cap. Types I and Ia assemblies shall be equipped with a button on the shorting cap. The button, when depressed, shall connect 5 V to analog ground (see table VIII) in order to activate the BIT (see 4.3.4.7.1).

3.4.7.2 J1 connector cap (type II). Type II assembly shall have a connector cap as specified in 3.4.7.1, except without the BIT activation button (see 4.3.4.7.2).

3.4.8 Thermal. The dissipation of assembly heat to the environment shall rely on the mechanical interfaces for a heat transfer path. Internal cooling shall be by conduction between heat generating components and the exterior of the assembly. Heat dissipation shall not exceed 35 W (see 4.3.4.8).

3.4.9 Electromagnetic radiation (EMR).

3.4.9.1 EMR (types I and Ia). Types I and Ia assemblies shall meet the requirements of conducted emissions (CE) 102 and radiated emissions (RE) 102 of MIL-STD-461 (see 4.3.4.9.1).

3.4.9.2 EMR (type II). Type II assemblies shall meet the requirements of CE102, conducted susceptibility (CS) 101, CS114, RS102 and RE103 of MIL-STD-461 (see 4.3.4.9.2).

3.5 Support and ownership requirements. Each DECU shall possess the following life cycle ownership characteristics.

3.5.1 Reliability. The assembly shall achieve an inherent reliability of 20 000 hours Mean-Time-Between-Failure (MTBF), and 15 000 hours all-cause. The supplier shall verify base levels of maturity on all production assemblies (see 4.3.5.1 through 4.3.5.1.3 and 6.5.2).

3.5.2 Maintainability. The assembly shall not require preventive maintenance. The Mean-Time-to-Repair (MTTR) shall not exceed 2.5 clock hours for fault isolation and replacement down to the DECU module level (see 4.3.5.2).

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3.5.3 Interchangeability. The components of the assembly shall be of modular design such that another component of the same modular design, with the same functional, physical, and performance characteristics, shall be interchangeable without alteration of the components, and without subsequent alignment or calibration (see 4.3.5.3).

3.5.4 Quality. All electronic components of the assembly shall meet an established minimum quality level. Micro-electronic integrated circuits shall meet the quality levels of MIL-STD-883 where available. Passive components, except for magnets, shall be established reliable devices of failure rate R (see MIL-HDBK-217 guidelines and 6.6). Discrete semiconductors shall be at least JANTX class conforming to MIL-PRF-19500. Unscreened components shall not be used unless a waiver is obtained from the Procuring Activity. If the above quality level parts are unavailable, a substitute upgrade to MIL-STD-38510 quality is acceptable. Any other substitute is unacceptable (see 4.3.5.4).

3.5.4.1 Software quality factors. The software quality factors to be addressed for each CSCI shall include reliability, efficiency, maintainability and ease of testing.

3.5.5 Diagnostics.

3.5.5.1 Continuous self-test. The assembly shall include hardware and/or software supporting embedded self-test diagnostics. Self-test diagnostics shall run continuously upon power-up or CPU reset, and shall test the assembly without interruption of normal operation. The self-test diagnostics shall monitor the performance parameters of, and take the action specified in 3.3.1.5, 3.5.5.1.1 through 3.5.5.1.8.3 and 4.3.5.5.1.

3.5.5.1.1 Analog-to-Digital (A/D) converter, multiplexer (MUX), Vref, and ground check. The assembly shall read ground and ± Vref/2 and compare them to a known bit pattern to check A/D and MUX conversion accuracy. The assembly shall read a set of voltages and compare it to a known bit pattern to check for stuck bits. The assembly shall run these tests every 20 ms and store the appropriate message in EEPROM (see 4.3.5.5.1.1).

a. Type I and Ia messages stored in EEPROM shall support tables III and XI.
b. Type II messages stored in EEPROM shall support AlliedSignal Engines document 21-9187.

3.5.5.1.1.1 Ground potential check. If measured ground potential deviates from the known ground bit pattern by more than ± 0.25 Vdc for 5 consecutive 20 ms-cycles, the assembly shall enter PM 0 (during a start) or PM II (after a completed start or while not in start mode). If measured ground potential deviates from the known ground bit pattern by more than ± 1.0 Vdc for 5 consecutive 20 ms-cycles, the assembly shall enter PM III (after completing a start) (see 4.3.5.1.1.1).

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3.5.5.1.1.2 ± Vref/2 check. If measured ± Vref/2 deviates from the known bit pattern by more than ± 1 Vdc for 5 consecutive 20 ms-cycles, the assembly shall enter PM 0 (during a start) or PM III (after a completed start or while not in start mode). If measured ± Vref/2 deviates from the known bit pattern by more than ± 0.25 Vdc for 5 consecutive 20 ms-cycles, the assembly shall enter PM II (after completing a start) (see 4.3.5.5.1.1.2).

3.5.5.1.1.3 Stuck bit. If a comparison of measured voltages to known bit patterns shows that any of the four least significant bits are stuck (see 3.5.5.1.1), the assembly shall enter PM II (after completing a start). If a comparison of measured voltages to known bit patterns shows that any of the eight most significant bits are stuck, the assembly shall enter PM 0 (during a start) or PM III (after a completed start or while not in start mode) (see 4.3.5.5.1.1.3).

3.5.5.1.2 CPU. The assembly shall check each instruction type by evaluating a default instruction calculation against a known result. If the calculation fails, the assembly shall enter the tight loop mode (see 3.3.5.4.2). The assembly shall run this evaluation at power-up, after CPU reset, and during each background cycle (see 4.3.5.5.1.2).

3.5.5.1.3 RAM function check. The assembly shall check all RAM locations for data retention by writing one word to a memory location, reading the location, and comparing the two values. If the values are not equal, the assembly shall enter the tight loop mode. The assembly shall perform the check at the rate of one word per 20 ms after power-up (see 4.3.5.5.1.3).

3.5.5.1.4 Programmable Read-Only Memory (PROM) check (checksum). The assembly shall sum the contents of every memory location of program EEPROM and compare against a known checksum. If the checksums are not equal, the assembly shall enter the tight loop mode. The assembly shall run this comparison within 7.5 seconds of power up (see 4.3.5.5.1.4).

3.5.5.1.5 Background iteration checks. The assembly shall run two iteration checks continuously. The first check requires the completion of at least one real-time cycle during each background cycle. If this fault occurs, the assembly shall set a flag. The second check requires the completion of at least one background cycle during 175 real-time cycles. If the fault occurs, the assembly shall enter the tight loop mode (see 3.3.5.4.2 and 4.3.5.5.1.5).

3.5.5.1.6 Real-time task time-out. The assembly shall monitor task completion every 20 ms. If the assembly fails to execute all real-time tasks within 20 ms, the assembly shall reset the CPU (see 4.3.5.5.1.6).

3.5.5.1.7 Watchdog timer. The assembly shall contain a Programmable Clock And Timer (PCAT) microcircuit that shall support the watchdog timer function. The assembly shall check all timers and interrupt circuits for function and synchronization on power-up or CPU reset and for synchronization every 10 ms (see 4.3.5.5.1.7). If this check fails, the assembly shall reset the CPU.

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3.5.5.1.7.1 Watchdog self-test. During power-up mode, the assembly shall issue early and late PCAT resets. If a watchdog fault is not detected, the assembly shall cause the CPU to reset (see 4.3.5.5.1.7.1).

3.5.5.1.7.2 Watchdog timer test. If the assembly fails to reset the PCAT every 10 to 10.5 ms, the assembly shall reset the CPU (see 4.3.5.5.1.7.2).

3.5.5.1.8 Fault EEPROM checks. The assembly shall allow a maximum of 600 writes to fault EEPROM per power-up or CPU reset. The assembly shall perform a data redundancy check (see 3.5.5.1.8.1) on power-up or CPU reset and a write verification check (see 3.5.5.1.8.2) when data is written to fault EEPROM (see 4.3.5.5.1.8).

3.5.5.1.8.1 Data redundancy check. During power-up or CPU reset, the assembly shall read all calibration data, all diagnostic fault data, and all engine monitor data from fault EEPROM, and shall compare identical data stored in three consecutive locations. If at least two of the three values do not agree, the assembly shall default to the nominal value (see 4.3.5.5.1.8.1).

a. Types I and Ia fault data stored in EEPROM shall support tables III and XI. For calibration and diagnostic fault data, if the redundancy test fails, the assembly shall enter PM IV and store the fault in EEPROM.
b. Type II fault data stored in EEPROM shall support AlliedSignal Engines document 21-9187. For calibration and diagnostic fault data, if the redundancy test fails, the assembly shall store the fault in EEPROM.

3.5.5.1.8.2 Fault EEPROM write verification. The assembly, when writing a value to fault EEPROM, shall read the location and compare the two values. If the values do not match, the assembly shall prevent further writes to EEPROM. For types I and Ia only, if the write verification check fails, the assembly shall enter PM IV (see 4.3.5.5.1.8.2).

3.5.5.1.8.3 Digital-to-Analog (D/A) converter check. The assembly shall write 3 bit patterns to the A/D and read them back through the D/A at the rate of one pattern each 20 ms. If the deviation between the written and read patterns is greater than 20% for 8 successive checks (0.48 seconds), the assembly shall enter PM 0 (during a start) and PM III (after a completed start or while not in start mode), and store the appropriate fault in EEPROM (see 4.3.5.5.1.8.3).

a. For types I and Ia faults stored in EEPROM shall support table XI.
b. For type II faults stored in EEPROM shall support AlliedSignal Engines document 21-9187.

3.5.5.2 BIT. BIT circuits shall be provided to check all inputs and outputs by sending a current through each circuit and measuring the impedance to determine the validity of the signal and identify any Line Replacement Units (LRU) hard faults (see 4.3.5.5.2).

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a. Type I and Ia assemblies shall display the identified hard faults in the diagnostic window (see tables III and XI). These checks shall be performed by depressing the switch on the shorting cap of the J1 connector (see 3.4.7).
b. Type II assemblies shall set the appropriate data bit for the identified hard faults on the 1553 data bus in accordance with AlliedSignal Engines document 21-9187. These checks shall be initiated by a command over the 1553 data bus.

3.5.5.2.1 LRU. LRU checking shall include the following signals (see 4.3.5.5.2.1):

a. PLA RVDT PRI/SEC
b. IGV RVDT PRI/SEC
c. PTS RVDT PRI/SEC
d. WF LVDT PRI/SEC
e. IGN Exciter
f. IGV Solenoid
g. PTS Solenoid

3.5.5.2.2 EEPROM stored faults. The assembly shall classify EEPROM faults as diagnostic messages (see tables III and XI). These messages are stored in EEPROM (see 4.3.5.5.2.2).

a. Types I and Ia assemblies shall display the messages specified above in the diagnostic window during BIT.
b. Type II assemblies shall use the messages stored in EEPROM and set the appropriate BIT on the MIL-STD-1553 data bus in accordance with AlliedSignal Engines document 21-9187.

3.5.5.2.2.1 Clearing EEPROM faults.

3.5.5.2.2.1.1 Clearing EEPROM faults (types I and Ia). The 45-second stop timer shall be interrupted and the EEPROM fault memory shall be cleared when the assembly detects the following conditions after the timer has been initiated (see 4.3.5.5.2.2.1.1):

a. NH < 5%.
b. Tactical idle true (see 3.3.1.7.6.1.5).
c. PLA ≥ 50°.

After the memory is cleared, the stop timer shall resume.

3.5.5.2.2.1.2 Clearing EEPROM faults (type II). The clearing procedure for type II assemblies shall be activated over the 1553 data bus (see AlliedSignal Engines document 21-9187 and 4.3.5.5.2.2.1.2).

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3.5.5.2.2.2 Software identification (types I and Ia only). The assembly shall provide a coded message for software identification. The message shall be displayed as the first message after BIT is initiated. The message shall read "S/W ISS XXX" (XXX is a three-character software version number) (see 4.3.5.5.2.2.2).

3.5.5.2.3 Fuel flow restriction. A fuel flow restriction shall result when all of the following conditions exist for 20 seconds minimum (see 4.3.5.5.2.3):

a. T_7 limiter (see 3.3.2.5) or maximum NPT governor (see 3.3.2.2.1.3) does not control the assembly.
b. NH is a minimum of 6% below the NH trim line (see 3.3.6.6).
c. WFR exceeds 780 lb/h (354 kg/h).

3.5.5.2.3.1 Fuel flow restriction (types I and Ia). If the fuel flow restriction fault is indicated, the assembly shall enter PM IV and the fault shall be stored in EEPROM (see 4.3.5.5.2.3.1).

3.5.5.2.3.2 Fuel flow restriction (type II). If the fuel flow restriction fault is indicated, the fault shall be stored in EEPROM in accordance with AlliedSignal Engines document 21-9187 (see 4.3.5.5.2.3.2).

3.5.6 Nameplates and marking. Where possible, all components, wires, test points, and connectors shall be clearly labeled with applicable reference designations to facilitate identification in operating and maintenance documents. The assembly shall be marked in accordance with SAE-AS478, and identified in accordance with MIL-STD-130. Nameplates shall be in accordance with MIL-DTL-15024 (see 4.3.5.6).

3.5.7 Workmanship. All parts of the assembly shall be free of dirt or other extraneous material and shall be free of workmanship defects that would impair performance. Workmanship shall be in accordance with MIL-STD-252 (see MIL-HDBK-454, Requirement 9 guidelines, 4.3.5.7 and 6.6).

3.6 Operating environment requirements. Each assembly shall operate under the following environmental conditions, without degradation in performance.

3.6.1 Temperature.

3.6.1.1 Operating temperatures. The assembly, excluding the backup battery, shall meet the requirements of 3.3.1.7 through 3.3.1.10 during and after exposure to temperatures ranging from -60 to +120°F (-51 to +49°C). The backup battery shall meet the requirements of 3.3.1.7 through 3.3.1.10 during and after exposure to temperatures ranging from -40 to +120°F (-40 to +49°C) (see 4.3.6.1.1 through 4.3.6.1.1.2).

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3.6.1.2 Storage temperature. The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10 after exposure to temperatures ranging from -60 to +160°F (-51 to +71°C) (see 4.3.6.1.2). For types Ia and II only, the assembly shall meet the requirements with and without the backup battery installed.

3.6.2 Relative humidity. The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10 during and after exposure to a relative humidity of 95 ± 5% (see 4.3.6.2).

3.6.3 Fungus. All nonmetallic materials that are not resistant to moisture and fungus shall be treated and shall not support fungus growth during and after exposure to fungus testing in accordance with 4.3.6.3.

3.6.4 Vibration.

3.6.4.1 Vibration (types I and Ia). The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10, without malfunction during and after exposure to 180 minutes of vibration, (spectra specified in figure 25) in each of three mutually perpendicular axes (vertical, lateral, and longitudinal) (see 4.3.6.4 or 4.3.6.4.1).

3.6.4.2 Vibration (type II). The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10 without malfunction, during and after exposure to 180 minutes of vibration in each of three mutually perpendicular axes (vertical, lateral, and longitudinal). The spectra specified in figure 26 shall be used if the assembly is installed on the vibration-isolated vehicle mounting bracket. The spectra specified in figure 27 shall be used if the assembly is tested without the bracket (see 4.3.6.4 or 4.3.6.4.1).

3.6.5 Basic shock. The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10, without malfunction, during and after exposure to 3 half-sine wave shock pulses in each direction (+ and -) of three mutually perpendicular axes (vertical, lateral, and longitudinal) at 73 ± 18°F (23 ± 10°C). The shock pulses shall be 30 ± 3g amplitude and 11 ± 1.1 ms duration if the assembly is installed on the vibration-isolated vehicle mounting bracket. The shock pulses shall be of 32.4 ± 3.2g amplitude and 14.6 ± 1.4 ms if the assembly is not installed on the bracket (see 4.3.6.5 or 4.3.6.5.1).

3.6.6 Gun firing shock. The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10, without malfunction, during and after exposure to 3 half-sine wave shock pulses in each direction (+ and -) of 3 mutually perpendicular axes (vertical, lateral, and longitudinal) at 73 ± 18°F (23 ± 10°C). The shock pulses shall be 100 ± 10 g amplitude and 1.5 ± 0.2 ms duration if the assembly is installed on the vibration-isolated vehicle mounting bracket. The shock pulses shall be of 57 ± 5.7 g amplitude and 4.1 ± 0.4 ms if the assembly is not installed on the bracket (see 4.3.6.6 or 4.3.6.6.1).

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3.6.7 Ballistic shock.

3.6.7.1 Ballistic shock (types I and Ia). The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10, without malfunction, during and after exposure to 3 half-sine wave shock pulses in each direction (+ and -) of 3 mutually perpendicular axes (vertical, lateral, and longitudinal). The shock pulses shall be 200 ± 20 g amplitude and 0.5 ± 0.2 ms duration (see 4.3.6.7.1).

3.6.7.2 Ballistic shock (type II). The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10, without malfunction, after exposure to a minimum of 3 ballistic shocks as specified in figures 28 and 29 in the vertical direction to the base of the assembly with no shock mounts. Each shock shall meet the shock response spectrum within +9 decibels (db) and -6 db between 500 and 10 000 Hz, measured at the plane of the assembly base (see 4.3.6.7.2).

3.6.8 Cleaning spray. The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10 after exposure to a jet spray of tap water, with a nozzle pressure of 25 ± 2 psig (172 ± 14 kPa), applied perpendicular to the assembly surface at a distance of 1 ± 0.5 ft (0.30 ± 0.15 m). The duration of the exposure shall be 3.5 minutes for each surface (six sides) for a total exposure period of 21 minutes, at a coverage rate of 1 square foot per minute (ft^2/min) (0.1 m^2/min) (see 4.3.6.8). For type II only, the assembly shall meet the requirements with the backup battery removed (see 4.3.6.8 through 4.3.6.8.2).

3.6.9 Submergence.

3.6.9.1 Submergence (types I and Ia). Types I and Ia assemblies shall meet the requirements of 3.3.1.7 through 3.3.1.10 and there shall be no evidence of water leakage after a minimum of 5 minutes submergence to a depth of 1 in. (2.54 cm) between the water surface and the top of the assembly (see 4.3.6.9.1).

3.6.9.2 Submergence (type II). Type II assemblies shall meet the requirements of 3.3.1.7 through 3.3.1.10 and there shall be no evidence of water leakage after a minimum of 3 minutes submergence to a depth of 1 in between the water surface and the top of the assembly (see 4.3.6.9.2). Type II assemblies shall be internally pressurized to 6.0 + 0.5 pounds per square inch differential (psid) (41.4 ± 3.4 kilopascals (kPa)).

3.6.10 Salt fog. The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10 after exposure for 48 hours to a salt fog solution consisting of 5% by weight sodium chloride (NaC1) and 95% weight distilled water with a temperature range of 90 to 95°F (32 to 35°C). The fallout solution and related parameters shall be in accordance with MIL-STD-810 (see 4.3.6.10).

3.6.11 Chemicals. The assembly shall meet the requirements of 3.3.1.7 through 3.3.1.10 after exposure to vapors in direct contact (not submerged) with, the following materials for 48 hours at room temperature (see 4.3.6.11):

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a. Fuel:
   (1) A-A-52557 (DF 1, DF 2, or DF A), ASTM D975 (Commercial Diesel No. 1-D or No. 2-D), or MIL-F-16884 (Marine Diesel Fuel Oil).
   (2) MIL-DTL-5624 (Grade JP-4 or JP-5) or ASTM D1655 (Commercial Turbine Jet-A or Jet A-1).
   (3) MIL-DTL-83133 (Grade JP-8).
   (4) ASTM D4814 (MoGas) or regular automotive leaded gasoline.
b. Hydraulic fluid: MIL-PRF-6083 and MIL-PRF-46170.
c. Cleaning agents: A-A-59133 or equivalent.

3.6.12 Nuclear environment.

3.6.12.1 Electromagnetic pulse (EMP).

3.6.12.1.1 Radiated susceptibility, electric field (Army, RS103). The assembly shall meet the RS103 requirement of MIL-STD-461 and 3.3.1.7 through 3.3.1.10 and 3.6.12.3 after exposure to EMP. Evaluation of the radiated susceptibility of equipment and subsystems shall include connectors, cables, and interconnecting wiring to electromagnetic pulse fields (see 4.3.6.12.2).

3.6.12.1.2 Conducted susceptibility, damped sinusoidal transients, pins, and terminals (Army, CS101 and CS116). The assembly shall meet the CS101 and CS116 requirement of MIL-STD-461 and the requirements of 3.3.1.7 through 3.3.1.10 and 3.6.12.3, after exposure to initial nuclear radiation, which includes neutron fluency, gamma peak dose rate, and total radiation dose as specified in LS 198800000160(S). Conducted susceptibility of equipment and subsystems shall be evaluated when subjected to the injection of EMP onto equipment interface pins and terminals of power leads, control leads, signal leads, and grounds which are not internally grounded (see 4.3.6.12.2).

3.6.12.2 Nuclear radiation. Hardness Critical Items (HCIs) and Hardness Critical Processes (HCPs) shall be identified in technical data drawings and specifications in accordance with the HCIs and HCPs criteria of ASME-Y14.100. Selection of electronic piece part HCIs shall be source-controlled except when the generic military standard part has a design margin greater than ten with respect to nuclear radiation or EMP failure in the circuit application. Electronic piece parts which have a probability of survival in the circuit application of less than 99% at 90% confidence shall not be used. Repair parts or actions which are designated as HCIs or HCPs shall be identified as such in logistical documentation (see 4.3.6.12.2).

3.6.12.3 Transient response and recovery. Transient upset of the assembly is allowable during exposure to the nuclear radiation and EMP, provided the system automatically recovers to normal operational state within 2 seconds or recovers through manual operator intervention by removing and reapplying input power and a normal initiation procedure (see 4.3.6.12.3).

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4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.1.1).
b. Qualification inspection (see 4.1.2)
   (1) Acceptance tests (see 4.1.2.2).
   (2) Control tests (see 4.1.2.4)

4.1.1 First article inspection. Unless otherwise specified (see 6.2), the procuring activity shall select two assemblies out of the first ten produced under the production contract for first article inspection. First article samples shall be inspected as specified in table XIV. Approval of the first article samples by the procuring activity shall not relieve the supplier of the obligation to supply assemblies that are fully representative of those inspected as first article samples. Any changes or deviation of the production units from the first article samples shall be subject to the approval of the Procuring Activity. Unless otherwise specified, all testing shall be performed with facilities and resources furnished by the supplier.

4.1.1.1 First article failure. Deficiencies found during, or as a result of, first article inspection shall be cause for rejection of the sample(s) until evidence has been provide by the supplier that corrective action has been taken to eliminate the deficiency. Such deficiencies shall be considered as evidence that all items produced are deficient, unless contrary evidence is furnished by the supplier. Deficiencies in production and first article units shall be corrected by the supplier. First article inspection shall be repeated until the unit passes inspection.

4.1.2 Qualification inspection. Unless otherwise specified (see 6.2) the procuring activity shall select two assemblies out of the first ten produced under the production contract for qualification inspection. These samples shall be inspected as specified in table XIV. Software requirements verified in 4.2.2 and 4.2.2.1 during qualification shall be tested by a checksum on each assembly. Unless otherwise specified, all testing shall be performed with facilities and resources furnished by the supplier.

4.1.2.1 Qualification failure. Inspection shall cease when an assembly fails to meet any of the qualification requirements herein. The contractor shall provide corrective action and shall be approved by the Procuring Activity.

4.1.2.2 Acceptance tests (100%). Each assembly shall be subjected to the acceptance tests specified in table XIV. The requirements of 3.3.1.7 through 3.3.1.10 shall be tested in accordance with the limits of this specification to verify hardware input/output. Unless otherwise specified (see 6.2), all testing shall be performed with the supplier's facilities and resources.

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4.1.2.3 Acceptance examination. For examination purposes, defects shall be classified in accordance with table XV.

4.1.2.4 Control tests. Unless otherwise specified (see 6.2), the supplier shall select one representative sample out of 200 assemblies consecutively produced under the production contract for control testing. Control tests shall be performed no more than once in three months and no less than once in six months. Samples shall be inspected in accordance with table XIV. Approval of the samples by the Procuring Activity shall not relieve the supplier of the obligation to supply assemblies that are fully representative of those inspected as control test samples. Any changes or deviation of the production units from the control test samples shall be subjected to the approval of the Procuring Activity.

4.1.2.4.1 Control test failure. Deficiencies found during, or as a result of, control testing shall be cause for rejection of the sample(s) until evidence has been provided by the supplier that corrective action has been taken to eliminate the deficiency. Such deficiencies shall be considered as evidence that all items produced prior to completion of the testing are deficient, unless contrary evidence is furnished by the supplier. All deficiencies shall be corrected by the supplier.

4.2 Order of inspection. Perform operating environment tests first, followed by the remaining verifications in any sequence.

4.2.1 Functional testing. Functional testing shall be performed using the electrical connections and interfaces described in table VIII to verify the functional performance of hardware and control logic. A validation of the control logic detail shall be performed as specified in 4.2.2. The assembly shall be tested to verify conformance with 3.3.1.7 through 3.3.1.10 as specified in table XIV. Unless otherwise specified (see 6.2), testing shall be performed with facilities provided by the supplier.

4.2.2 Software testing. Testing shall be performed to verify that software functions meet design and performance requirements of 3.3.1. Each performance and design requirement shall be identified, with corresponding testing requirements. The software test description shall be developed in accordance with IEEE 12207, and shall specify tests which verify the specified requirements, including provisions for verifying the performance requirements described in this specification. The traceability of the tests to the requirements shall be documented. The following levels of testing shall be required:

a. Sub-program testing.
b. Program testing.
c. Software qualification testing.

Software qualification testing and system testing shall be documented.

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4.2.2.1 Software qualification tests. Software qualification tests shall be performed following a successful Test Readiness Review (TRR). Testing shall be monitored and audited by the supplier's Software Quality Assurance. The software, including a simulator, shall be used to perform software qualification testing. Techniques used during acceptance testing shall include the use of AGT 1500 engine simulation equipment. The supplier's laboratory shall be used for testing of the software with engine simulation equipment.

4.3 Verification methods. The types of verification methods included in this section are visual inspection, measurement, sample tests, full scale demonstration tests, simulation, modeling, engineering evaluation, component properties analysis, and similarity to previously approved or previously qualified designs.

4.3.1 Verification alternatives. The manufacturer may propose alternative test methods, techniques or equipment, including the application of statistical process control, tool control, or cost-effective sampling procedures, to verify performance. See the contract for alternatives that replace verifications required by this specification.

4.3.2 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified:

a. Temperature: 73 ± 18°F (23 ± 10°C).
b. Pressure: 25.5 to 30.5 in. (650 to 775 mm) mercury.
c. Relative humidity: 50 ± 30%.

4.3.3 Operating requirements verification.

4.3.3.1 Performance. Unless otherwise specified, each performance test specified herein shall be performed at 28 ± 1 Vdc with power applied to the assembly terminals as specified in 3.4.2.3. Verification of the internal hardware functions shall be made at 19 ± 1 Vdc. The assembly shall not produce voltage transients.

4.3.3.1.1 Real-time clock test. The assembly shall be tested to verify conformance to 3.3.1.1.

4.3.3.1.2 Backup battery. The assembly shall be tested to verify conformance to 3.3.1.2.

4.3.3.1.2.1 Backup battery during engine operation. The assembly shall be tested to verify conformance to 3.3.1.2.1.

4.3.3.1.2.2 Backup battery operation on start. The assembly shall be tested to verify conformance to 3.3.1.2.2.

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4.3.3.1.3 NED. The assembly shall be tested to verify conformance to 3.3.1.3.

4.3.3.1.3.1 NED (types I and Ia). To verify that no message is reported see 3.3.1.3.1.

4.3.3.1.3.2 NED (type II). The assembly shall be tested to verify conformance to 3.3.1.3.2.

4.3.3.1.4 Diagnostic window. The diagnostic window shall conform to the requirements of 3.3.1.4.

4.3.3.1.5 Power supply voltage. Apply input power and bring the assembly to a fully operational condition to meet the requirements of 3.3.1.5. Remove the energy source and verify that a fully charged backup battery supplies a backup power source of 20 ± 2 Vdc by measuring the voltage across Interlocks I and II and ground (see table VIII).

4.3.3.1.5.1 Power consumption. During steady state operation, verify that assembly power consumption does not exceed the value specified in 3.3.1.5.1.

4.3.3.1.6 Data bus communications.

4.3.3.1.6.1 EIA-232. The assembly shall be tested to verify conformance to 3.3.1.6.1.

4.3.3.1.6.2 1553 (type II only). The assembly shall be tested to verify conformance to 3.3.1.6.2.

4.3.3.1.7 Signal inputs.

4.3.3.1.7.1 Engine inlet air temperature (T_1). The assembly shall be tested to verify conformance to 3.3.1.7.1.

4.3.3.1.7.2 Power turbine inlet temperature (T_7). The assembly shall be tested to verify conformance to 3.3.1.7.2.

4.3.3.1.7.3 LVDT and RVDT secondary signals. The assembly shall be tested to verify conformance to 3.3.1.7.3.

4.3.3.1.7.3.1 PTS position feedback. The assembly shall be tested to verify conformance to 3.3.1.7.3.1.

4.3.3.1.7.3.2 IGV position feedback. The assembly shall be tested to verify conformance to 3.3.1.7.3.2.

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4.3.3.1.7.3.3 WF metering valve position. The assembly shall be tested to verify conformance to 3.3.1.7.3.3.

4.3.3.1.7.3.4 PLA. The assembly shall be tested to verify conformance to 3.3.1.7.3.4.

4.3.3.1.7.4 Frequency signal inputs. The assembly shall be tested to verify conformance to 3.3.1.7.4.

4.3.3.1.7.5 1553 data bus input (type II only). The assembly shall be tested to verify conformance to 3.3.1.7.5.

4.3.3.1.7.6 Discrete signal inputs. The assembly shall be tested to verify conformance to 3.3.1.7.6.

4.3.3.1.7.6.1 Discrete input signal processing.

4.3.3.1.7.6.1.1 Purge. The assembly shall be tested to verify conformance to 3.3.1.7.6.1.1.

4.3.3.1.7.6.1.1.1 Auto cooldown logic activation (type II only). The assembly shall be tested to verify conformance to 3.3.1.7.6.1.1.1.

4.3.3.1.7.6.1.2 Start.

4.3.3.1.7.6.1.2.1 Start (types I and Ia). The assembly shall be tested to verify conformance to 3.3.1.7.6.1.2.1.

4.3.3.1.7.6.1.2.2 Start (type II). The assembly shall be tested to verify conformance to 3.3.1.7.6.1.2.2.

4.3.3.1.7.6.1.3 Stop. The assembly shall be tested to verify conformance to 3.3.1.7.6.1.3.

4.3.3.1.7.6.1.4 Fault reset.

4.3.3.1.7.6.1.4.1 Manual fault reset. The assembly shall be tested to verify conformance to 3.3.1.7.6.1.4.1.

4.3.3.1.7.6.1.4.2 Automatic fault reset. The assembly shall be tested to verify conformance to 3.3.1.7.6.1.4.2.

4.3.3.1.7.6.1.5 Tactical idle. The assembly shall be tested to verify conformance to 3.3.1.7.6.1.5.

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4.3.3.1.7.6.1.6 Pivot neutral. The assembly shall be tested to verify conformance to 3.3.1.7.6.1.6.

4.3.3.1.7.6.1.7 NBC.

4.3.3.1.7.6.1.7.1 NBC (types I and Ia). The assembly shall be tested to verify conformance to 3.3.1.7.6.1.7.1.

4.3.3.1.7.6.1.7.2 NBC (type II). The assembly shall be tested to verify conformance to 3.3.1.7.6.1.7.2.

4.3.3.1.8 Output signals.

4.3.3.1.8.1 Output drivers. The assembly shall be tested to verify conformance to 3.3.1.8.1.

4.3.3.1.8.1.1 PTS driver (PTS solenoid HI and PTS solenoid LO). The assembly shall be tested to verify conformance to 3.3.1.8.1.1.

4.3.3.1.8.1.2 IGV driver (IGV solenoid HI, IGV solenoid LO). The assembly shall be tested to verify conformance to 3.3.1.8.1.2.

4.3.3.1.8.1.3 WF driver (fuel solenoid LO). The assembly shall be tested to verify conformance to 3.3.1.8.1.3.

4.3.3.1.8.2 NPT meter. The assembly shall be tested to verify conformance to 3.3.1.8.2.

4.3.3.1.8.3 Discrete output. The assembly shall be tested to verify conformance to 3.3.1.8.3.

4.3.3.1.8.3.1 Engine starter relay. The assembly shall be tested to verify conformance to 3.3.1.8.3.1.

4.3.3.1.8.3.2 Main fuel cutoff solenoid. The assembly shall be tested to verify conformance to 3.3.1.8.3.2.

4.3.3.1.8.3.3 Engine run-time meter. The assembly shall be tested to verify conformance to 3.3.1.8.3.3.

4.3.3.1.8.3.4 Vehicle indicator lamps. The assembly shall be tested to verify conformance to 3.3.1.8.3.4.

4.3.3.1.8.3.4.1 NBC prioritization. The assembly shall be tested to verify conformance to 3.3.1.8.3.4.1.

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4.3.3.1.8.3.5 Ignition exciter. The assembly shall be tested to verify conformance to 3.3.1.8.3.5.

4.3.3.1.8.3.6 Backup fuel cutoff solenoid. The assembly shall be tested to verify conformance to 3.3.1.8.3.6.

4.3.3.1.9 Differential transformer excitation. The assembly shall be tested to verify conformance to 3.3.1.9.

4.3.3.1.10 1553 data bus output (type II only). The assembly shall be tested to verify conformance to 3.3.1.10.

4.3.3.2 Fuel scheduling.

4.3.3.2.1 Starting. The assembly shall be tested to verify conformance to 3.3.2.1.

4.3.3.2.2 NPT governing. The assembly shall be tested to verify conformance to 3.3.2.2.

4.3.3.2.2.1 NPT idle (pivot/neutral). The assembly shall be tested to verify conformance to 3.3.2.2.1.

4.3.3.2.2.1.1 NPT idle (drive). The assembly shall be tested to verify conformance to 3.3.2.2.1.1.

4.3.3.2.2.1.2 NPT idle (tactical idle). The assembly shall be tested to verify conformance to 3.3.2.2.1.2.

4.3.3.2.2.1.3 Maximum NPT governing. The assembly shall be tested to verify conformance to 3.3.2.2.1.3.

4.3.3.2.3 NH governing.

4.3.3.2.3.1 Standard NH idle. The assembly shall be tested to verify conformance to 3.3.2.3.1.

4.3.3.2.3.2 T1 biased NH idle. The assembly shall be tested to verify conformance to 3.3.2.3.2.

4.3.3.2.3.3 Maximum NH governing. The assembly shall be tested to verify conformance to 3.3.2.3.3.

4.3.3.2.4 Acceleration schedule. The assembly shall be tested to verify conformance to 3.3.2.4.

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4.3.3.2.4.1 Acceleration command. The assembly shall be tested to verify conformance to 3.3.2.4.1.

4.3.3.2.5 Maximum (T_7) control. The assembly shall be tested to verify conformance to 3.3.2.5.

4.3.3.2.6 Deceleration schedule. The assembly shall be tested to verify conformance to 3.3.2.6.

4.3.3.2.7 Temperature gradient control (T_7 lag function). The assembly shall be tested to verify conformance 3.3.2.7.

4.3.3.2.8 NPO. The assembly shall be tested to verify conformance to 3.3.2.8.

4.3.3.2.9 Secondary WF. The assembly shall be tested to verify conformance to 3.3.2.9.

4.3.3.3 PTS control. The assembly shall be tested to verify conformance to 3.3.3.

4.3.3.4 IGV control. The assembly shall be tested to verify conformance to 3.3.4.

4.3.3.4.1 IGV prioritization. The assembly shall be tested to verify conformance to 3.3.4.1.

4.3.3.5 PMs. The assembly shall be tested to verify conformance to 3.3.5.

4.3.3.5.1 PM 0 - no crank (type II only). The assembly shall be tested to verify conformance to 3.3.5.1.

4.3.3.5.1.1 PM 0 fault conditions. The assembly shall be tested to verify conformance to 3.3.5.1.1.

4.3.3.5.2 PM I - engine abort. The assembly shall be tested to verify conformance to 3.3.5.2.

4.3.3.5.2.1 PM I fault conditions. The assembly shall be tested to verify conformance to 3.3.5.2.1.

4.3.3.5.3 PM II. The assembly shall be tested to verify conformance to 3.3.5.3.

4.3.3.5.3.1 PM II fault conditions. The assembly shall be tested to verify conformance to 3.3.5.3.1.

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4.3.3.5.4 PM III. The assembly shall be tested to verify conformance to 3.3.5.4.

4.3.3.5.4.1 PM III fault conditions. The assembly shall be tested to verify conformance to 3.3.5.4.1.

4.3.3.5.4.2 Tight loop mode. The assembly shall be tested to verify conformance to 3.3.5.4.2.

4.3.3.5.4.2.1 Tight loop mode conditions. The assembly shall be tested to verify conformance to 3.3.5.4.2.1.

4.3.3.5.4.3 PM III fault removal. The assembly shall be tested to verify conformance to 3.3.5.4.3.

4.3.3.5.5 PM IV. The assembly shall be tested to verify conformance to 3.3.5.5.

4.3.3.5.5.1 PM IV fault conditions. The assembly shall be tested to verify conformance to 3.3.5.5.1.

4.3.3.5.6 PM V (type II only). The assembly shall be tested to verify conformance to 3.3.5.6.

4.3.3.5.7 CPU reset mode. The assembly shall be tested to verify conformance to 3.3.5.7.

4.3.3.5.7.1 CPU reset mode conditions. The assembly shall be tested to verify conformance to 3.3.5.7.1.

4.3.3.5.8 Transformer fault protection. The assembly shall be tested to verify conformance to 3.3.5.8.

4.3.3.6 Algorithms.

4.3.3.6.1 PTS test cell logic. The assembly shall be tested to verify conformance to 3.3.6.1.

4.3.3.6.2 PTS adaptive routine logic. The assembly shall be tested to verify conformance to 3.3.6.2.

4.3.3.6.3 Engine health check. The assembly shall be tested to verify conformance to 3.3.6.3.

4.3.3.6.3.1 Engine health check (types I and Ia). The assembly shall be tested to verify conformance to 3.3.6.3.1.

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4.3.3.6.3.2 Engine health check (type II). The assembly shall be tested to verify conformance to 3.3.6.3.2.

4.3.3.6.4 Cooldown logic.

4.3.3.6.4.1 Cooldown logic types I and Ia (M1 and M1A1 vehicles). The assembly shall be tested to verify conformance to 3.3.6.4.1.

4.3.3.6.4.2 Cooldown logic (type II).

4.3.3.6.4.2.1 Cooldown logic type II (M1A2 Abrams vehicle). The assembly shall be tested to verify conformance to 3.3.6.4.2.1.

4.3.3.6.4.2.2 Cooldown logic type II (HAB). The assembly shall be tested to verify conformance to 3.3.6.4.2.2.

4.3.3.6.5 External stop and LOP command. The assembly shall be tested to verify conformance to 3.3.6.5.

4.3.3.6.5.1 External stop and LOP command (types I and Ia). The assembly shall be tested to verify conformance to 3.3.6.5.1.

4.3.3.6.5.2 External stop and LOP command (type II). The assembly shall be tested to verify conformance to 3.3.6.5.2.

4.3.3.6.6 Manual NH trim.

4.3.3.6.6.1 Manual NH trim (types I and Ia). The assembly shall be tested to verify conformance to 3.3.6.6.1.

4.3.3.6.6.2 Manual NH trim (type II). The assembly shall be tested to verify conformance to 3.3.6.6.2.

4.3.3.6.7 Rollback protection. The assembly shall be tested to verify conformance to 3.3.6.7.

4.3.3.6.8 Flameout detection and auto relight. The assembly shall be tested to verify conformance to 3.3.6.8.

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4.3.3.6.8.1 WF trimming. The assembly shall be tested to verify conformance to 3.3.6.8.1.

4.3.4 Interface requirements verification. Complete each verification under 4.3.4.

4.3.4.1 Mechanical. The assembly shall be tested to verify conformance to 3.4.1.

4.3.4.1.1 Weight. The assembly shall be tested to verify conformance to 3.4.1.1.

4.3.4.2 Electrical. The assembly shall be tested to verify conformance to 3.4.2.

4.3.4.2.1 Connectors and receptacles. The assembly shall be tested to verify conformance to 3.4.2.1.

4.3.4.2.2 Pins. The assembly shall be tested to verify conformance to 3.4.2.2.

4.3.4.2.3 Power. The assembly shall be tested to verify conformance to 3.4.2.3.

4.3.4.3 1553 data bus (type II only). The assembly shall be tested to verify conformance to 3.4.3.

4.3.4.4 1553 RT interface board (type II only). The assembly shall be tested to verify conformance to 3.4.4.

4.3.4.5 EIA-232. The assembly shall be tested to verify conformance to 3.4.5.

4.3.4.6 Bonds and grounds. The assembly shall be tested to verify conformance to 3.4.6. The dc resistance of each bond shall be measured in accordance with MIL-STD-202, Method 307.

4.3.4.6.1 Chassis isolation. The assembly shall be tested to verify conformance to 3.4.6.1.

4.3.4.6.1.1 Chassis isolation – acceptance test. The assembly shall be tested to verify conformance to 3.4.6.1.1.

4.3.4.6.1.2 Chassis isolation – qualification, first article test and control tests. The assembly shall be tested to verify conformance to 3.4.6.1.2.

4.3.4.7 J1 connector cap.

4.3.4.7.1 J1 connector cap (types I and Ia). The assembly shall be tested to verify conformance to 3.4.7.1.

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4.3.4.7.2 J1 connector cap (type II). The assembly shall be tested to verify conformance to 3.4.7.2.

4.3.4.8 Thermal (heat transfer). Verify that the dissipation of the assembly's heat to the environment meets the requirement of 3.4.8, without forced circulation of interior ambient air.

4.3.4.9 EMR.

4.3.4.9.1 EMR (types I and Ia). The assembly shall be tested to verify conformance to 3.4.9.1.

4.3.4.9.2 EMR (type II). The assembly shall be tested to verify conformance to 3.4.9.2.

4.3.5 Ownership and support requirements verification.

4.3.5.1 Reliability. The MTBF requirement specified in 3.5.1 shall be verified by a Failure Mode Effects and Criticality Analysis (FMECA) of test data collected from vehicle tests and/or laboratory tests. Use of the test and failure data on similar components is permitted, providing allowances are made for differences in design, usage, and environment. If test history data is insufficient to provide evidence of the assembly MTBF, compliance shall be determined on the basis of the FMECA failure rates (see MIL-HDBK-217 guidelines and 6.6).

4.3.5.1.1 Analysis method. A combination of hardware and functional methods shall be used for FMECA analysis. The functional block diagram of the assembly shall be constructed to provide the ability for tracing failure mode effects through all levels of indenture. In order to reflect implementation and input/output, the functional block diagram shall identify hardware assemblies. Hardware functions shall be analyzed at the component level. All unused parts, connectors, and pins or chips shall be included in the reliability analysis. Ground rules and analysis assumptions shall be identified and recorded prior to initiation of the analysis. Additional ground rules and assumptions may be added for any assembly if requirements change (see 3.5.1).

4.3.5.1.2 Classes of defects severity. Classes of severity shall be included with the classes of defects in accordance with table XV. The sum of all classes of defects severity shall be no less than the MTBF specified in 3.5.1.

4.3.5.1.3 Fault tree analysis. Fault tree analysis shall be performed for the following system failure states:

a. NH overspeed
b. T_7 overtemperature (run and start modes)
c. NPT overspeed
d. Flame out

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e. Surge
   (1) Run mode
   (2) Start mode
f. Reduced performance
   (1) Major power loss
   (2) Minor power loss, loss of efficiency, and peak power on dynamic response
   (3) Start malfunction
   (4) Stop malfunction
   (5) Start actuation during operation
   (6) Loss of data busses

4.3.5.2 Maintainability. Compliance to 3.5.2 shall be on the basis of formal maintainability analysis and predictions, using approved failure rates and MTTR data sources and prediction procedures.

4.3.5.3 Interchangeability. Verification of conformance to 3.5.3 shall be accomplished by inspection of technical data supported by analysis.

4.3.5.4 Quality. Verification shall be by industry and/or Government approved documentation or by test (see 3.5.4 and 4.1.1 through 4.1.2.3.1).

4.3.5.4.1 Software quality. The supplier shall conduct software quality control in accordance with good commercial standards which are approved by the Procurement Activity (see 3.5.4.1).

4.3.5.5 Diagnostics.

4.3.5.5.1 Continuous self-test. The assembly shall be tested to verify conformance to 3.5.5.1.

4.3.5.5.1.1 A/D converter, MUX, Vref, and ground check. The assembly shall be tested to verify conformance to 3.5.5.1.1.

4.3.5.5.1.1.1 Ground potential check. The assembly shall be tested to verify conformance to 3.5.5.1.1.1.

4.3.5.5.1.1.2 ± Vref/2 check. The assembly shall be tested to verify conformance to 3.5.5.1.1.2.

4.3.5.5.1.1.3 Stuck BIT test. The assembly shall be tested to verify conformance to 3.5.5.1.1.3.

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4.3.5.5.1.2 CPU test. The assembly shall be tested to verify conformance to 3.5.5.1.2.

4.3.5.5.1.3 RAM function check. The assembly shall be tested to verify conformance to 3.5.5.1.3.

4.3.5.5.1.4 PROM check (checksum). The assembly shall be tested to verify conformance to 3.5.5.1.4.

4.3.5.5.1.5 Background iteration checks. The assembly shall be tested to verify conformance to 3.5.5.1.5.

4.3.5.5.1.6 Real-time task time-out. The assembly shall be tested to verify conformance to 3.5.5.1.6.

4.3.5.5.1.7 Watchdog timer. The assembly shall be tested to verify conformance to 3.5.5.1.7.

4.3.5.5.1.7.1 Watchdog self-test. The assembly shall be tested to verify conformance to 3.5.5.1.7.1.

4.3.5.5.1.7.2 Watchdog timer test. The assembly shall be tested to verify conformance to 3.5.5.1.7.2.

4.3.5.5.1.8 Fault EEPROM checks. The assembly shall be tested to verify conformance to 3.5.5.1.8.

4.3.5.5.1.8.1 Data redundancy check. The assembly shall be tested to verify conformance to 3.5.5.1.8.1.

4.3.5.5.1.8.2 Fault EEPROM write verification. The assembly shall be tested in accordance with 3.5.5.1.8.2.

4.3.5.5.1.8.3 D/A converter check. The assembly shall be tested to verify conformance to 3.5.5.1.8.3.

4.3.5.5.2 BIT. The assembly shall be tested to verify conformance to 3.5.5.2.

4.3.5.5.2.1 LRU. The assembly shall be tested to verify conformance to 3.5.5.2.1.

4.3.5.5.2.2 EEPROM stored faults. The assembly shall be tested to verify conformance to 3.5.5.2.2.

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4.3.5.5.2.2.1 Clearing EEPROM faults.

4.3.5.5.2.2.1.1 Clearing EEPROM faults (types I and Ia). The assembly shall be tested to verify conformance to 3.5.5.2.2.1.1.

4.3.5.5.2.2.1.2 Clearing EEPROM faults (type II). The assembly shall be tested to verify conformance to 3.5.5.2.2.1.2.

4.3.5.5.2.2.2 Software identification (types I and Ia only). The assembly shall be tested to verify conformance to 3.5.5.2.2.2.

4.3.5.5.2.3 Fuel flow restriction. The assembly shall be tested to verify conformance to 3.5.5.2.3.

4.3.5.5.2.3.1 Fuel flow restriction (types I and Ia). The assembly shall be tested to verify conformance to 3.5.5.2.3.1.

4.3.5.5.2.3.2 Fuel flow restriction (type II). The assembly shall be tested to verify conformance to 3.5.5.2.3.2.

4.3.5.6 Nameplates and marking. The assembly shall be tested to verify conformance to 3.5.6.

4.3.5.7 Workmanship. Workmanship examinations shall be performed at all phases of fabrication, assembly, and test to verify conformance to 3.5.7.

4.3.6 Operating environment requirements verification. Complete each of the following environmental verifications:

a. Assemblies undergoing qualification and first article inspection of table III shall be subjected to the performance tests specified in the applicable paragraphs before, during (where required in the test method), and immediately after performance of each environmental test.
b. Assemblies undergoing control testing of table III shall first pass the acceptance tests indicated therein. After all environmental tests have been completed, the assemblies need testing only once for post-environmental performance tests specified.
c. Compliance with the selected performance tests during exposure to environment as specified herein provides evidence of conformance to the requirements.

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4.3.6.1 Temperature.

4.3.6.1.1 Operating temperature. The assembly shall be tested in accordance with 4.3.6.1.1.1 and 4.3.6.1.1.2 to verify conformance with 3.6.1.1.

4.3.6.1.1.1 High temperature. High temperature requirements shall be verified by the following tests:

a. Types I, Ia and II with battery installed. Subject the assembly, in a non-operating state, to a temperature of no less than the maximum specified in 3.6.1.2 for 48 hours. At the conclusion of the 48 hours, reduce the temperature to no less than the maximum temperature specified in 3.6.1.1. Allow the assembly temperature to stabilize for one hour, then perform the tests of 4.2.1.
b. Types Ia and II with battery removed. Remove the backup battery. Subject the assembly, in a non-operating state, to no less than the maximum temperature specified in 3.6.1.2 for 48 hours. At the conclusion of the 48 hours, reduce the temperature to 73 ± 18°F. Allow the assembly temperature to stabilize for 1 hour. Install the backup battery and perform the tests of 4.2.1.

4.3.6.1.1.2 Low temperature. Low temperature requirements shall be verified by the following tests:

a. Types I, Ia and II with battery installed. Subject the assembly, in a non-operating state, to a temperature of no higher than -60°F for 24 hours. At the conclusion of the 24 hours, perform the tests of 4.2.1. At the conclusion of testing, turn off the assembly and increase the temperature to 73 ± 18°F. Allow the assembly temperature to stabilize for one hour, then perform the tests of 4.2.1.
b. Types Ia and II with battery removed. Remove the backup battery. Subject the assembly, in a non-operating state, to a temperature of no higher than -60°F for 24 hours. At the conclusion of the 24 hours, increase the temperature to 73 ± 18°F. Allow the assembly temperature to stabilize for one hour. Install the backup battery and perform the tests of 4.2.1.

4.3.6.1.2 Storage temperature. The assembly shall be tested to verify conformance to 3.6.1.2.

4.3.6.2 Relative humidity. With no power applied to the assembly, raise the internal chamber temperature to 140°F (60°C) and the relative humidity to 95 ± 5% over a 2 hour period. Maintain for 6 hours. Reduce the internal temperature to 86°F (30°C) over a period of 8 hours and maintain this temperature for 8 hours. This process shall be considered 1 cycle. Repeat for 10 cycles. At the end of the fifth and tenth cycles, while at 86°F, operate the assembly at 29 ± 1V for 5 minutes. After completion of 10 cycles, perform an acceptance test at 73 ± 18°F. After completion of this test, visually inspect and verify conformance to 3.6 2.

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4.3.6.3 Fungus. The assembly shall be subjected to fungus exposure in accordance with MIL-STD-810, Method 508.4 for 28 days minimum, with the following exceptions:

a. Ambient temperature: 82 ± 2°F (28 ± 1°C)
b. Relative humidity: 98 ± 2%

The assembly shall be examined for damage to verify conformance to 3.6.3.

4.3.6.4 Vibration. Mount the assembly using the installed interface configuration for its intended application, or equivalent, and subject the assembly to the requirements of 3.6.4. The vibration shall be imposed at a logarithmic sweep rate of 15 minutes from 5 to 500 to 5 Hz at 73 ± 18°F. During exposure, monitor the assembly to verify that intermittent operation does not occur and perform the tests of 4.1.2.2 at 29 ± 1 Vdc in each axis. At the conclusion of this test, examine the assembly for damage, and perform the tests of 4.1.2.2 (see 3.6.4.1 and 3.6.4.2).

4.3.6.4.1 Vibration control test. Subject the assembly to the tests of 4.3.6.4, except that the exposure time shall be 75 minutes.

4.3.6.5 Basic shock. Mount the assembly using the installed interface configuration for its intended application, or equivalent, and subject the assembly to the shock impulses of 3.6.5. During exposure, monitor the assembly to verify that intermittent operation does not occur and perform the tests of 4.1.2.2 at 29 ± 1 Vdc in each axis. At the conclusion of this test, examine the assembly for damage and perform the tests of 4.1.2.2.

4.3.6.5.1 Basic shock control test. Subject the assembly to the test of 4.3.6.5, except that the amplitude of the shock pulses shall be 85% of the peak amplitudes specified in 3.6.5. The shock pulse duration shall not be diminished from the values specified in 3.6.5.

4.3.6.6 Gun firing shock. Mount the assembly using the installed interface configuration for its intended application, or equivalent, and subject the assembly to the shock impulses specified in 3.6.6. During exposure, monitor the assembly to verify that intermittent operation does not occur and perform the tests of 4.1.2.2 at 29 ± 1 Vdc in each axis. At the conclusion of this test, examine the assembly for damage and perform the tests of 4.1.2.2.

4.3.6.6.1 Gun firing control test. Subject the assembly to the tests of 4.3.6.6, except that the amplitude of the shock pulses shall be 75% of the peak amplitudes specified in 3.6.6. The shock pulse duration shall not be diminished from the values specified in 3.6.6.

4.3.6.7 Ballistic shock.

4.3.6.7.1 Ballistic shock (types I and Ia). Mount the assembly using the installed interface configuration for its intended application, or equivalent. Subject the assembly to the shock pulses specified in 3.6.7.1. At the conclusion of this test, examine the assembly for damage and perform the tests of 4.1.2.2.

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4.3.6.7.2 Ballistic shock (type II). Mount the assembly using the installed interface configuration for its intended application, or equivalent. Subject the assembly to the shock pulses specified in 3.6.7.2. At the conclusion of this test, examine the assembly for damage and perform the tests of 4.1.2.2. This requirement may be satisfied by an analysis of the assembly which demonstrates that exposure to the type II basic shook will not lead to mechanical failure of subassemblies or discrete components. The analysis, submitted by the supplier, shall exclude internal packaging of the individual electronic piece parts. For the analysis, the ballistic input shall be the same time history shown in Figure 29, applied three times to the base of the assembly in the vertical direction.

4.3.6.8 Cleaning spray. The assembly shall be subjected to a cleaning jet spray test as specified in 3.6.8.

4.3.6.8.1 Cleaning spray (types I and Ia). After exposure to the spray, types I and Ia assemblies shall be visually examined for deterioration and subjected to the tests of 4.1.2.2. Upon completion, the assembly backup battery shall be removed, and a visual inspection shall be performed to verify that there is no water leakage (see 3.6.8).

4.3.6.8.2 Cleaning spray (type II). After exposure to the spray, type II assemblies shall have the backup battery removed. The assembly shall again be exposed to a cleaning spray test as specified in 3.6.8. The auxiliary cover plate shall then be removed, and a visual inspection shall be performed to verify that there is no evidence of water leakage or deterioration. The backup battery shall be reinstalled and the assembly subjected to the tests of 4.1.2.2 (see 3.6.8).

4.3.6.9 Submergence.

4.3.6.9.1 Submergence (types I and Ia). Types I and Ia assemblies shall be subjected to a submergence test as specified in 3.6.9.1. At the conclusion of this test, subject the assembly to the tests of 4.1.2.2. Upon completion, the assembly backup battery shall be removed, and a visual inspection shall be performed to verify that there is no water leakage.

4.3.6.9.2 Submergence (type II). Type II assemblies shall be subjected to the submergence as specified in 3.6.9.2. After submergence, the assemblies shall be visually examined to verify that there is no evidence of water leakage and subjected to the tests of 4.1.2.2.

4.3.6.10 Salt fog. The assembly shall be subjected to a salt fog test exposure as specified in 3.6.10. At the conclusion of this exposure, examine the assembly for damage or deterioration and perform the tests of 4.1.2.2.

4.3.6.11 Chemicals. The assembly shall be exposed to chemicals as specified in 3.6.11. After exposure examine the assembly for damage or deterioration and perform the tests of 4.1.2.2.

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4.3.6.12 Nuclear environment.

4.3.6.12.1 EMP. The assembly shall be tested by the Government for compliance with the requirements of 3.6.12.1.1 and 3.6.12.1.2 using the test methods of MIL-STD-461.

4.3.6.12.2 Nuclear radiation. Testing, inspection, and analysis shall be conducted by the Government in accordance with the requirements of 3.6.12.2. Each electronic piece part shall be analyzed by the Government to determine the radiation survivability design margin. When generic part design margins are less than ten, statistical analysis which provides a survival probability of 99% in the circuit application, at 90% probability confidence of actual electronic device test data, shall be performed by the Government (see 6.5.7).

4.3.6.12.3 Assembly testing. Testing, inspection, and analysis of a complete assembly for transient response and damage shall be performed by the Government to ensure adequacy of the analysis and to ensure that no latent or synergistic failure modes exist.

4.3.6.12.4 Transient response and recovery. Testing, inspection, and analysis shall be conducted by the Government as specified in 4.3.6.12.1 and 4.3.6.12.2 to verify conformance to 3.6.12.3.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When actual packaging of material is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department of Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

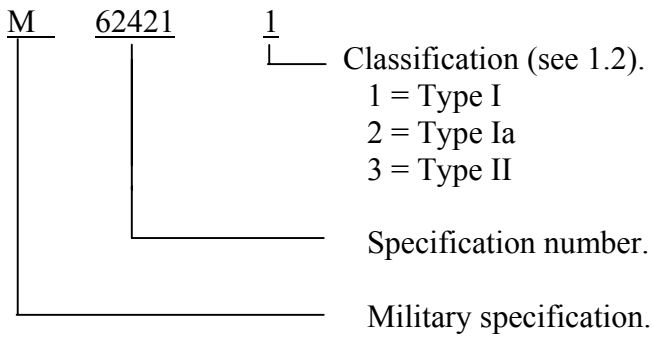
6.1 Intended use. The assemblies in this specification are intended to provide fuel handling and interface capabilities for the ACT 1500 gas turbine engines used in the M1, M1A1, M1A2 and HAB vehicles. The assembly converts engine, vehicle, and driver electrical input to control signals to the (HMU) and driver instrument panel, and receives electrical feedback signals from the HMU and driver (PLA).

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6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Issue of DoDISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1 and 2.3).
c. If first article sample is required (see 3.1 and 4.1.2).
d. If surface treatment and painting is other than as specified (see 3.2.4.1).
e. If first article inspection conditions are other than specified (see 4.1.1).
f. If qualification inspection is required (see 4.1.2).
g. If 100% acceptance tests are to be performed by other than the supplier (see 4.1.2.2).
h. If control testing is required (see 4.1.2.4).
i. If function testing is to be performed by other than the supplier (see 4.2.1).
j. Packaging requirements (see 5.1).

6.3 Part or Identifying Number (PIN). The PINs to be used for the assemblies acquired to this specification are created as shown below:



6.4 Subject term (key word) listing.

AGT 1500
Digital engine controls
Engine controller
Gas turbine engine

6.5 Definitions.

6.5.1 Software language. ADA is the name of the language, not an acronym (see 3.2.5.2).

6.5.2 Failure. A failure is defined as any end-item malfunction requiring corrective action or maintenance (see 3.5.1).

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6.5.3 NBC prioritization. Originally intended to signal a T7 over temperature condition, this function has been altered to accommodate the addition of the NBC system to the vehicle (see 3.3.1.8 3.4.1).

6.5.4 Deceleration schedule. Deceleration should be any negative NH rate (see 3.3.2.6).

6.5.5 PTS control. The PTS position specified in figure 13 should be called the PTS integrator position (see 3.3.3).

6.5.6 IGV prioritization. This should be referred to as IGV prioritization logic (see 3.3.4.1).

6.5.7 Design margin. The design margin should be the radiation level at which electronic piece part circuit parameter failure occurs compared to the specified radiation criteria (see 4.3.6.12.2).

6.6 Supplemental publications. MIL-HDBK-217 should be used to provide general guideline information for a common basis of reliability predictions for military electronic system and equipment (see 3.5.4). MIL-HDBK-454, Guideline 9, should be used, as it establishes the acceptable criteria for military electronic equipment (see 3.5.7).

TABLE I. Analog input parameters.

Input parameters	Device	Range	Device accuracy	Res./Bit	Assembly conv. accuracy
Power level angle PLA	RVDT	4 - 64° (±1°) pos (0.074 - 1.184 Vac)	See table 12	0.0466°pos	±2.75° pos
Inlet guide vane angle IGV	RVDT	4 - 64° (±1°) pos (0.074 - 1.184 Vac)	See table 5	0.0366°pos	±2.75° pos
Power turbine stator angle PTS	RVDT	4 - 64° (±1°) pos (0.074 - 1.184 Vac)	See table 5	0.0496°pos	±2.75° pos
Fuel flow measurement WF	LVDT	0.0 - 0.2 in pos (0.055 - 1.11) Vac	See figure 5	0.122 Mils	±0.0025 in. (±10.0 lb/hr)
T_7 measurement	Type K chromel/ alumel	700 - 1600°F (15.6 - 35.6 mV)	±11°F at 1112 ±16°F	0.468°F	±15°F
T_1 measurement	Platinum res. probe	-65 - 125°F	0.217 Ω/°F	0.163°F	±4.5°F
NH	Mag. pickup	11 - 2164 Hz (5 V peak)	Not applicable	(0.003 Hz)	±0.5%
NPT	Mag. pickup	11 - 2200 Hz (7 V peak)	Not applicable	(0.003 Hz)	±0.5%

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TABLE II. Discrete input parameters.

Input type	Type	Signal char.	Thresholds
Purge	Discrete	Open or 28 Vdc	9.25 ±1.25 Vdc
Start	Discrete	Open or 28 Vdc	17.5 ±1.5 Vdc
Stop	Discrete	Open or 28 Vdc	9.25 ±1.25 Vdc
Fault reset	Discrete	Open or 28 Vdc	17.5 ±1.5 Vdc
Tactical idle	Discrete	Open or 28 Vdc	4.4 ±1 Vdc
Pivot neutral	Discrete	Open or 28 Vdc	4.4 ±1 Vdc
External cutoff	Discrete	Open or 28 Vdc	17.5 ±1.5 Vdc

TABLE III. Detection criteria.

			Fault messages (see table IV)
D1	T_1 window	T1 < -100°F T1 > +200°F	5
D2	T_7 window	T7 <250°F at 30% NH or T7 >1800°F	19
D3	T_7 start	T7 ≥ 1360°F; 5% < NH < 50% T7 < 250°F at 30% NH	19
D4	NH level	NH < 1%	1, 2
D5	NPT level	NPT < 1%	3, 4
D7	WF error	Difference between WF request and WF actual < -100 lb/h or difference > +100 lb/h for more than 0.5 seconds	31
D8	WF level	WF actual <15 lb/h for 100 ms	16
D9	IGV error	See figure 25	24
D10	PTS level	PTS actual >70° RVDT	18
D11	Supply voltage during start	Less than 8 Vdc when backup battery >18 V less than 10.75 Vdc when backup battery < 18 V	
D12	Start timer	Time > 60 seconds for NH <50%	
D13	Fuel cutoff valve continuity level	Resistance <5 or > 500 Ω	8
D14	DECU	DECU hardware failure	26
	DECU4	EEPROM failure	27
D15	PTS error	Absolute difference between PTS request and PTS actual > 30°RVDT for > 5 seconds	25
D16	Battery voltage	Battery voltage < 18 V	28
D17	Backup solenoid	Current <250 ma	9
D18	Supply voltage while running	Supply voltage < 10.75 volts and backup battery voltage > 17 volts	29
D19	WF restrict	WFR, A > 780 pph, NH < (NHTRIM – 6%), NPT < 103%, $T_{7i} < T_{7\ -wf}$	32

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TABLE IV. Resistance temperature relationship.

Temperature	Resistance (Ω)
-100 ± 0.89	70.63 ± 0.20
-50 ± 0.67	81.84 ± 0.15
0 ± 0.54	92.95 ± 0.12
10 ± 0.54	95.15 ± 0.12
20 ± 0.54	97.36 ± 0.12
30 ± 0.54	99.56 ± 0.12
40 ± 0.46	101.76 ± 0.10
50 ± 0.46	103.95 ± 0.10
60 ± 0.54	106.14 ± 0.12
70 ± 0.54	108.33 ± 0.12
80 ± 0.54	110.51 ± 0.12
90 ± 0,69	112.69 ± 0.15
100 ± 0,69	114.87 ± 0.15
110 ± 0,69	117.04 ± 0.15
120 ± 0,69	119.21 ± 0.15
130 ± 0.83	121.38 ± 0.18

TABLE V. PTS and IGV RVDT characteristics.

Accuracy: as per table below (over the temperature range of -20°F to +150°F)					
Displ. degrees	Nominal VRMS	Minimum VRMS	Maximum VRMS	Neg. error m VRMS	Pos. error m VRMS
4	0.0769	0.717	0.0821	-5.2	+5.2
10	0.1923	0.1871	0.1975	-5.2	+5.2
20	0.3846	0.3706	0.3898	-14.0	+5.2
30	0.5769	0.5576	0.5821	-19.3	+5.2
40	0.7692	0.7447	0.7744	-24.5	+5.2
50	0.9615	0.9370	0.9667	-24.5	+5.2
60	1.1538	1.1439	1.1614	-9.9	+7.6
64	1.2307	1.2255	1.2359	-5.2	+5.2
70	1.3461	1.3362	1.3537	-9.9	+7.6

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TABLE VI. PLA RVDT characteristics.

Accuracy: as per table below (over the temperature range of -25°F to +140°F)		
Displ. (degrees)	Nominal V out (VRMS)	Error (MV RMS)
-5	0.0962	±4.0
-10	0.1923	5.2
-20	0.3846	7.6
-30	0.5769	9.9
-40	0.7692	12.2
-50	0.9615	14.6
-60	1.1538	16.9
-70	1.3461	19.3

TABLE VII. Magnetic speed pickup characteristics.

Coil volts resistance	Engine speed	Frequency	Min. output (neutral to peak)
NH	5%	108 Hz	0.3 V
100 ±15 Ω	100%	2164 Hz	5.0 V
NPT	5%	110 Hz	0.4 V
80 to 460 Ω	100%	2200 Hz	7.0 V
With 4.7K ohm load			

TABLE VIII. Connector pin assignments.

Diagnostic connector J1			
Pin no.	Signal label	Type	Signal characteristics
A	RXD 232	Input digital	Digital access port
B	NH_1	AC input speed pickup	60V peak differential and 10 V common mode maximum
C	NH_1 return	Signal analog	
D	Analog ground	Ref., 0 V	Do not use for power return greater than 0.010-A
E	-Vref	Ref., -10 V	Each has 10 K ohm series resistor and will not supply power
F	+Vref/2	Ref., +5 V	
G	20 Vp-p	Output analog	Power oscillator output to LVDT excitation and RVDT's (5.55 Vac com.) not protected
H	WF req. test	Internal analog	Not protected –0.4 V (40 lb/h) to normal mode –8 V (800 lb/h)
J	Backup fuel cutoff solenoid LO	Output discrete	Driven - not protected
K	Starter pilot	Output discrete	Driven side – not protected relay HI

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TABLE VIII. Connector pin assignments – Continued.

Diagnostic connector J1			
Pin no.	Signal label	Type	Signal characteristics
L	Main fuel cutoff solenoid	Output discrete	Driven side not protected solenoid HI
M	Ignition	Output discrete	Driven side – not protected exciter HI
N	WF solenoid HI	Output discrete	Driven side – not protected
P	TXD 232	Output digital	Digital access port
R	Interlock 1	Power source	28V supply - do not draw more than 0.5-A (from J3-B) from each (2-A total)
S	Interlock I	Power supply	28V supply - do not draw more than 0.5-A from each (2-A total)
T	Interlock II	Power source	28V supply - do not draw more than 0.5-A (from J3D) from each (2-A total)
U	Interlock II	Power source	28 V supply - do not draw more than 0.5-A from each (2-A total)
V	NPT meter	Output analog	2000 Ω output. 0 V (0%) to 5 V (120% NPT)
W	NH test	Internal analog	10K Ωoutput, 0 V (0%) to 10 V (100%), 1 V = 10%; 5 V = 50%
X	T_{7i}	Internal analog	Not protected, 10 V (0°F) to -10 V (20000°F), 0 V = 1000°F + sub V from 1000; add 100/volt
Y	WF actual	Internal analog	Not protected, 0.4 V (40 lb/h) to 8 V (800 lb/h)
Z	DNLD 232	Input digital	Digital access port Trans
a	Theta (T1)	Internal analog	Not protected, 0 to 8 V; 6 V = 125°F
b	PLA DC	Internal analog	Not protected, 0.40 V (4° RVDT) to 6.4 V (64° RVDT) Idle 0.50 - 0.60. Full 6.50 - 6.70 Driven side - not protected
c	PTS solenoid HI	Output analog	Driven side – not protected
d	-28 Vdc ret.	Power return	Power return only. Do not exceed 2 amps
e	IGV solenoid HI	Output analog	Driven side – not protected
f	PTS request	Internal analog	Not protected, 0.3V (65° RVDT open) to -5.7V (5° RVDT closed)
g	PTS actual	Internal analog	Not protected, -6.5 V (65° RVDT open) to -0.5 V (5°RVDT closed)
h	IGV request	Internal analog	Not protected, 0 V (4° RVDT open) to 10 V (64° RVDT closed)
i	IGV actual	Internal analog	Not protected, -0.67 V (4° RVDT open) to -10.67 V (64° RVDT closed)

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TABLE VIII. Connector pin assignments – Continued.

Engine connector - J2			
Pin no.	Signal label	Type	Signal characteristics
A	NH_2 HI	Input analog	0% = 0 Hz, 100% = 2164 Hz Mag. Pickup 5 V peak
B	NH_2 LO	Input analog	0% = 0 Hz, 100% = 2164 Hz Mag. Pickup 5 V peak
C	Spare		
D	NH_1 HI	Input analog	0% = 0 Hz, 100% = 2164 Hz Mag. Pickup 5 V peak
E	NH_1 LO	Input analog	0% = 0 Hz, 100% = 2164 Hz Mag. Pickup 5 V peak
F	Shield speed sensors	None	
G	NPT2 HI	Input analog	0% = 0 Hz, 100% = 2200 Hz Mag. Pickup 7 V peak
H	NPT2 LO	Input analog	0% = 0 Hz, 100% = 2200 Hz Mag. Pickup 7 V peak
J	Spare		
K	NPT1 HI	Input analog	0% = 0 Hz, 100% = 2200 Hz Mag. Pickup 7 V peak
L	NPT1 LO	Input analog	0% = 0 Hz, 100% = 2200 Hz Mag. Pickup 7 V peak
M	Spare		
N	Spare		
P	Spare		
R	Spare		
S	Spare		
T	Spare		
U	WF primary HI	Output analog	20 Vp-p, 5.55 Vac average
V	WF primary LO	Output analog	20 Vp-p, 5.55 Vac average
W	Spare		
X	Spare		
Y	WF secondary HI	Input analog	40 lb/h = 0.055 Vac average., 800 lb/h = 1.11 Vac average
Z	WF secondary LO	Input analog	40 lb/h = 0.055 Vac average., 800 lb/h = 1.11 Vac average
a	Spare		
b	Spare		
c	Shield primary	Ground	Power ground only
d	PTS primary HI	Output analog	20 Vp-p 5.55 Vac average
e	PTS primary LO	Output analog	20 Vp-p 5.55 Vac average

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TABLE VIII. Connector pin assignments – Continued.

Engine connector - J2			
Pin no.	Signal label	Type	Signal characteristics
f	PTS secondary HI	Input analog	4° = 0.074 Vac average, 64° = 1.184 Vac average
g	PTS secondary LO	Input analog	4° = 0.074 Vac average, 64° = 1.184 Vac average
h	IGV primary HI	Output analog	20 Vp-p, 5.55 Vac average
i	IGV primary LO	Output analog	20 Vp-p, 5.55 Vac average
j	IGV secondary HI	Input analog	4° = 0.074 Vac average, 64° = 1.184 Vac average
k	IGV secondary LO	Input analog	4° = 0.074 Vac average, 64° = 1.184 Vac average
m	Shield secondary	Ground	Power ground only
n	T_1 HI	Input analog	32°F = 100 Ω, 212°F = 139Ω
p	T_1 LO	Input analog	32°F = 100Ω 212°F = 139Ω
q	Spare		
r	T_1 shield	Ground	
s	Monitor +5 V	Output discrete	+5 Vdc (10K Ω output)
t	Monitor +15 V	Output discrete	+15 Vdc (10K Ω output)
u	Monitor –15 V	Output discrete	-15 Vdc (10K Ω output)
v	Shield solenoids	Ground	Power ground only
w	Main fuel cutoff solenoid HI	Output discrete	Open or 28 Vdc
x	Main fuel cutoff solenoid LO	Output discrete	Open or 28 Vdc
y	Backup fuel cutoff solenoid HI	Output discrete	Open or 28 Vdc
z	Backup fuel cutoff solenoid LO	Output discrete	Open or 28 Vdc
AA	PTS solenoid HI	Output analog	0 to 900 mA
BB	PTS solenoid LO	Output analog	0 to 900 mA
CC	IGV solenoid HI	Output analog	0 to 900 mA
DD	IGV solenoid LO	Output analog	0 to 900 mA
EE	WF solenoid HI	Output analog	0 to 900 mA
FF	WF solenoid LO	Output analog	0 to 900 mA
GG	Continuity 1		Internally connected to J3-F
HH	Continuity 2		Internally connected to J3-S

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TABLE VIII. Connector pin assignments – Continued.

Vehicle connector - J3			
Pin no.	Signal label	Type	Signal characteristics
A	28 Vdc return	Input discrete	Power supply return
B	28 Vdc	Input discrete	Power supply
C	28 Vdc return	Input discrete	Power supply return
D	28 Vdc	Input discrete	Power supply
E	Starter pilot	Output discrete	Open or 28 Vdc (on type II, sent relay signal across 1553 on signal label 25-01-02 Bit 1)
F	Continuity 1		Internally connected to J2-GG
G	Main fuel solenoid cutoff	Output discrete	Open or 28 Vdc (on type II, also sent solenoid HI across 1553 on signal abel 25-01-02 Bit 2)
H	Run time meter	Output discrete	Open or 28 Vdc (on type II, also sent across 1553 on signal label 25-01-04)
J	NPT meter	Output analog	0% = 0 Vdc, 120% = 5 Vdc (On type II, also sent across 1553 on signal label 25-01-05)
K	NPT meter	Output analog	0% = 0 Vdc, 120% = 5 Vdc (On type II, Rtn. Same as J)
L	NPO lamp	Output discrete	Open or 28 Vdc return (on type II, also sent across 1553 on signal label 25-01-02, Bit 0)
M	Fault alarm	Output discrete	Open or 28 Vdc return (on type II, also sent across 1553 on signal label 25-01-02, Bit 3)
N	T_7 flag	Output discrete	Open or 28 Vdc (on type II, also sent across 1553 on signal label 25-01-02, Bit 4)
P	Spare		
R	Shield, ignition exciter	Ground	Power ground only
S	Continuity 2		Internally connected to J2-HH
T	Shield, power	Ground	Power ground only
U	Purge switch	Input discrete	Open or 28 Vdc
V	Start switch	Input discrete	Open or 28 Vdc
W	Stop switch	Input discrete	Open or 28 Vdc
X	Spare		
Y	Fault reset switch	Input discrete	Open or 28 Vdc

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TABLE VIII. Connector pin assignments – Continued.

Vehicle connector - J3			
Pin no.	Signal label	Type	Signal characteristics
Z	Tactical idle switch	Input discrete	Open or 28 Vdc (On type II, also sent across 1553 24-01-03, Bit 4)
a	Pivot neutral switch	Input discrete	Open or 28 Vdc (On type II, also sent across 1553 24-01-03, Bit 0)
b	External cutoff switch (low oil pressure)	Input discrete	Open or 28 Vdc (On type II, also sent across 1553 on signal label 24-01-03, Bit 3)
c	Shield, PLA	Ground	Power ground only
d	Shield return		
e	Spare		
f	Ignition exciter	Output discrete	Open or 28 Vdc (On type II, indication sent across 1553 on signal label 25-01-02, Bit 6)
g	Ignition exciter return	Output discrete	Power ground only
h	Spare		
i	NED test	Input discrete	Open or 28 Vdc
j	PLA primary HI	Output analog	20 Vp-p, 5.55 Vac average
k	PLA primary LO	Output analog	20 Vp-p, 5.55 Vac average
m	PLA secondary HI	Input analog	4° = 0.074 Vac average, 64° = 1.184 Vac average
n	PLA secondary LO	Input analog	4° = 0.074 Vac average, 64° = 1.184 Vac average
p	Spare		
q	Starter pilot	Output discrete	Open or 28 Vdc
r	Starter pilot return	Output discrete	Open or 28 Vdc
s	Mon. 28V	Output discrete	
t	Spare		
A	T_7 alumel	Input analog	500°F = 10.560 mV chromel. 1000°F = 22.251 mV alumel. 1500°F = 33.913 mV thermocouple (chromel/alumel)
B	T_7 chromel	Input analog	500°F = 10.560 mV chromel. 1000°F = 22.251 mV alumel. 1500°F = 33.913 mV thermocouple (chromel/alumel)

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TABLE VIII. Connector pin assignments – Continued.

Engine thermocouple connector - J4			
C	Shield, T_7	None	
D	External shutoff	Internal digital	Digital
E	External shutoff return	Ground	Power ground
F	Spare		
1553 connector - J5 and J6 - type II assemblies			
A	1553 low (-)	Bus signal	Transformer isolated signals
B	1553 high (=)	Bus signal	
C	Spare		

TABLE IX. Analog output parameters.

Output parameters	Device	Min. To max. range	Steady-state-device accuracy	Steady-state assembly accuracy
WF driver	Solenoid	0 – 900 mA	275 ±5 mA	275 ±15 mA
PTS driver	Solenoid	0 – 900 mA	275 ±50 mA	350 ±34 mA
IGV driver	Solenoid	0 – 900 mA	350 ±50 mA	350 ±34 mA
NPT	Meter	1 – 5 Vdc	Not applicable	Not applicable

TABLE X. Discrete output parameters.

Output parameters	Signal char.	Maximum continuous current
Starter relay	Open or 28 Vdc	1.4-A
Main fuel cutoff solenoid	Open or 28 Vdc	1.4-A
Ignition exciter	Open or 28 Vdc	2.8-A
Backup fuel cutoff solenoid	Open or 28 Vdc	0.7-A
Engine runtime meter	Open or 28 Vdc	0.7-A
WF primary excitation	20 Vp-p 5.55 Vac	Not applicable
PLA primary excitation	20 Vp-p 5.55 Vac	Not applicable
PTS primary excitation	20 Vp-p 5.55 Vac	Not applicable
IGV primary Excitation	20 Vp-p 5.55 Vac	Not applicable
NPO lamp	Open or 28 Vdc	Not applicable
Fault alarm lamp	Open or 28 Vdc	Not applicable
NBC prioritization signal	Open or 28 Vdc	Not applicable

TABLE XI. Fault messages (type I).

	Hard faults	Diagnostic
1. NH1		X
2. NH2		X
3. NPT1		X
4. NPT2		X

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TABLE XI. Fault messages (type I) - Continued.

	Hard faults	Diagnostic
5 T1		X
6. IGV SOL	X	
7. PTS SOL	X	
8. CO SOL		X
9. BU SOL		X
10. PTS PRI	X	
11. PTS SEC	X	
12. IGV PRI	X	
13. IGV SEC	X	
14. WF PRI	X	
15. WF SEC	X	
16. WF LVDT INT		X
17. IGV RVDT INT		X
18. TS RVDT INT		X
19. T_7		X
20. STR PT RLY	X	
21. IGN EXC	X	
22. PLA PRI	X	
23. PLA RVDT INT		X
24. IGV SOL INT		X
25. PTS SOL INT		X
26. DECU		X
27. DECU4		X
28. DECU VOLTS		
29. SYS VOLTS		X
30. PLA SEC	X	
31. WF SOL INT		X
32. WF RESTRICT		X
33. O/P SHUTDOWN		X
34. ROLLBACK		X
35. FLAMEOUT		X
36. ADJ OIL PUMP		X

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TABLE XII. Auto relight ignition exciter activation schedule.

NH (%)	NH dot	NH double dot	T_7 dot
50	-0.1797	-0.1777	-1.9
60	-0.1797	-0.1777	-1.9
65	-0.1797	-0.1796	-3.2
70	-0.1797	-0.1596	-3.6
75	-0.25	-0.1596	-3.6
80	-0.27	-0.10	-4.5
85	-0.34	-0.12	-6.0
90	-0.45	-0.14	-6.5
95	-0.60	-0.14	-7.0
100	-0.60	-0.14	-7.5

TABLE XIII. Connectors/receptacles.

Name	Receptacle on unit	Mating connector
J1	621908	Cap (MS3476L18-32PX and M83/23-14R1832X) MS3116F18-32PX(SR)
J2	2993178-001	MS3475W22-55PN
J3	2993178-002	MS3476W20-41S
J4	2993178-003	MS3475W10-6SN
J5*	2993179-001	D38999/26WA98PA
J6*	2993179-002	D38999/26WA98PB

*Type II only

TABLE XIV. Classification of inspections.

Title	Reqmt.	Insp.	Qual	First article	Accept (100%)	Ctrl test
Configuration	3.2.3		X	X		X
Power interrupt	3.3.1.5	4.7.1.3.1	X	X	X	
Power consumption	3.3.1.5.1	4.7.1.2	X	X		X
232 Data bus communication	3.3.1.6.1	4.7.1.3.2	X			
1553 Data bus communication	3.3.1.6 2	4.7.1.4	X			
Signal inputs	3.3.1.7	4.3.3.1.7	X	X	X	
Frequency inputs	3.3.1.7.4	4.3.3.1.7.4	X	X	X	
Data bus inputs	3.3.1.7.5	4.3.3.1.7.5	X	X	X	
Discrete inputs	3.3.1.7.6	4.3.3.1.7.6	X	X	X	
Input processing	3.3.1.7.6.1	4.3.3.1.7.6.1	X	X	X	

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TABLE XIV. Classification of inspections – Continued..

Title	Reqmt.	Insp.	Qual	First article	Accept (100%)	Ctrl test
Output signal	3.3.1.8	4.3.3.1.8	X	X	X	
LVDT/RVDT excitation	3.3.1.9	4.3.3.1.9	X	X	X	
Data bus output	3.3.1.10	4.3.3.1.10	X	X	X	
Fuel schedule	3.3.2.9	4.3.3.2.9	X			
PTS	3.3.3	4.3.3.3	X			
IGV	3.3.4	4.3.3.4	X			
Protective modes	3.3.5	4.3.3.5	X			
Algorithms	3.3.6	4.3.3.6	X			
Weight	3.4.1.1	4.3.4.1.1	X	X		X
Performance	3.3.1	4.3.3.1				
Bonds & grounds	3.4.6	4.3.4.6	X	X		X
Chassis isolation	3.4.6.1.1	4.3.4.6.1.1			X	
Chassis isolation at 130°F	3.4.6.1.2	4.3.4.6.1.2	X	X		X
EMI/EMR	3.4.9	4.3.4.9	X	X		
Reliability	3.5.1	4.3.5.1	X			
Maintainability.	3.5.2	4.3.5.2	X			
Interchangeability	3.5.3	4.3.5.3	X			
Diagnostics	3.5.5	4.3.5.5	X			
A/D testing	3.5.5.1.1	4.3.5.5.1.1	X			
ID marking	3.5.6	4.3.5.6	X	X		X
Workmanship	3.5.7	4.3.5.7	X	X		X
Operating temp	3.6.1.1					
High temp		4.3.6.1.1.1	X	X		X
Low temp		4.3.6.1.1.2	X	X		X
Storage temp	3.6.1.2	4.3.6.1.2	X	X		X
Humidity	3.6.2	4.3.6.2	X	X		
Fungus	3.6.3	4.3.6.3	X	X		
Vibration	3.6.4	4.3.6.4	X	X		
Vibration control		4.3.6.4.1				X
Basic shock	3.6.5	4.3.6.5	X	X		
Basic shock control		4.3.6.5.1				X
Gun fire	3.6.6	4.3.6.6	X	X		
Gun fire control		4.3.6.6.1				X
Ballistic	3.6.7	4.3.6.7	X	X		

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TABLE XIV. Classification of inspections – Continued.

Title	Reqmt.	Insp.	Qual	First article	Accept (100%)	Ctrl test
Cleaning spray	3.6.8	4.3.6.8	X	X		X
Submergence	3.6.9	4.3.6.9	X	X	X*	X**
Salt fog	3.6.10	4.3.6.10	X	X		
Chemicals	3.6.11	4.3.6.11	X	X		
EMP	3.6.12.1	4.6.12.2.1	X			
Nuclear radiation	3.6.12.2	4.6.12.2.2	X			
Transient response	3.6.12.3	4.3.6.12.3	X			
Assembly testing		4.3.6.12.4	X			
Packaging		5.1	X	X		X

*Type II only
**Type I only

TABLE XV. Classification of severity defects.

Severity	Defect	Inspection method
Catastrophic		
101	NH overspeed (see 3.3.2.3.3)	Visual & SIE 1/
102	NPT overspeed (3.3.2.2.1.3)	Visual & SIE
103	T_7 Over temperature (run mode)	Visual
104	T_7 Over temperature (start mode)	Visual
105	Loss of NED protection (see 3.3.1.3.2)	SIE
Critical		
201	Severe power loss	Visual
202	Flameout (see 3.3.6.8)	Visual
	Reduced performance:	
203	Major power loss	Visual
204	Start malfunction (see 3.3.1.7.6.1.2)	Visual
205	Stop malfunction (see 3.3.1.7.6.1.3)	Visual
206	Start actuation during start	Visual
207	Loss of data bus (see 3.3.1.7.5)	Visual
Major		
301	Assembly incomplete (see 3.2.4)	Visual
302	Interfaces adversely affecting performance (see 3.4)	SIE
303	Faulty workmanship affecting performance (see 3.5.7)	Visual
Minor		
	Marginal:	
401	Minor power loss	Visual
402	Loss of efficiency	Visual & SIE
403	Loss of peak power on dynamic response	Visual
404	All hardware faults (see 4.2.1)	Visual

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TABLE XV. Classification of severity defects-Continued.

Severity	Defect	Inspection method
Minor		
405	Software faults that effect performance (see 4.2.2)	Visual & SIE
406	Non-interface dimensions out of tolerance (see 3.5.3)	SIE
407	Improper marking (see 3.5.6)	Visual
408	Faulty workmanship not affecting performance (see 3.5.7)	Visual
	Negligible:	
409	Software faults that do not effect performance	Visual & SIE
410	False NED detection	Visual
411	All faults requiring maintenance that do not effect performance	Visual & SIE

SIE 1/ = Standard Inspection Equipment.

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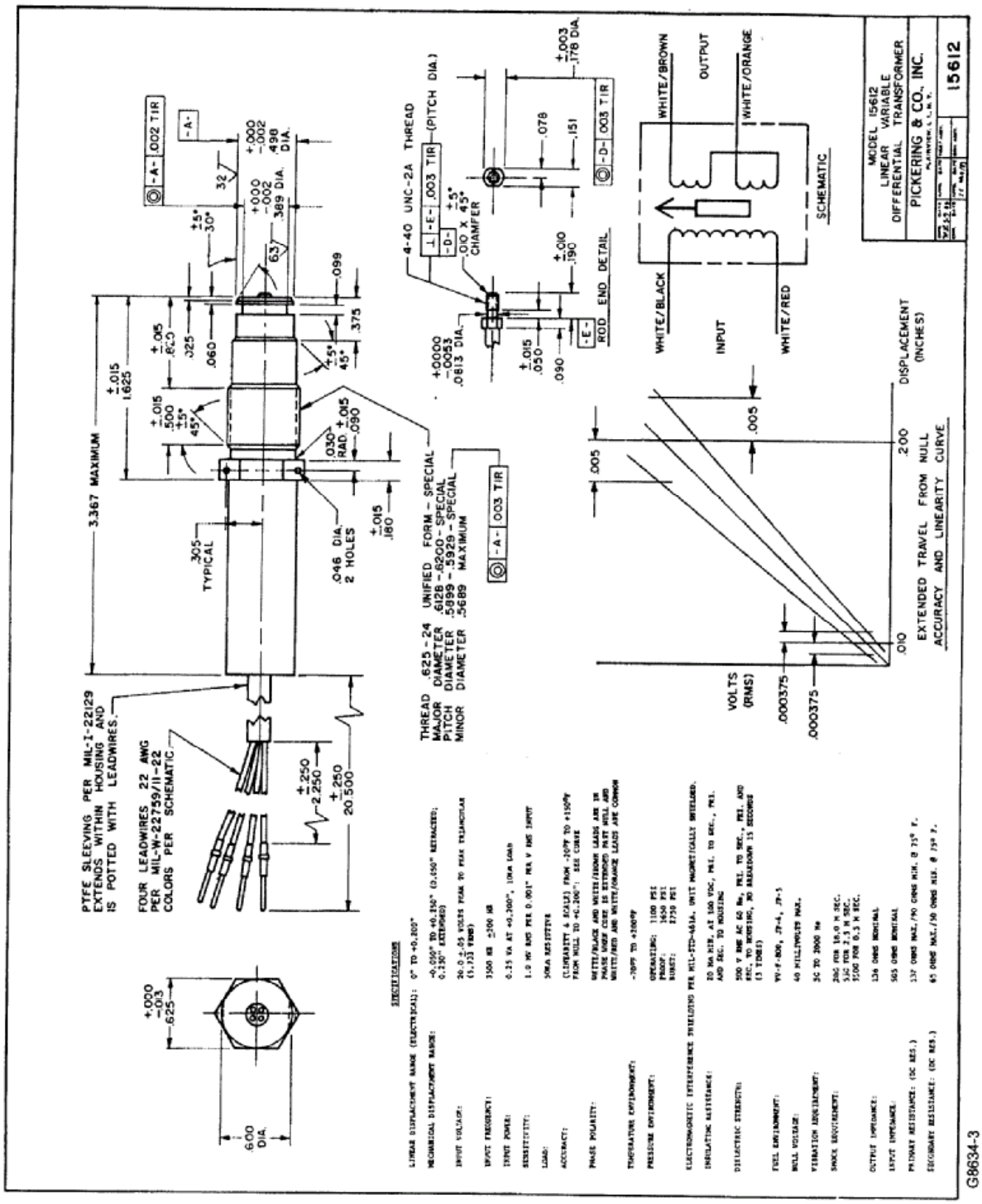


FIGURE 1. LVDT characteristics

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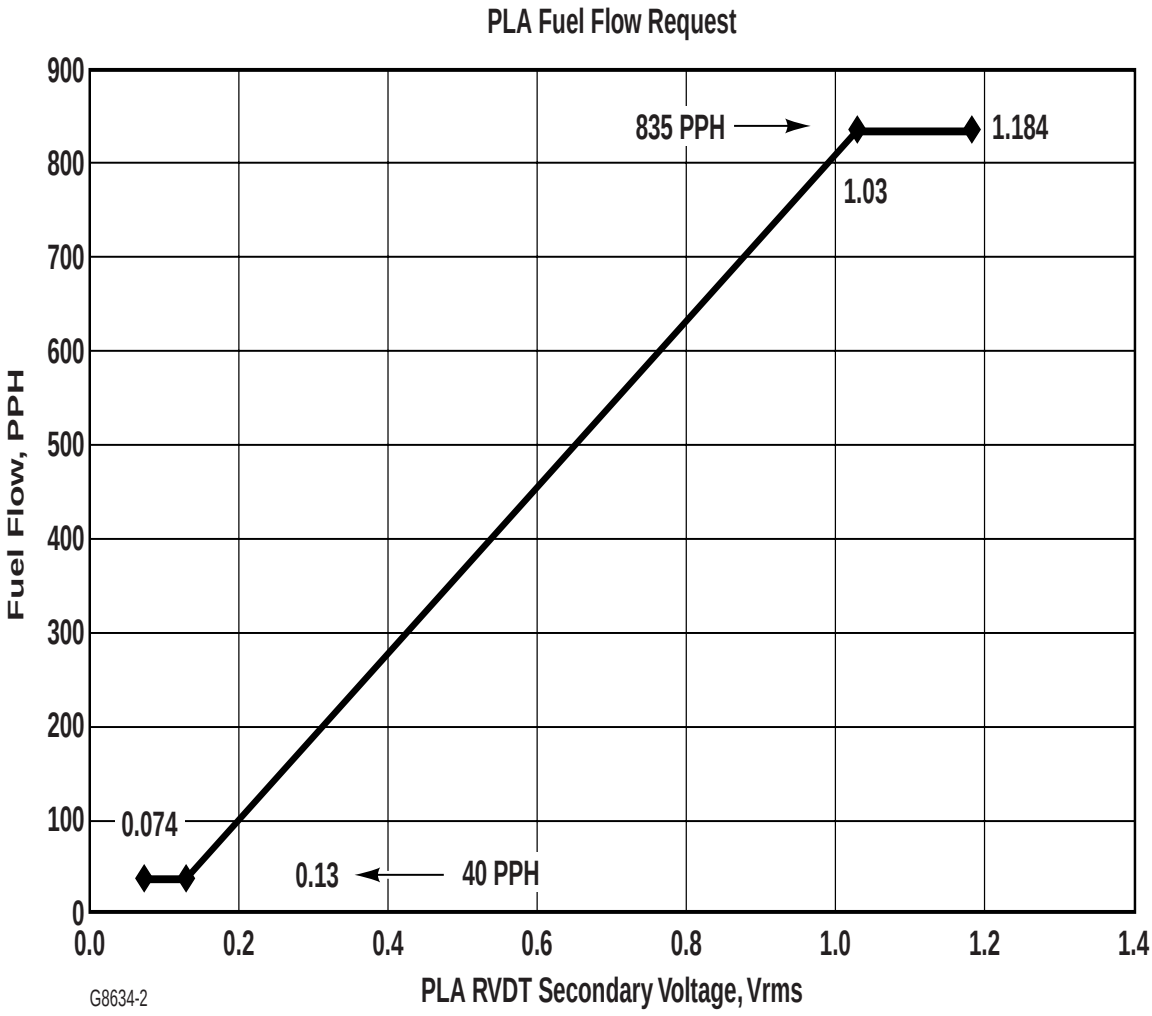


FIGURE 2. PLA RVDT output.

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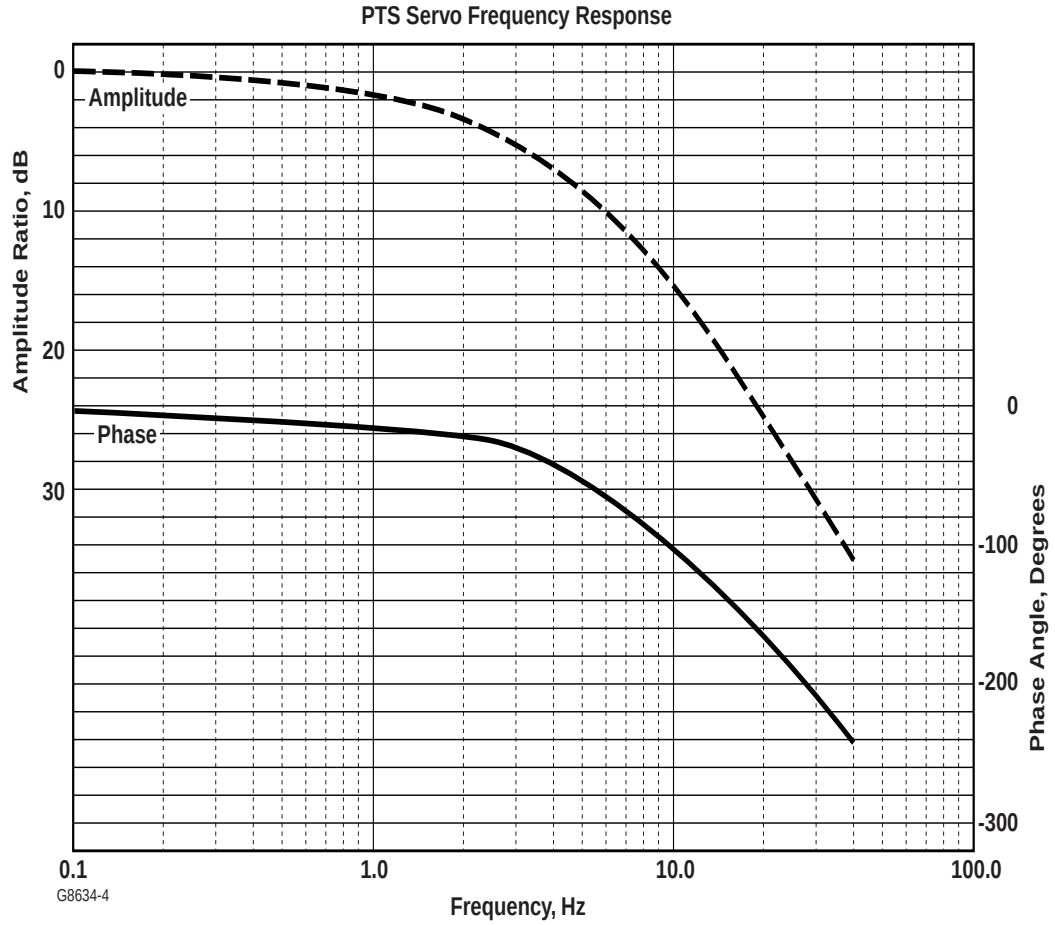


FIGURE 3. PTS solenoid frequency response.

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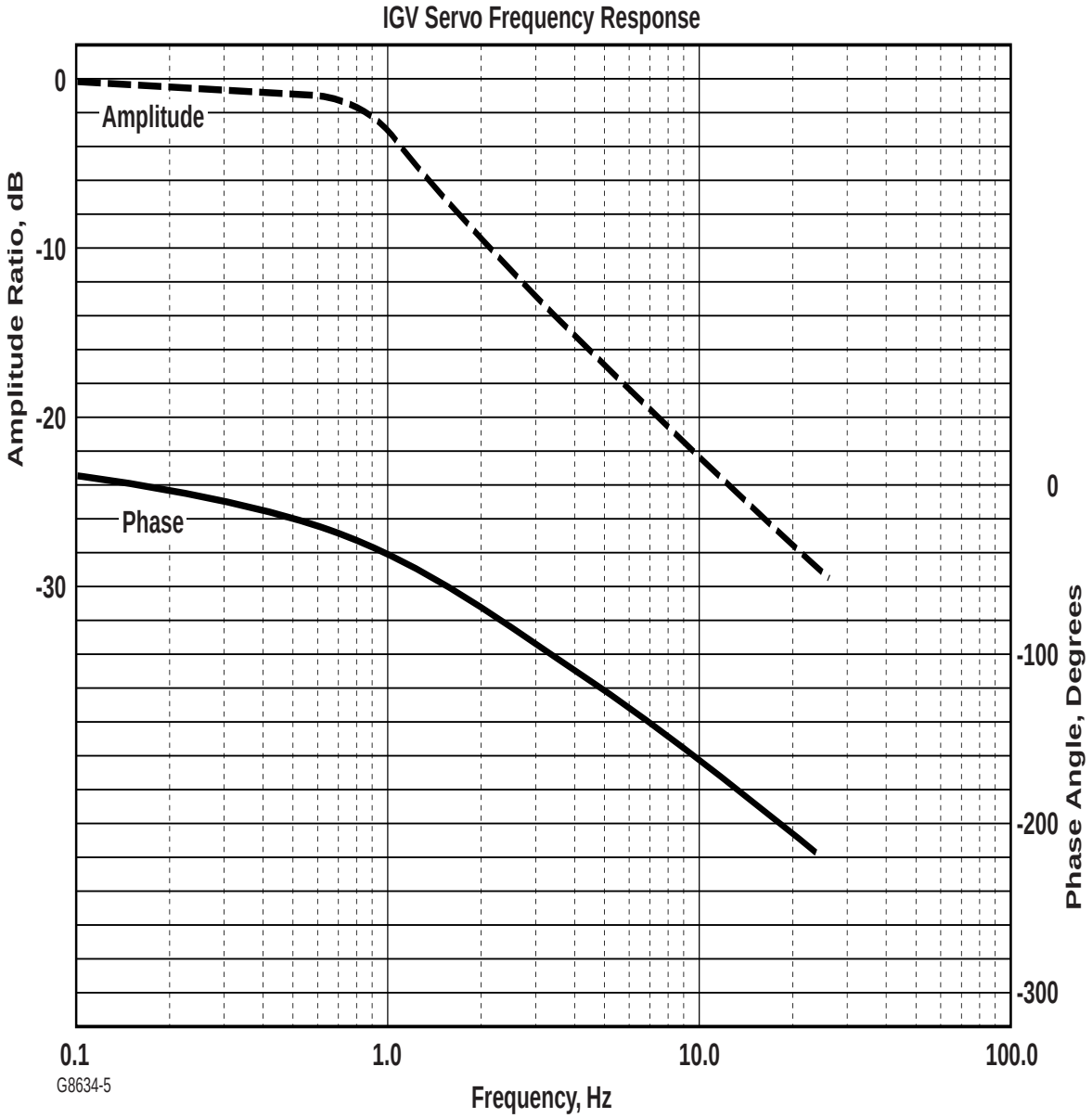


FIGURE 4. IGV solenoid frequency response.

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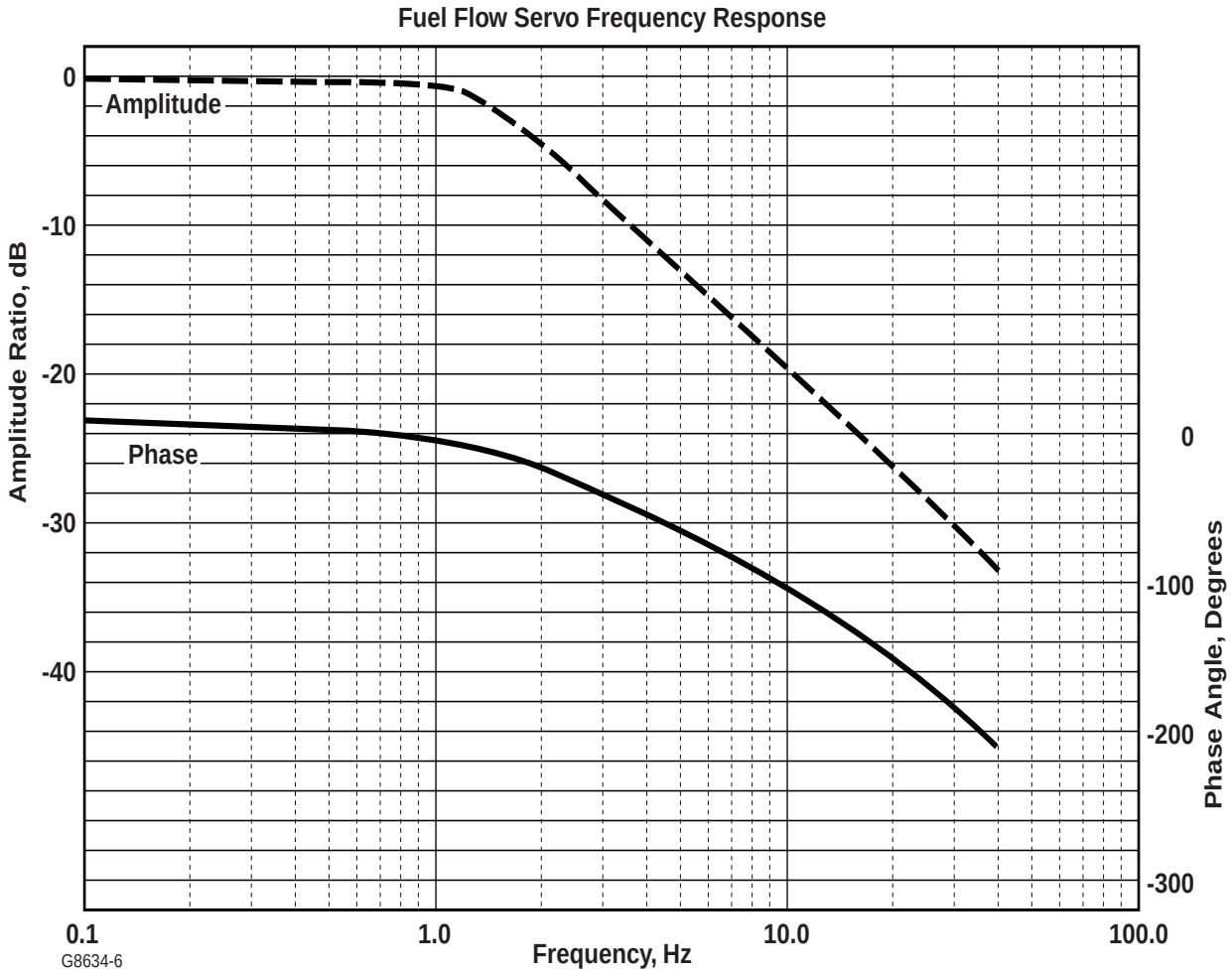


FIGURE 5. WF solenoid frequency response.

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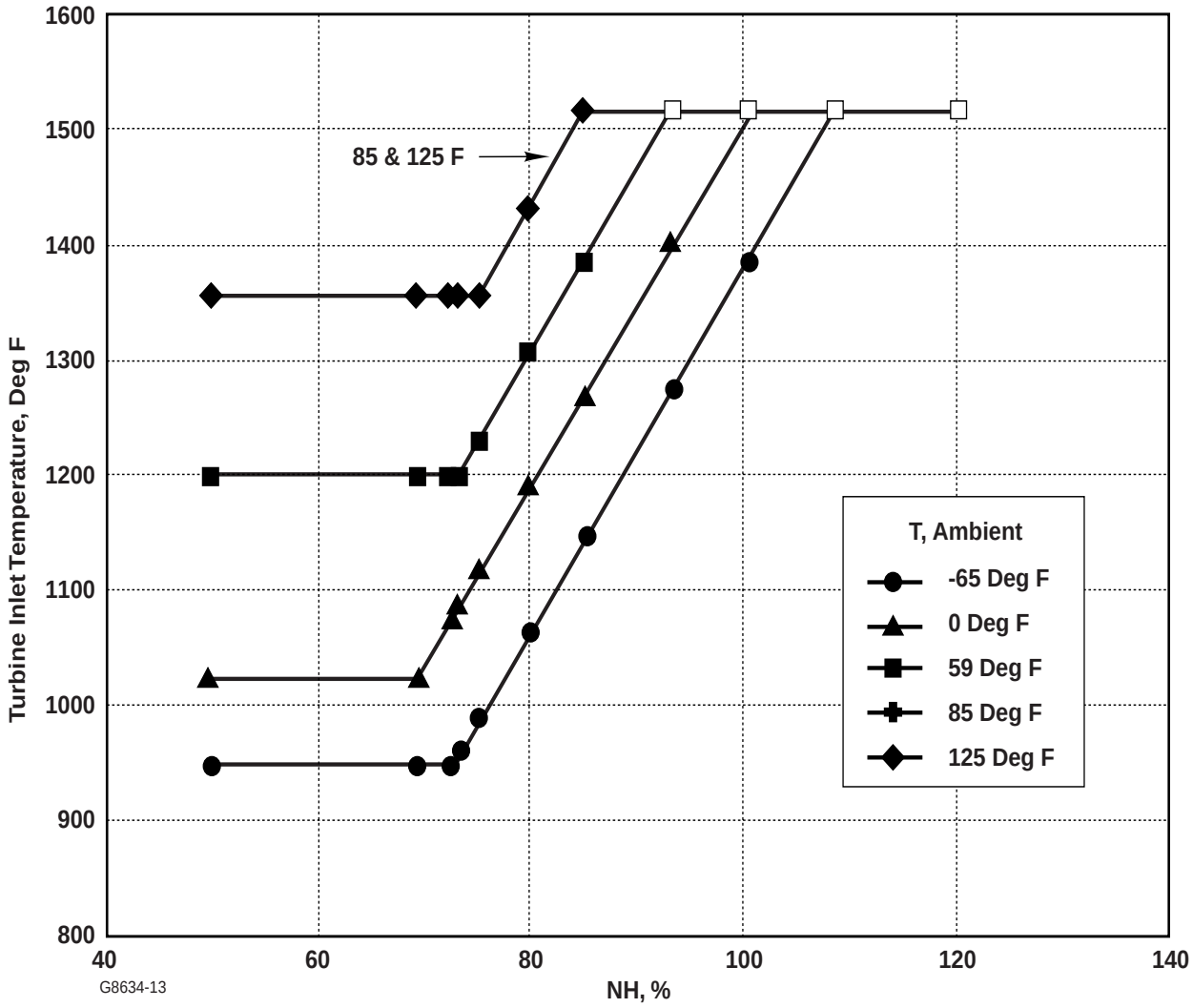


FIGURE 6. Maximum power inlet temperature schedule.

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Check the source to verify that this is the current version before use.

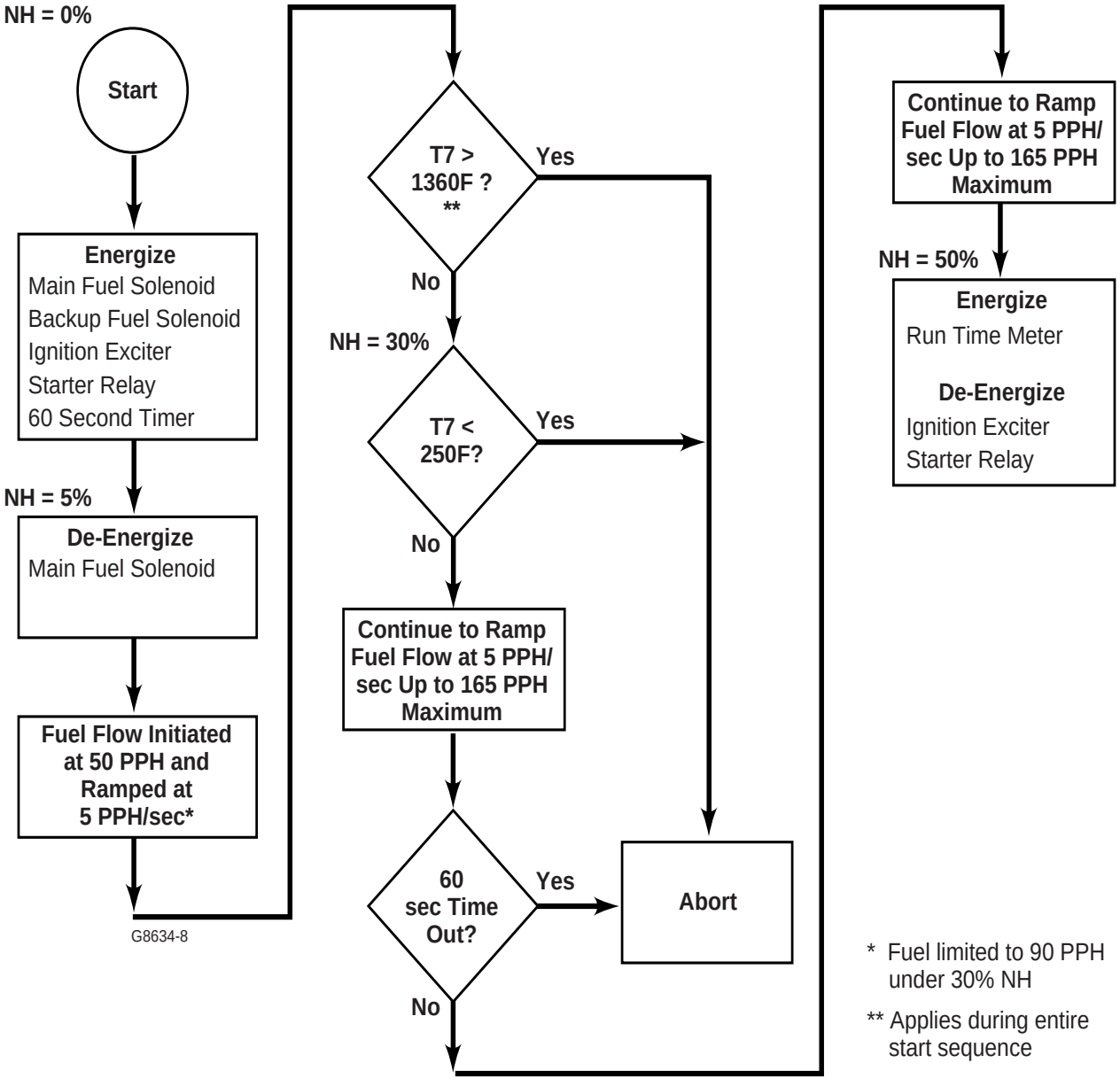


FIGURE 7. Start sequence.

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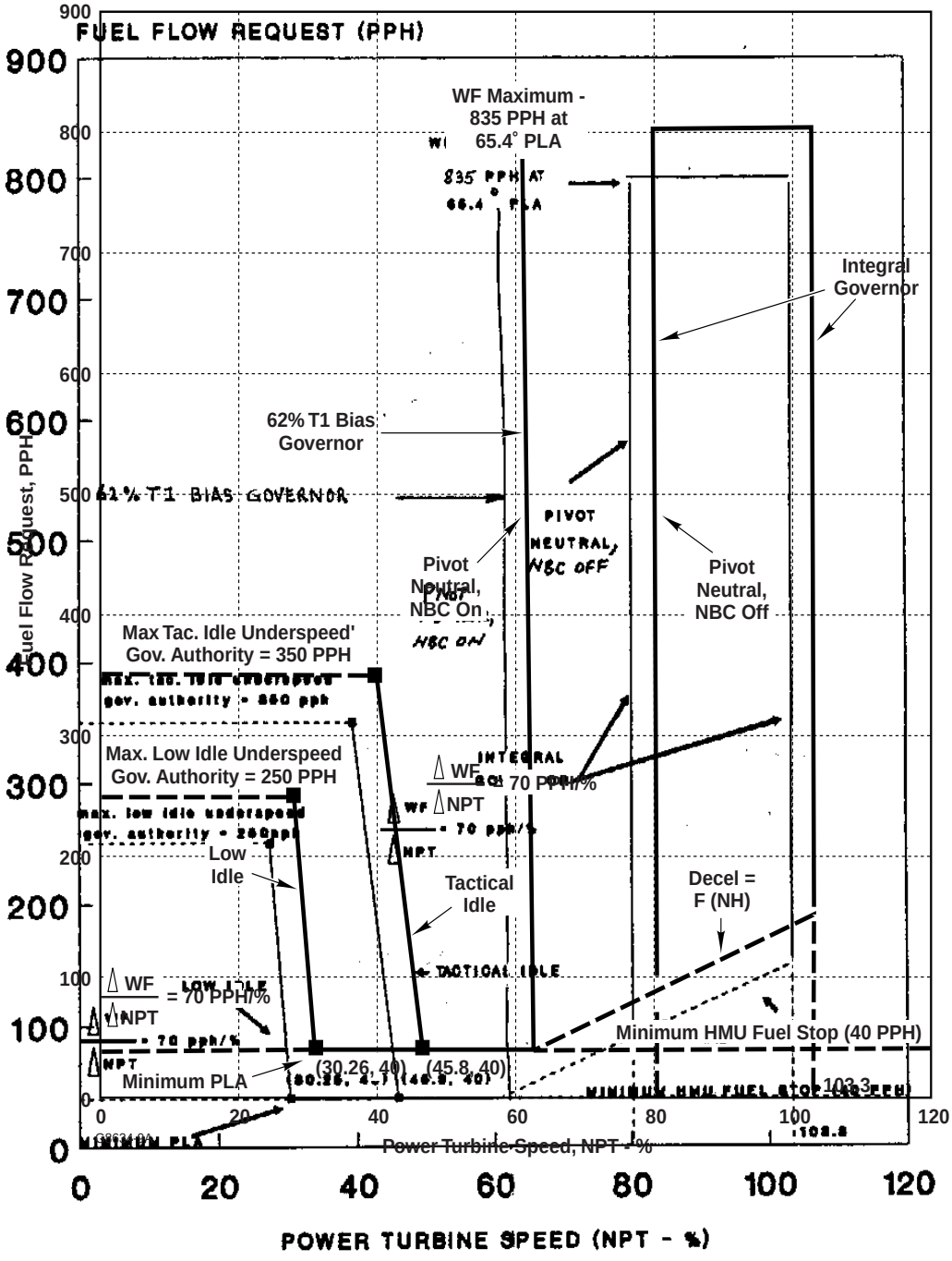


FIGURE 8. Power turbine governor schedule.

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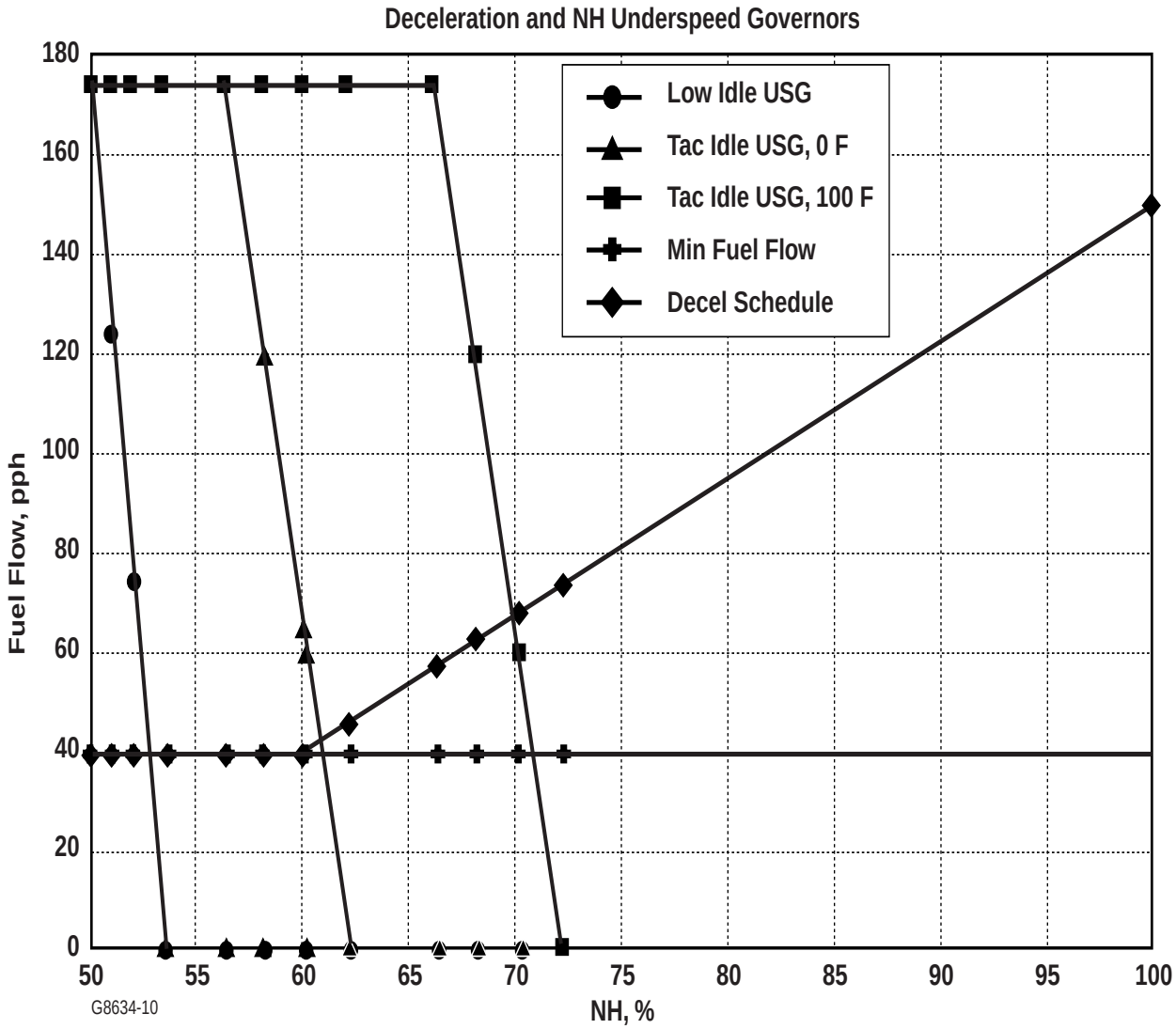


FIGURE 9. Deceleration and NH bottoming governor.

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Check the source to verify that this is the current version before use.

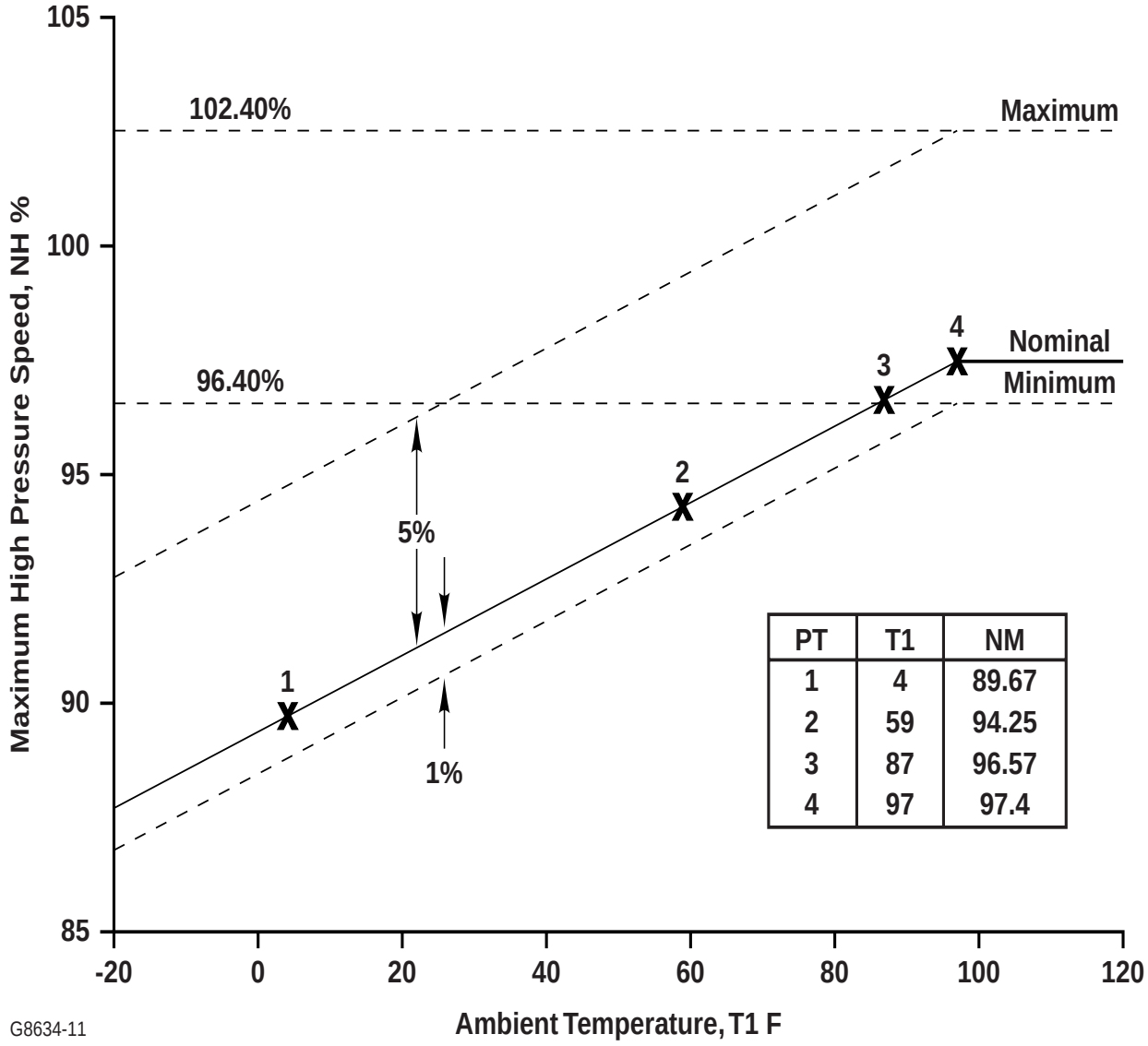


FIGURE 10. Maximum NH limiter.

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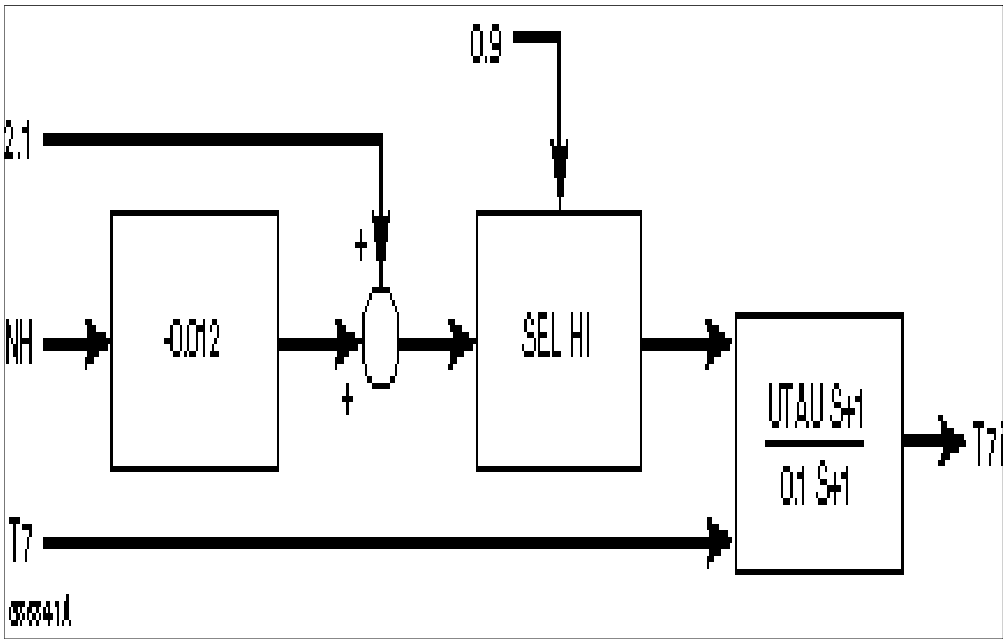
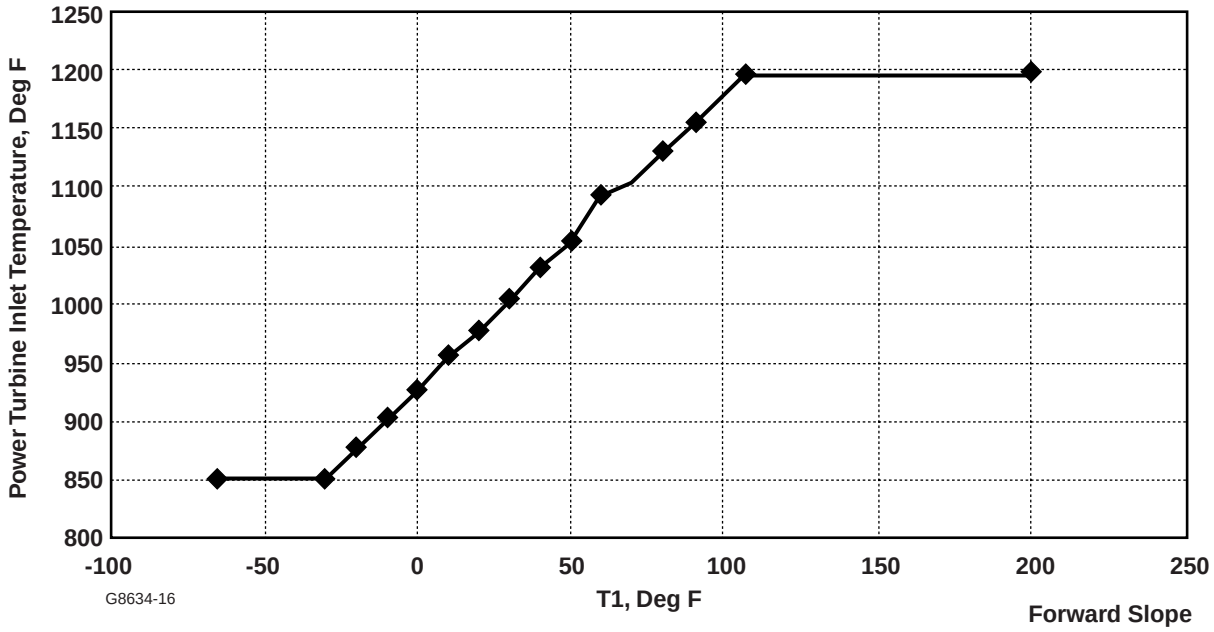


FIGURE 11. T_7 signal conditioning.

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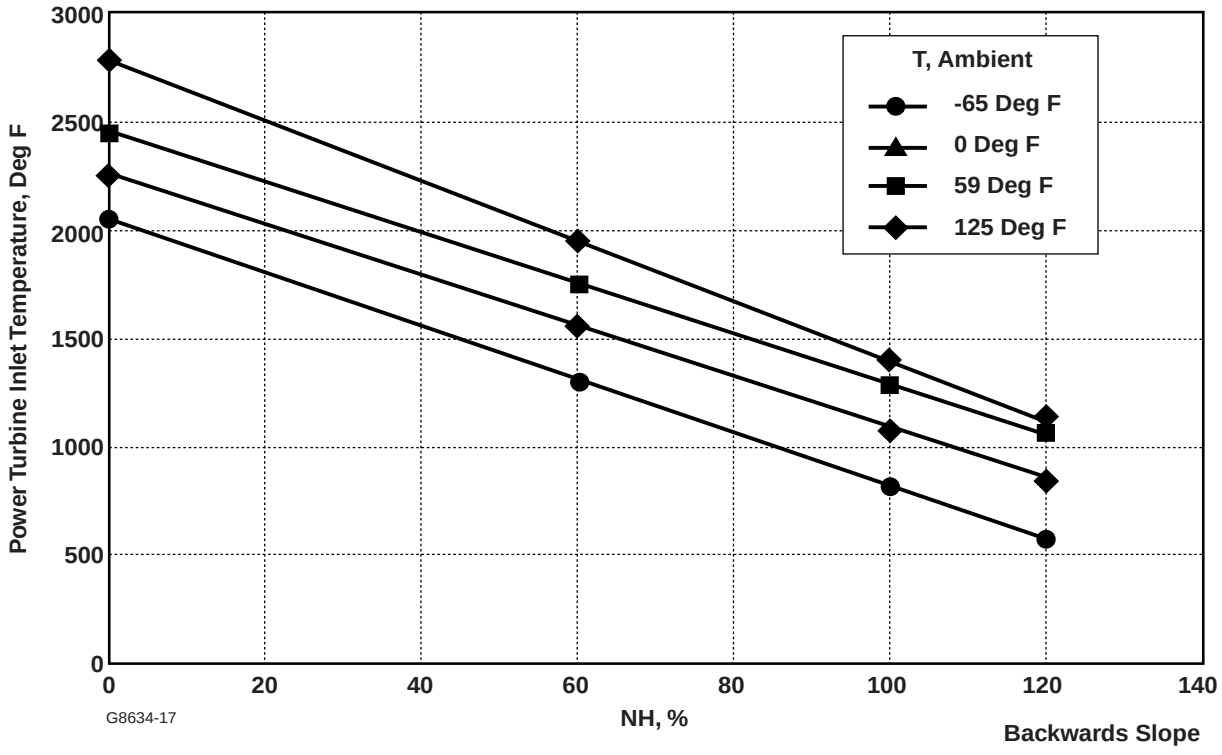


FIGURE 12. PTS schedule.

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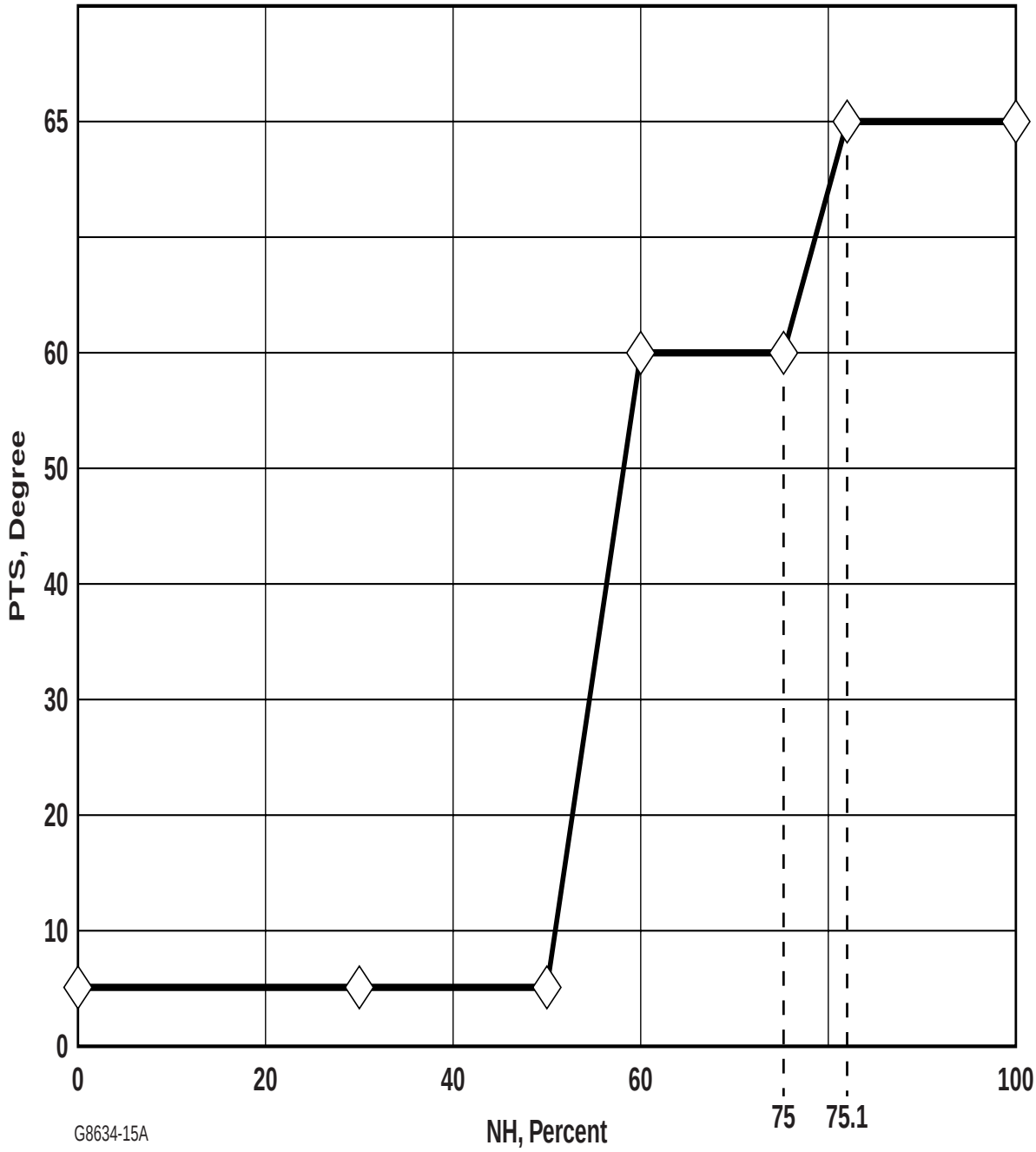


FIGURE 13. PTS – NH schedule.

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Check the source to verify that this is the current version before use.

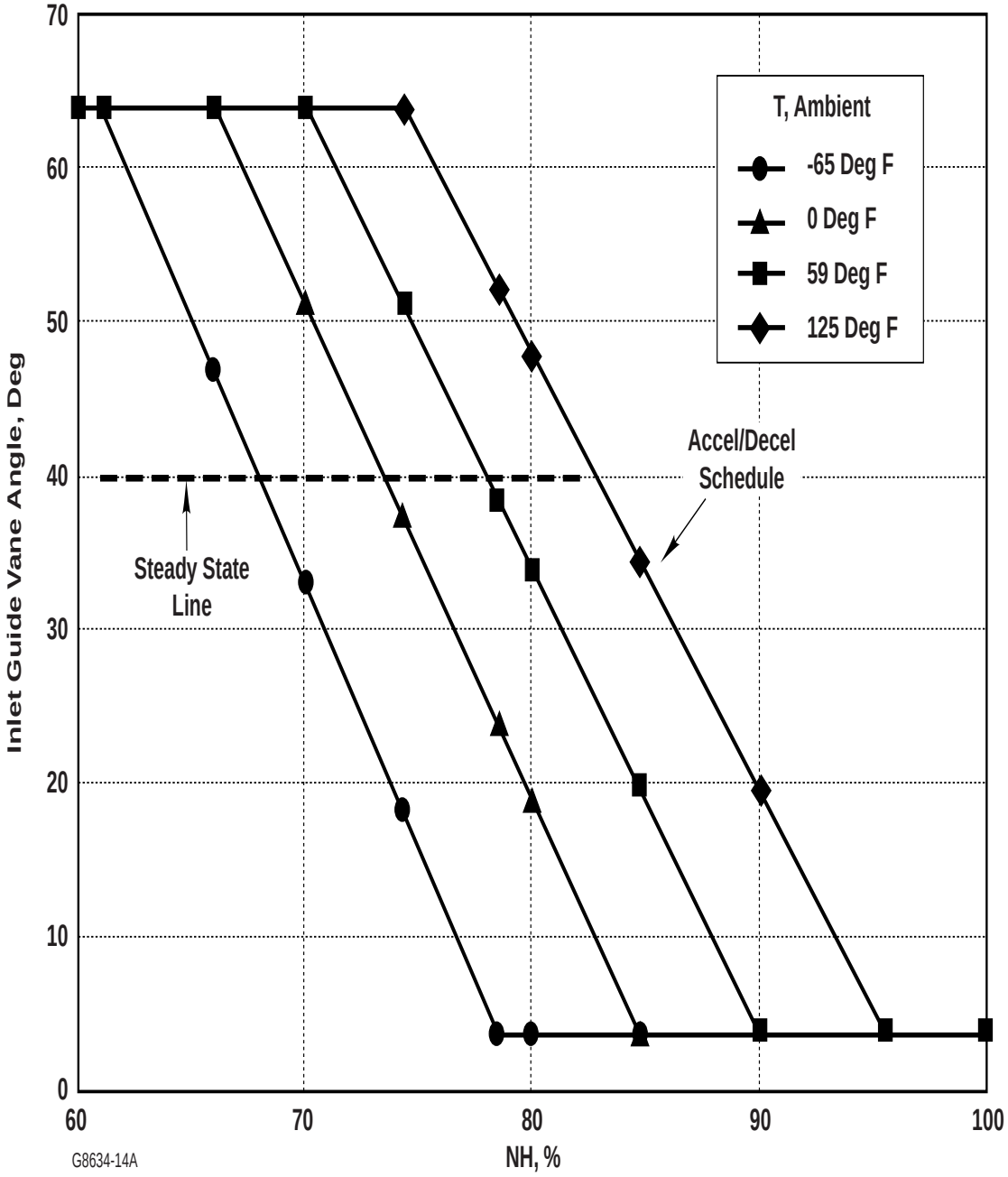
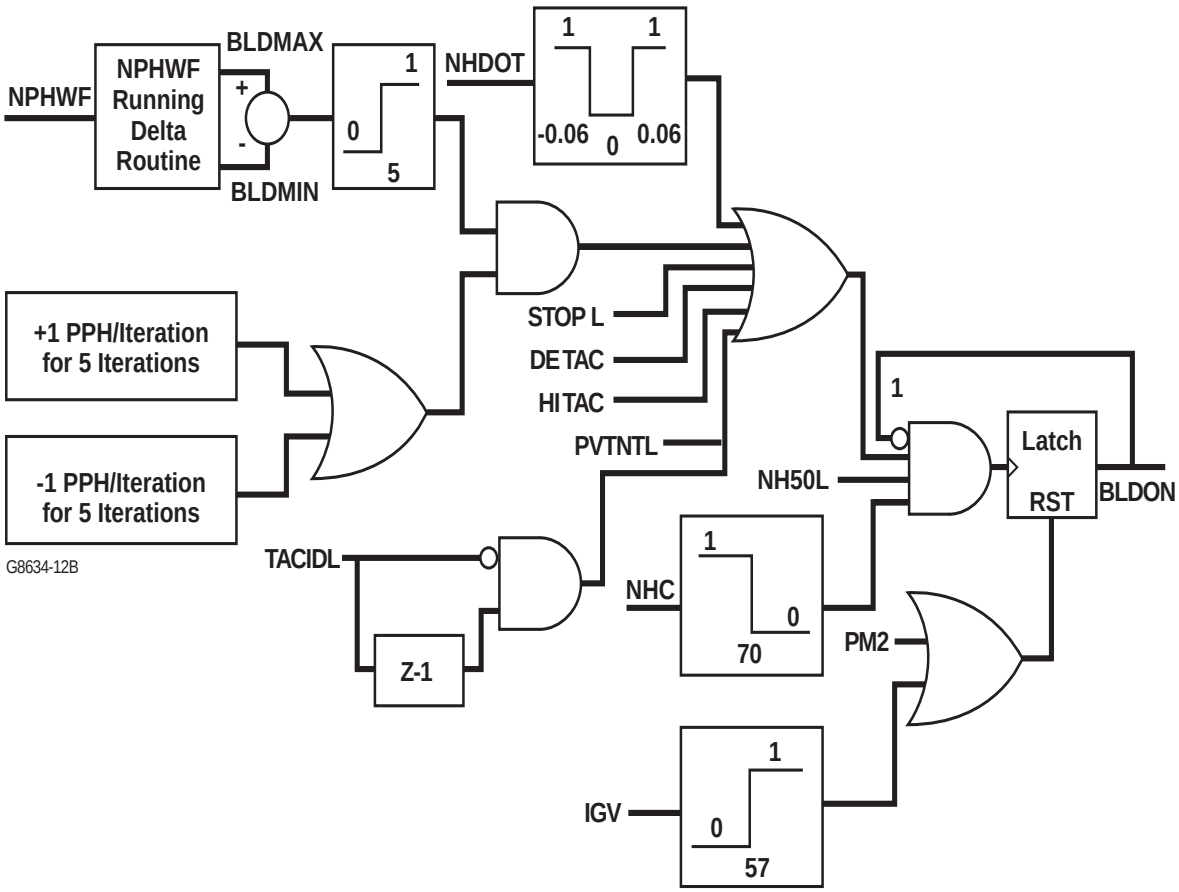


FIGURE 14. Inlet guide schedule.

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The Activation of BLDON Sets IGVR to 64° and Holds WFR at Its Current Level

FIGURE 15. IGV prioritization logic.

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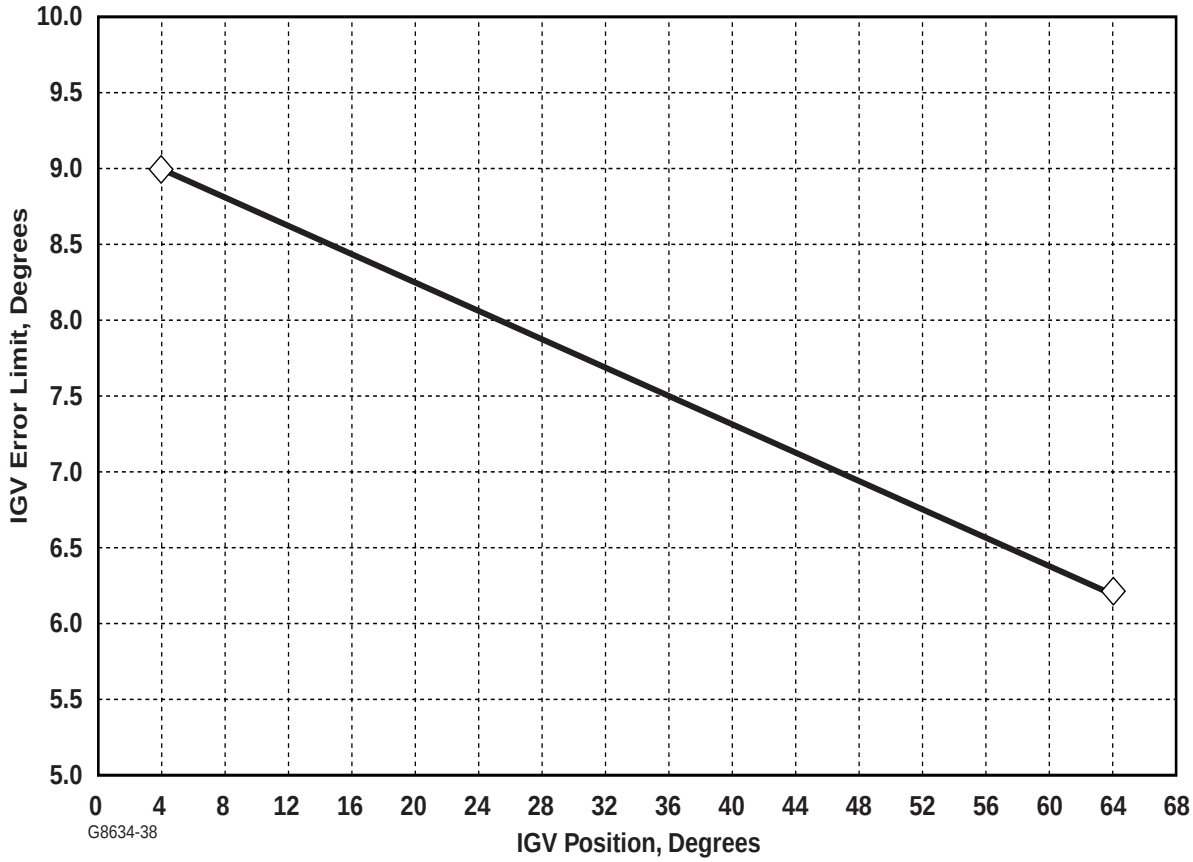


FIGURE 16. IGV error limits.

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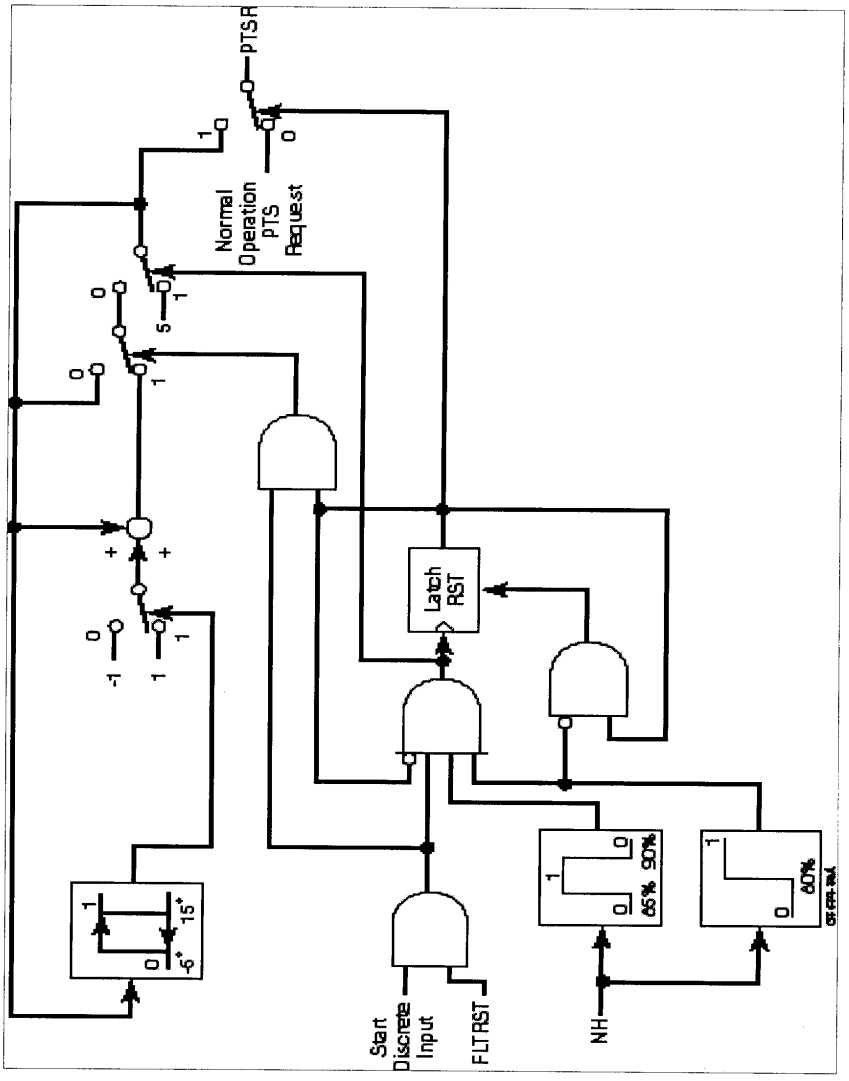


FIGURE 17. Test cell PTS logic.

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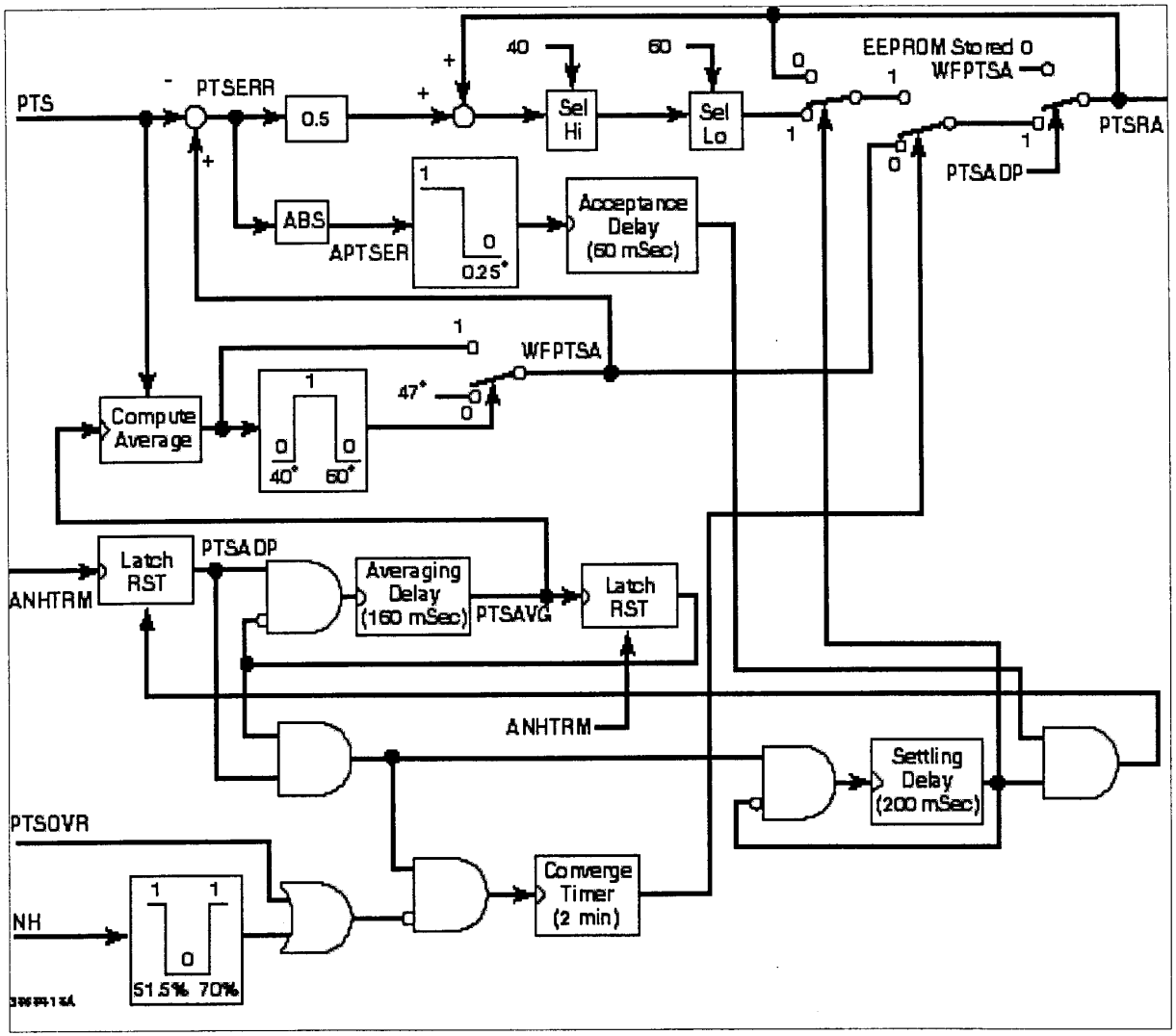


FIGURE 18. PTS adaptive routine.

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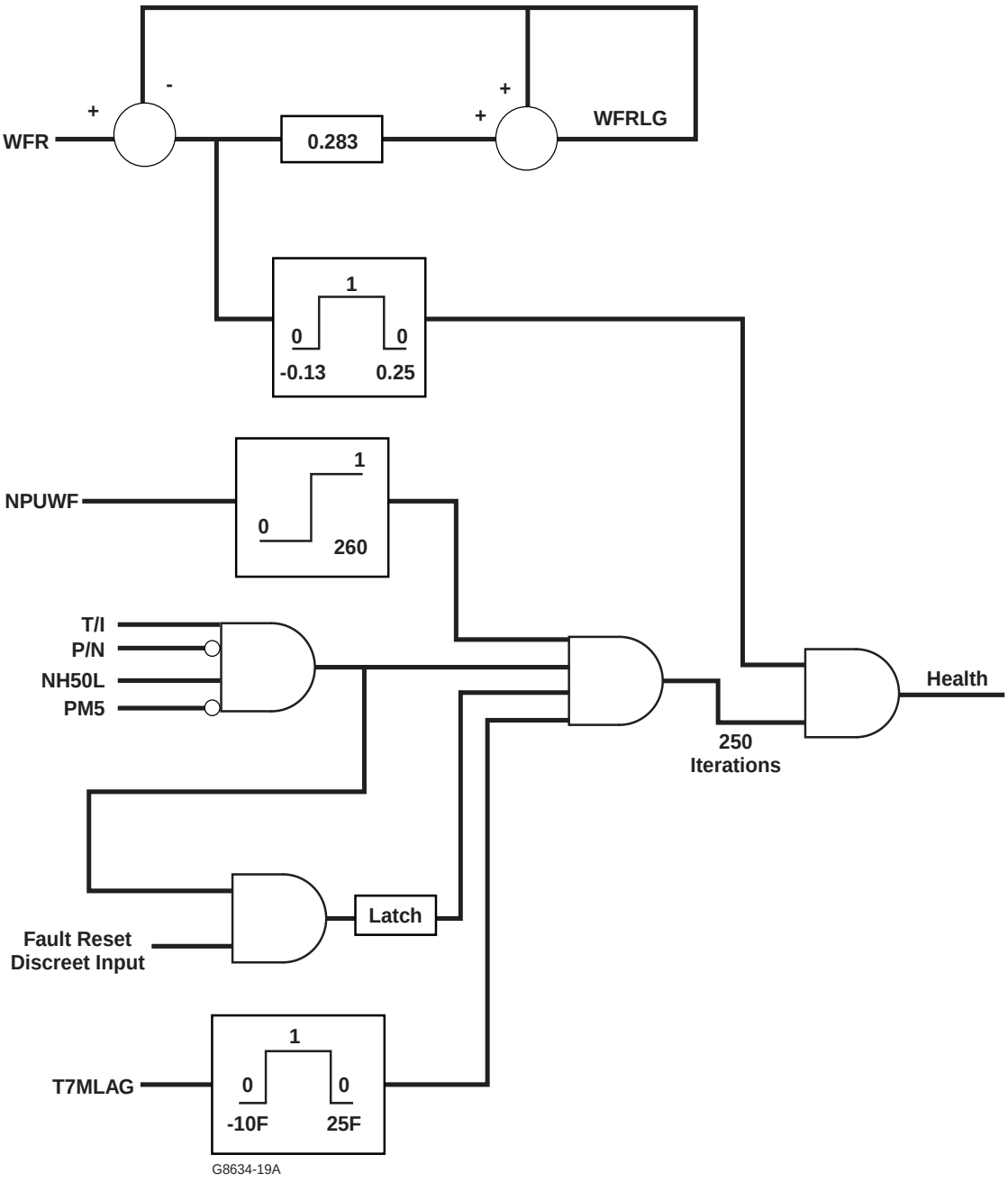


FIGURE 19. Engine health check algorithm.

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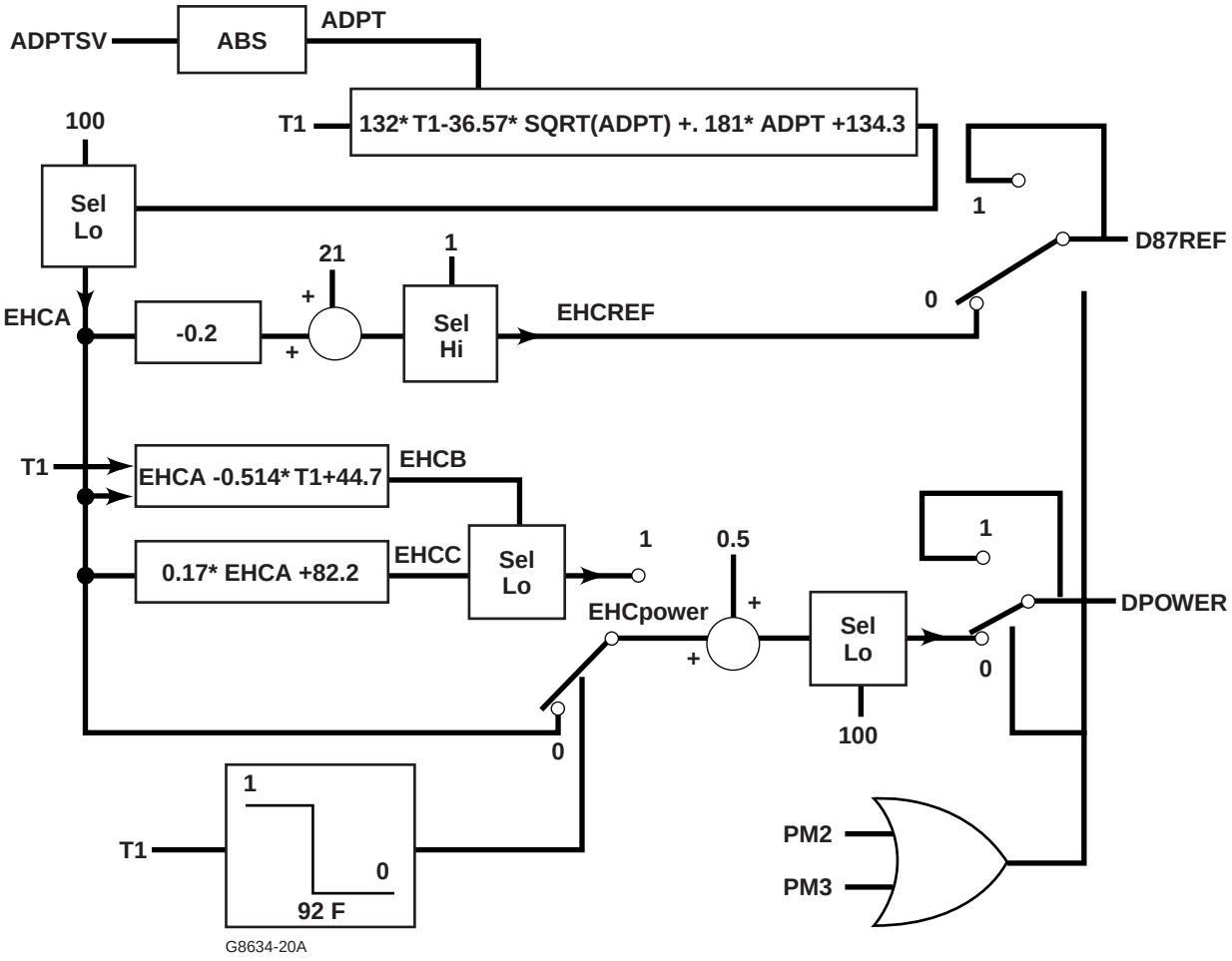


FIGURE 20. Engine health check algorithm.

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Check the source to verify that this is the current version before use.

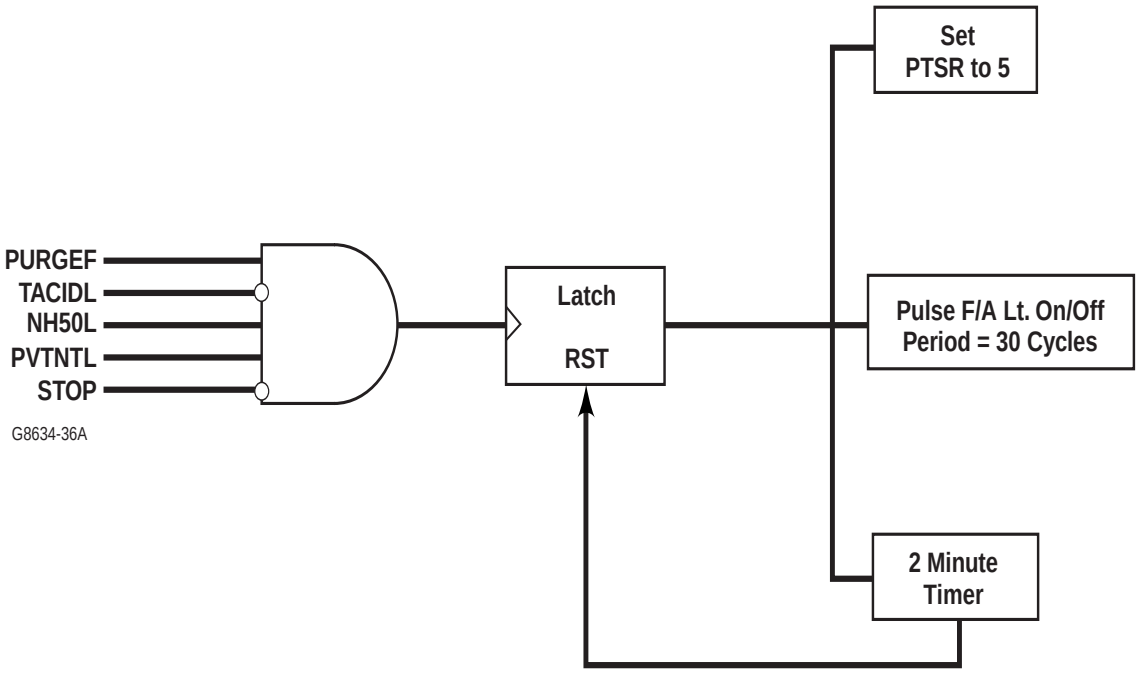


FIGURE 21. Cooldown logic, type I and Ia.

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Check the source to verify that this is the current version before use.

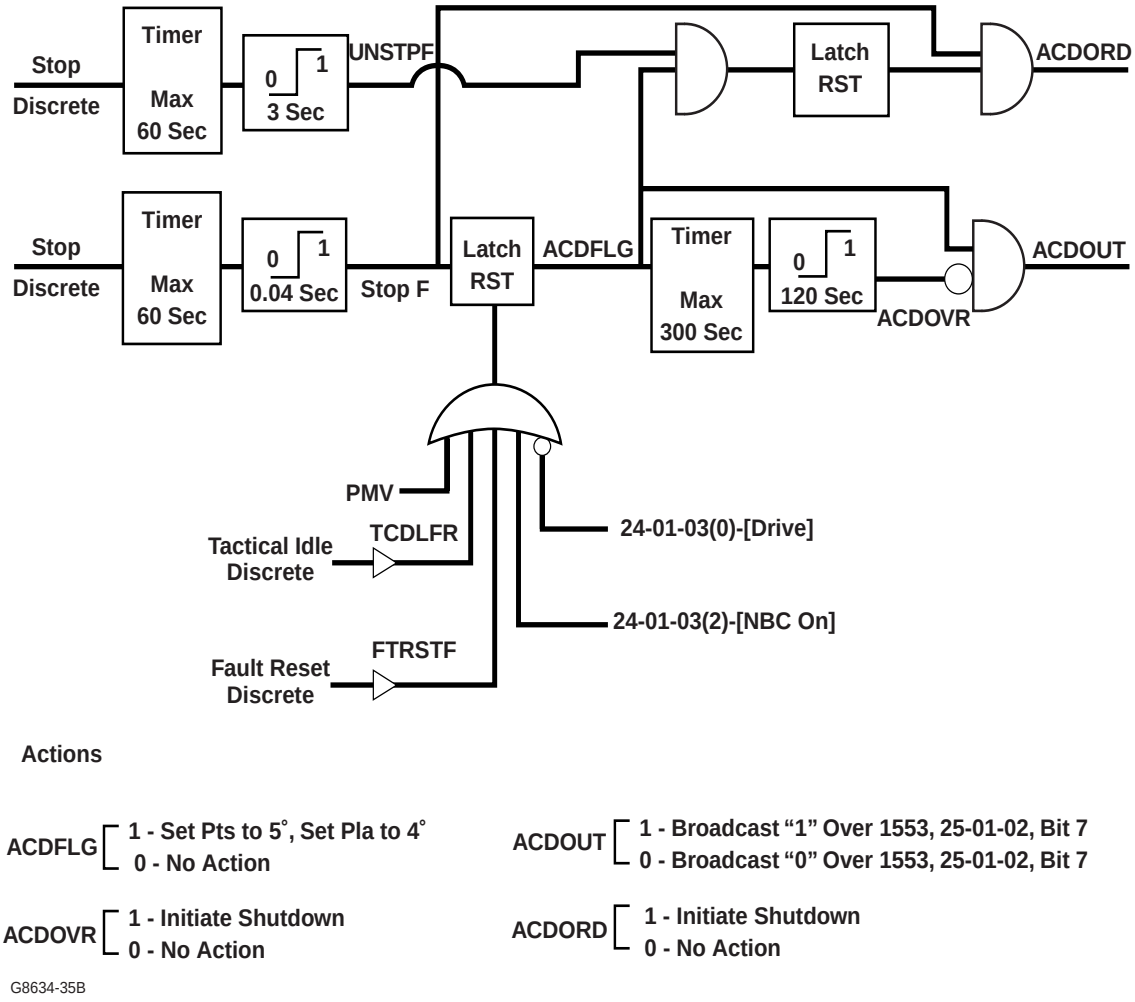


FIGURE 22. Cooldown logic, type II.

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Check the source to verify that this is the current version before use.

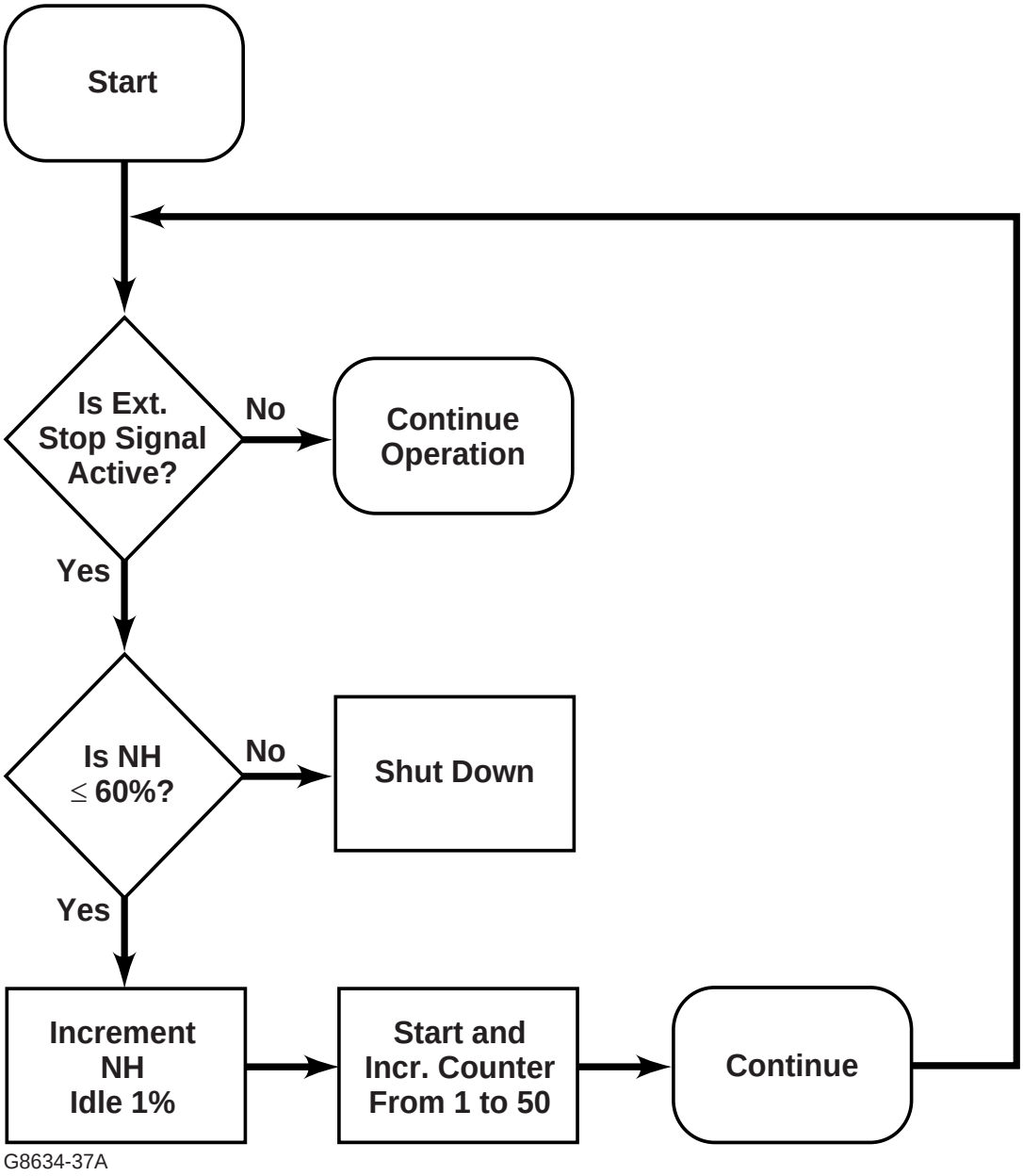


FIGURE 23. Low oil pressure shutdown logic.

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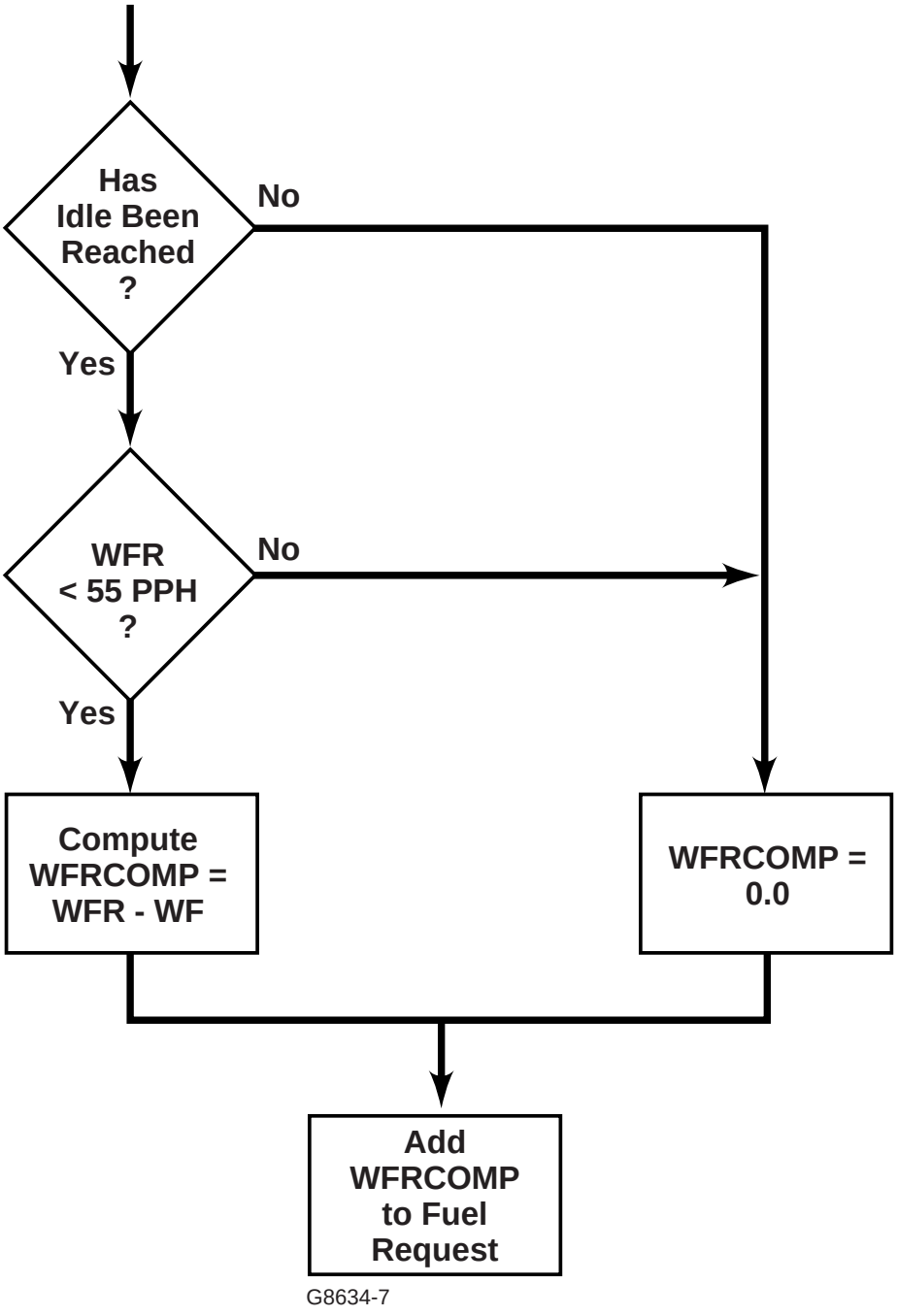


FIGURE 24. WF trimming procedure.

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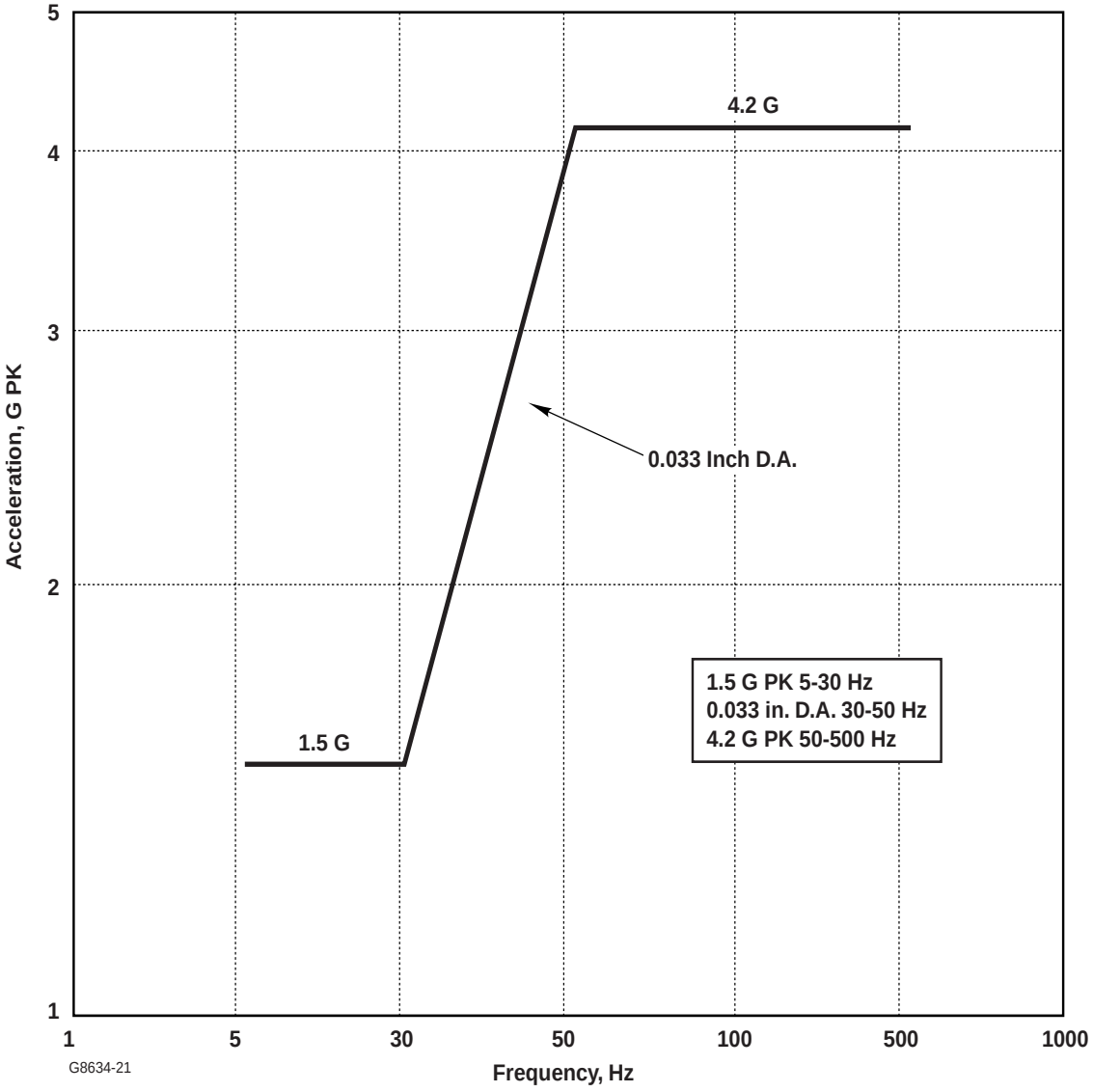


FIGURE 25. Peak vibration, type I and Ia.

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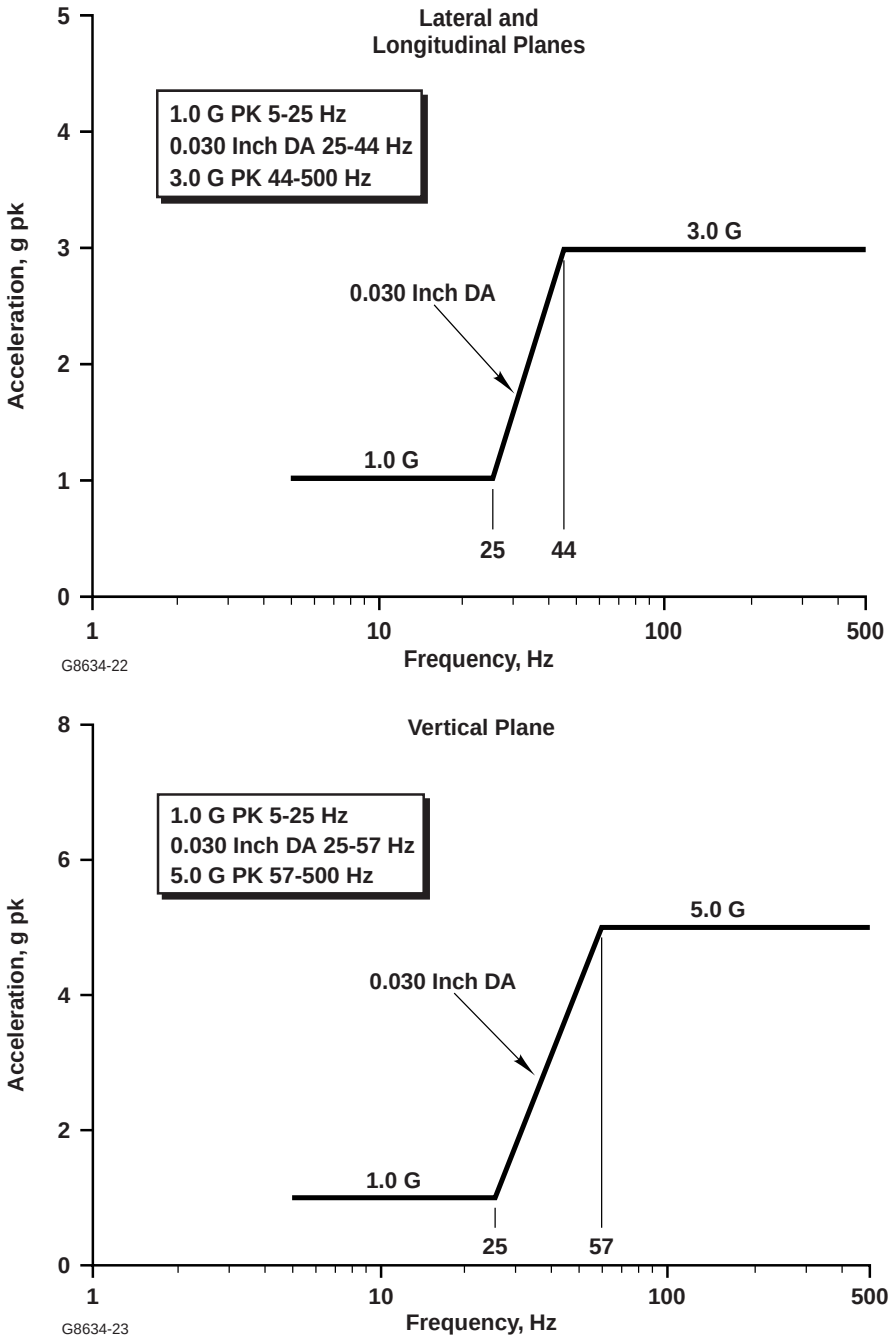


FIGURE 26. Peak vibration, type II with isolation mount.

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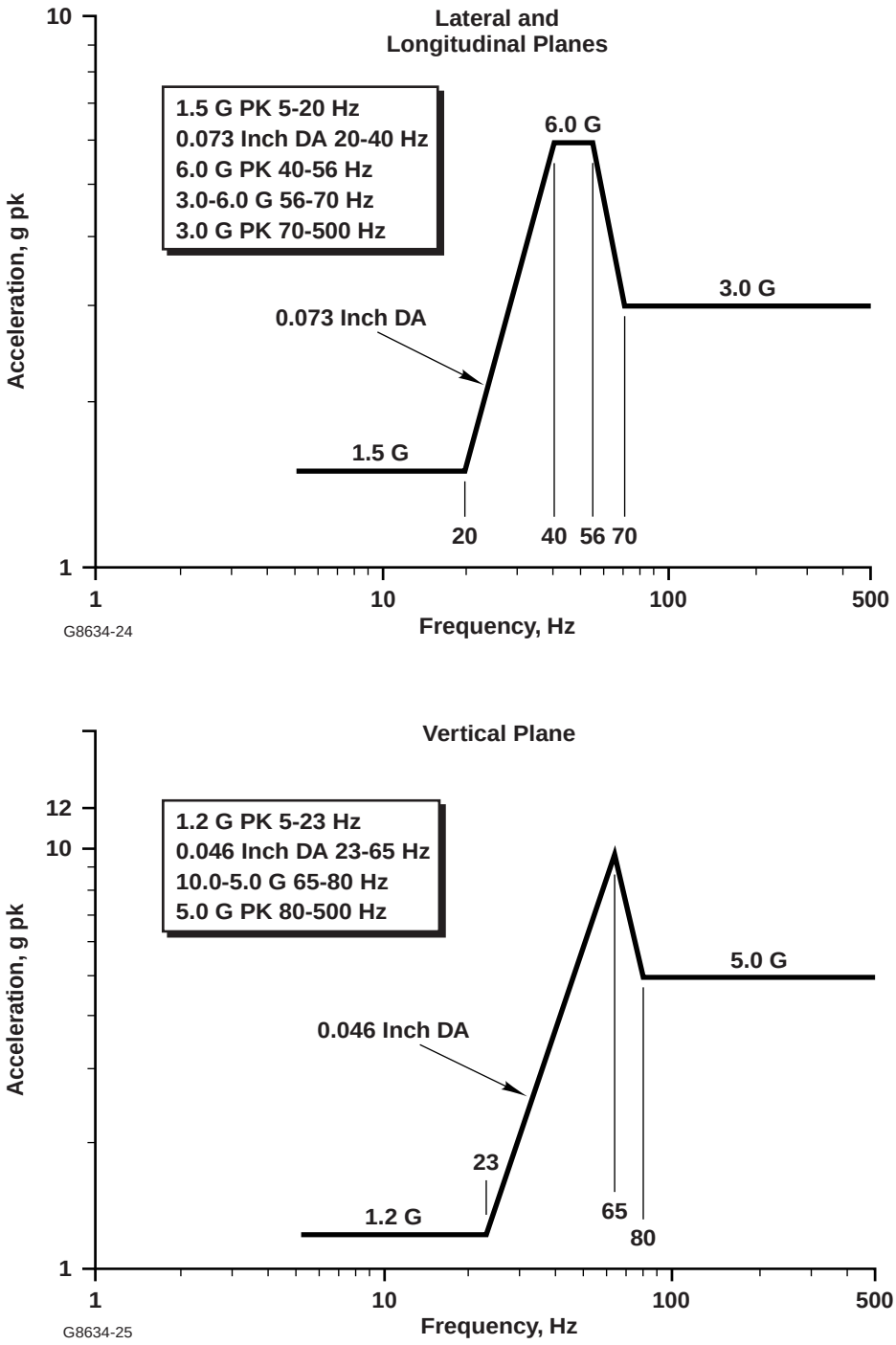


FIGURE 27. Peak vibration, type II without isolation mount.

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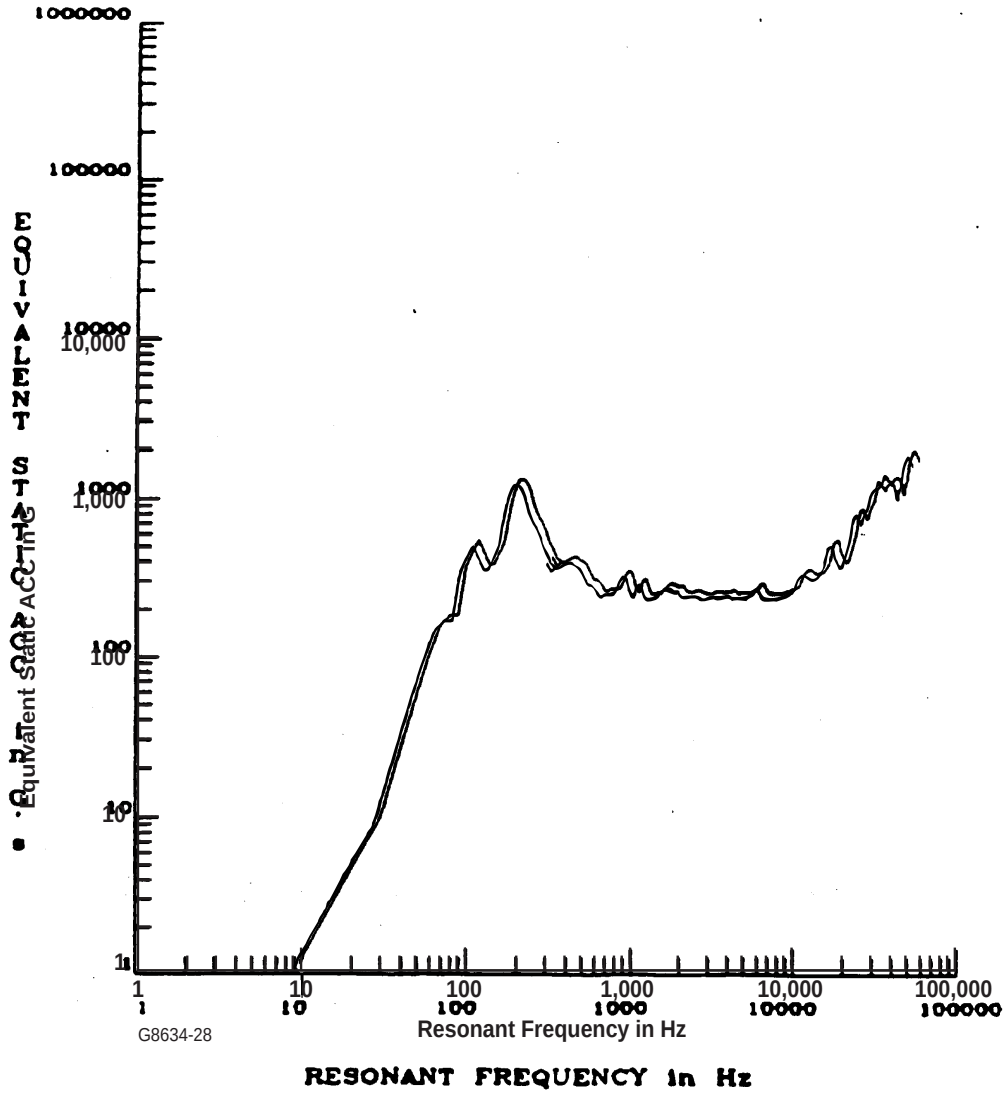


FIGURE 28. Ballistic shock spectrum.

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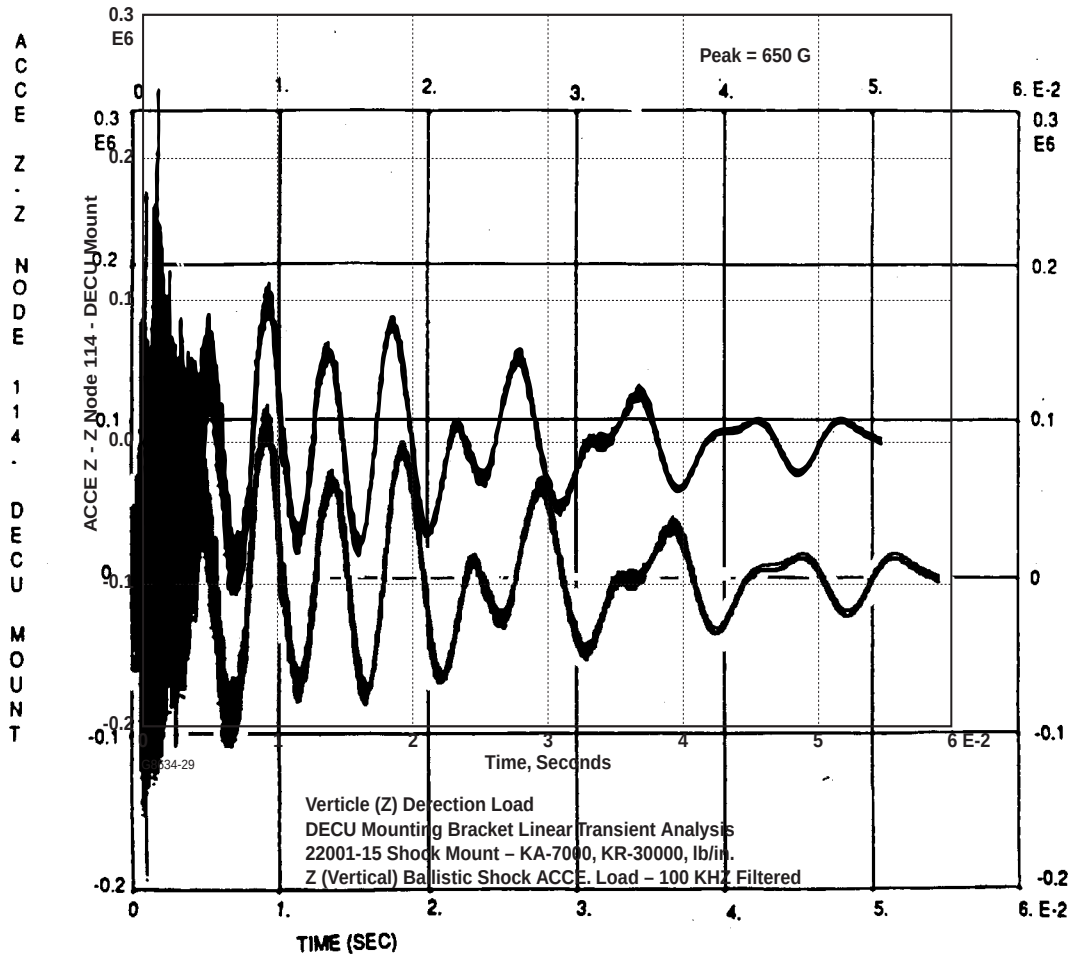


FIGURE 29. Ballistic shock output.

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APPENDIX A

CONTROL, UNIT, ELECTRONIC, DIGITAL (DECU)

A.1 Scope. This Appendix is a mandatory part of the specification. The information contained herein is intended for compliance. The assembly shall be able to communicate using RS232 serial port protocol.

A.2 References. None.

A.3 Requirements. The assembly shall be able to respond to commands issued from external devices via the RS232 serial port.

A.3.1 Read request. The assembly shall be able to decipher the following information:

a. Source of the command (header)
b. Message type
c. Data length and check sum
d. Identifier for request status

The assembly shall echo back the same header byte when it returns the requested information. The block request/status byte shall support the following:

TABLE A-I. Speed and clock words.

Word number	Word name	Description	Units	Range
1	NH1	HPC speed	Hz	0 to 2597
2	NH2	HPC speed	Hz	0 to 597
3	NPT1	PT speed	Hz	0 to 640
4	NPT2	PT speed	Hz	0 to 2640
5	SNH1	NH1 BIT voltage	V	0 to 10
6	SNH2	NH2 BIT voltage	V	0 to 10
7	SNPT1	NPT1 BIT voltage	V	0 to 10
8	SNPT2	NPT2 BIT voltage	V	0 to 10
9	RTCVLT	Clock battery	V	0 to 5
10	CTYEAR	Year		0 to 99
11	DTDATE	Date		1 to 31
12	CTMON	Month		1 to 12
13	CTHOUR	Hour		1 to 24

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APPENDIX A

TABLE A-I. Speed and clock words - Continued.

Word number	Word name	Description	Units	Range
14	CTMIN	Minute		1 to 59
15	CTSEC	Second		1 to 59
16	CTHUND	Hundredth		1 to 99
17	DISIN	Discrete inputs		Binary bits

TABLE A-II. Analog input words.

Word number	Word name	Description	Units	Range
1	PLA	Power lever angle	Degrees	0 to 100
2	PTS	Power turbine stators	Degrees	0 to 70
3	IGV	Inlet guide vanes	Degrees	0 to 70
4	WF	Fuel flow	LB/H	20 to 900
5	SVPLA	PLA BIT voltage	V	-10 to 0
6	SVPTS	PTS BIT voltage	V	-10 to 0
7	SVIGV	IGV BIT voltage	V	-10 to 0
8	SVWF	WF BIT voltage	V	-10 to 0
9	T7	PTIT	F	0 to 2000
10	ST7P	T7 BIT voltage +	V	1 to 7
11	ST7M	T7 BIT voltage -	V	6.5 to 10
12	TCZERO	T7 zero voltage	V	-9.93 to -9.59
13	CJUNCT	Cold junction temperature	F	-65 to 178
14	T1	Inlet temperature	F	-100 to 200
15	ST1	T1 BIT voltage	V	-2.65 to –2.45
16	VP1OBT	+10V reference	V	0 to 2.5
17	VRM10R	- 10V reference	V	-5 to 0
18	VRAGND	Analog ground ref.	V	-.1 to .1
19	VRP15R	+ 15V reference	V	9.95 to 10
20	VRM15R	- 15V reference	V	-10 to –9.96

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APPENDIX A

TABLE A-III. Digital to analog words.

Word number	Word name	Description	Units	Range
1	PBVWF	WF primary voltage	V	0 to 6
2	PBVPLA	PLA primary voltage	V	0 to 6
3	PBVIGV	IGV primary voltage	V	0 to 6
4	PBVPTS	PTS primary voltage	V	0 to 6
5	DACHK0	D/A BIT check voltage	V	-5
6	DACHK1	D/A BIT check voltage	V	0
7	DACHK2	D/A BIT check voltage	V	5
8	TRIVLT	Tri wave voltage	V	0 to 10
9	BCKVLT	Backup battery voltage	V	16 to 23
10	PSVLT	Primary input voltage	V	0 to 30

TABLE A-IV. Output driver words.

Word number	Word name	Description	Units	Range
1	OSTRC	Start relay current	V	0 to 1
2	ODSTRV	Start relay voltage	V	0 to 4
3	ODRTMC	Runtime meter current	V	0 to 10
4	ODRTMV	Runtime meter voltage	V	0 to 10
5	ODIGNC	Ignition exciter current	V	0 to 1
6	ODIGNV	Ignition exciter voltage	V	0 to 4
7	ODBCKC	Backup solenoid current	V	0 to 1
8	ODBCKV	Backup solenoid voltage	V	0 to 4
9	ODCOSC	Cutoff solenoid current	V	0 to 3
10	ODCOSV	Cutoff solenoid voltage	V	0 to 4
11	ODPTSC	PTS current	V	0 to 2
12	ODPTSV	PTS voltage	V	0 to 4
13	ODIGVC	IGV current	V	0 to 2
14	ODIGVV	IGV voltage	V	0 to 4
15	ODWFC	WF current	V	0 to 2
16	ODWFV	WF voltage	V	0 to 4

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APPENDIX A

TABLE A-V. Fault EEPROM words.

Word number	Word name	Description	Units	Range
1	NPIDLE	NPT idle point	%	25 to 30
2	INVAL	NH trim index		0 to 25
3	NHTRM	NH trim value	%	3.8 to 8.8
4	D87REF	Engine health check		1 to 14
5	DPOWER	Percent day power	%	0 to 100
6	WFPTSA	PTS trim value	Degrees	40 to 60
7	EEFLT	Diagnostic word		Binary bits
8	HFLT0	BIT message 1		Binary bits
9	HFLT1	BIT message 2		Binary bits
10	IRATE	Input rate fault word		Binary bits
11	IRNGE	Input range fault word		Binary bits
12	IFAULT	Input fault word		Binary bits
13	ENGMON	Engine hours	Hours	0 to 32 767
14	T7MON	T7 hours	Hours	0 to 32 767
15	NHMON	NH hours	Hours	0 to 32 767
16	NPMON	NPT hours	Hours	0 to 32 767
17	STMON	Number of starts		0 to 32 767
18	FM2MON	Number of PMII		0 to 32 767
19	FM3MON	Number of PMIII		0 to 32 767

TABLE A-VI. Built in test messages.

Number	BIT message	Description
1	NH1	NH1 sensor failure
2	NH1 INT	NH1 sensor intermittent
3	NH2	NH2 sensor failure
4	NH2 INT	NH2 sensor intermittent
5	NPT1	NPT1 sensor failure
6	NPT1 INT	NPT1 sensor intermittent
7	NPT2	NPT2 sensor failure
8	NPT2 INT	NPT2 sensor intermittent
9	T1	T1 sensor failure

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TABLE A-VI. Built in test messages - Continued.

Number	BIT message	Description
10	T1 INT	T1 sensor intermittent
11	T7	T7 sensor failure
12	T7 INT	T7 sensor intermittent
13	CO SOL	Cutoff solenoid failure
14	BU SOL	Backup solenoid failure
15	PT RLY	Starter pilot relay failure
16	IGN EXC	Ignition exciter failure
17	WF SOL	WF solenoid failure
18	WF PRI	WF primary failure
19	WF SEC	WF secondary failure
20	WF LVDT INT	WF sensor intermittent
21	IGV SOL	IGV solenoid failure
22	IGV PRI	IGV primary failure
23	IGV SEC	IGV secondary failure
24	IGV RVDT INT	IGV sensor intermittent
25	PTS SOL	PTS solenoid failure
26	PTS PRI	PTS primary failure
27	PTS SEC	PTS secondary failure
28	PTS RVDT INT	PTS sensor intermittent
29	PLA PRI	PLA primary failure
30	PLA SEC	PLA secondary failure
31	PLA RVDT INT	PLA sensor intermittent
32	FLAMEOUT	Engine flameout
33	ADJ IGV	Adjust IGV
34	ADJ PTS	Adjust PTS
35	LOP SHUTDOWN	Low oil pressure shutdown
36	DECU VOLTS	Low backup battery voltage
37	SYS VOLTS	Low system voltage
38	WF RESTRICT	Fuel flow restricted
39	WF ERROR	Fuel flow error
40	ROLLBACK	Engine rollback
41	DECU	DECU EEPROM error

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A.3.2 Write request. The assembly shall be able to change some of the parameters upon a write request with the following information:

a. Source of the command (header)
b. Message type
c. Data length and check sum
d. Identifier for request status

The word descriptions are as follows:

TABLE A-VII. Display word.

Number	Word name	Description	Range
1	DSPFLG(DSPCHK)	Display on or off	0 = Off, >0 = On

TABLE A-VIII. Digital to analog words.

Number	Word name	Description	Range
1	NPT	NPT meter output	0 to 120%
2	NH	NH test output	0 to 120%
3	T_7I	T_7I	0 to 2000°F
4	DISO	Discrete outputs	Discrete bits

TABLE A-IX. Output driver words.

Number	Word name	Description	Range
1	WFR	Fuel flow request	0 to 800 lb/h
2	IGVR	IGV request	0 to 70 degrees
3	PTSR	PTS request	0 to 70 degrees
4	DISO	Discrete outputs	Discrete bits

TABLE A-X. Power converter word.

Number	Word name	Description	Range
1	DRVRSL (BATCOM)	Backup battery disable	0 = enable

TABLE A-XI. Run BIT word.

Number	Word name	Description	Range
1	DSPFLG (BITOVR)	Run built in test	>0 = run

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TABLE A-XII. Date and time words setting.

Number	Word name	Description	Range
1	CIYEAR	Real time year	0 to 99
2	CIDATE	Real time date	1 to 31
3	CIMON	Real time month	1 to 12
4	CIHOUR	Real time hour	0 to 23
5	CIMIN	Real time minute	0 to 59
6	CISEC	Real time second	0 to 59

A.3.3 Programming EEPROM. The assembly memory shall be accessible to the RS232 for both read and write functions. The vendor shall define the protocol use for programming and verifying the memory.

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a. NAME	b. TELEPHONE *(Include Area Code)* (1) Commercial (586) 574-8745 (2) DSN 786-8745
c. ADDRESS *(Include Zip Code)* Commander U.S. Army Tank-automotive and Armaments Command ATTN: AMSTA-TR-E/AEIT 6501 E. 11 MILE ROAD Warren, MI 48397-5000	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Standardization Program Office (DLSC-LM) 8725 John J. Kingman Road, Suite 2533 Fort Belvoir, Virginia 22060-6221 Telephone (703) 767-6888 DSN 427-6888

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